Forecasting Intraday Realized Variance with Exogenous Features and Block Ridge Regression

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Abstract

We forecast thirty-minute-ahead S&P 500 realized variance on a near-24-hour panel of over a quarter million bars (1998–2024), asking what forty-one exogenous channels add to a HAR backbone, which estimator can extract it, and what structure the nonlinearity has. Three discipline findings precede the model findings: the representation of missing and stale data moves accuracy more than any modeling decision; the back-transform from the fitting scale to the variance scale is a modeling choice, and a calibrated, strictly causal second-moment correction beats the retrospective Duan convention on every arm; and causal hyperparameter tuning on short validation tails is repeatedly worse than measured constants. The exogenous information is real but dense and weak: invisible to selection and unpenalized joint fits, recoverable only by shrinkage, and optimally handled by a two-block ridge with penalties differing by two orders of magnitude. The recoverable nonlinearity is small beside that information effect and sits at pairwise order — products of columns the panel already contains, gated on the session clock. The two-block ridge — one exact rolling solve, two fixed penalties — improves the benchmark at strong paired significance.

1 Introduction

This paper forecasts thirty-minute-ahead realized variance for the S&P 500 complex on a near-twenty-four-hour panel running from January 1998 to April 2024 — over a quarter million scored half-hour bars, most of them outside regular trading hours. The question is not whether volatility is predictable; on this panel the target's own lag ladder carries the overwhelming share of the predictable variation; the lag ladder is the benchmark throughout. The questions are what, if anything, forty-one exogenous channels — higher-order moments, liquidity, cross-sectional aggregates, sentiment, implied volatility, and options order flow — plus four scheduled FOMC calendar channels overlaying that design, add to that backbone; which estimator class can extract it; and what structure the extractable nonlinearity has.

Every answer is measured out of sample, bar by bar, under one common evaluation protocol, and every claim is paired against the same per-bar least-squares benchmark (QLIKE 0.22521).

Four findings organize the paper. The first is about the data itself. On a mixed-frequency panel, the representation of missing and stale data is a larger determinant of forecast accuracy than any modeling decision we report: at any given bar, over a third of the exogenous design is forward-filled rather than freshly printed, and a construction that cannot distinguish the two produces single-window failures orders of magnitude larger than every model-class effect in the paper. The preparation pipeline — availability indicators computed before the fill, masked scaling statistics, neutral substitution — is therefore presented in full, with the incident record attached.

The second finding is about evaluation. Models are fit on a variance-stabilized scale and scored on the raw variance scale, and the back-transform between them is not neutral: squaring a calibrated forecast is biased low by exactly the model's own residual variance, so the correction is part of each arm's forecast. We replace the standard static scalar correction with a calibrated, strictly causal retransformation protocol — a Mincer–Zarnowitz mean map re-estimated per window and an adaptive second moment seeded by the previous window and updated bar by bar — which beats the retrospective Duan [1983] convention it replaces on every arm measured, and beats it causally. A mean-reverting GARCH$(1,1)$ refinement [Bollerslev, 1986, Engle and Mezrich, 1996] of the same recursion — the covariance-stationary case, not the integrated EWMA of Engle and Bollerslev [1986] — improves further still and is reported as an available, not adopted, upgrade.

The third finding is about the exogenous information and the estimator jointly — they cannot be separated. The exogenous channels carry real, statistically significant information, but none of it is recoverable by an unshrunk fit at this width: selection and unpenalized joint fits fail, and only shrinkage recovers the signal. The head-to-head that follows locates the structure precisely: ridge beats lasso on every design, the lasso's zeroes fall where zeroing is free, and no small set of columns or principal directions concentrates the signal. The coefficient structure is *dense but weak*, the regime of Giannone et al. [2021] and Shen and Xiu [2025] — many nonzero coefficients, none large — and the estimator built for it is ridge with a block-diagonal Tikhonov penalty: the strong low-dimensional backbone barely penalized, the wide exogenous block shrunk two orders of magnitude harder, with fixed penalties and no selection step anywhere.

The fourth finding is about nonlinearity. Gradient-boosted trees beat the best linear arm on all nine feature sets, but by a margin an order of magnitude smaller than the information effect, and decompositions of the tree fits attribute the bulk of that margin to additive-plus-pairwise structure — products of columns the panel already contains, gated on the session clock. That structure is a specification statement, not yet an estimator: the paper's linear ladder ends at the block-diagonal ridge — one solver, two fixed penalties, refit exactly at every bar — which scores 0.22350 at DM -3.1 against the benchmark's 0.22521. The paper closes by reading the forecasts against the last thirty minutes of the expiration-day chain. The last-30-min trade buys or sells one straddle at 15:30 ET on the sign of the gap between the forecast of the half hour's variance and the variance the straddle's quotes imply, and holds it to the cash settlement (Section 5.4). Scored under one recalibration on the same 866 days, every one of the paper's forecasts improves on selling the straddle every

day in point estimate, one of the eight with an interval above zero. The section's headline forecast is not the paper's block-diagonal ridge but a ridge fitted for the traded bar alone, on the 16 inputs a forecaster can rebuild at 15:30, chosen for that feasibility and not as the best of the table: a Sharpe ratio of 1.90 (1.44 at the crossed spread), above the paper's forecast by +0.90 [+0.19, +1.66] and above always short by +1.70 [+0.09, +3.11]. Its forecast is not significantly more accurate than the paper's ($p = 0.156$), no single change from the paper's forecast to it is resolved on its own, and it is not separated from the lasso on the same inputs. The edge is a tail trade — its 20 best days, all of them buys, carry 68% of its P&L — and the trade has shown no edge after the sample ends in April 2024. Sizing on the magnitude of the gap does not improve on its sign.

A methodological theme runs through all four: discipline beats tuning. Every hyperparameter in the campaign is either a fixed constant or selected causally on a short forward validation tail, and the campaign measures what that selection buys — repeatedly, less than it costs. Tuned elastic nets lose to tuned ridge not because the data prefer dense coefficients but because penalty mistakes are cheap for ridge and destructive for selectors; a fixed forgetting half-life helps and a tuned one eliminates the gain; fixed block penalties outperform their causally tuned counterparts. The final model's constants are justified individually by measurement and never optimized on evaluation loss.

Roadmap. Section 2 places the paper in the literature. Section 3 defines the panel, the preparation pipeline, and the exogenous buckets. Section 4 fixes the evaluation framework — the benchmark, the testing machinery, and the back-transformation protocol — and then the methods: the estimators and the block constructions. Section 5 reports the results in the order summarized above. Section 6 concludes. The appendix carries the full arm-level master table and the protocol notes; Appendix D collects side analyses of the last-30-min trade, its P&L by cell, and the diagnostics parked from Section 5.4.

2 Related Works

Realized-volatility modeling. The paper's target and backbone descend from the literature on realized measures: realized variance as a sum of squared intraday returns [Andersen and Bollerslev, 1997, Barndorff-Nielsen and Shephard, 2004], its approximate long memory documented by Andersen et al. [2003], and the heterogeneous autoregressive model of Corsi [2009], whose daily/weekly/monthly averages we adapt to a base-2 intraday ladder. Refinements of the HAR backbone are numerous: measurement-error-dependent coefficients [Bollerslev et al., 2016], state-dependent time variation [Bekierman and Manner, 2018], decoupled short- and long-run behavior [Bennedsen et al., 2022], joint models of returns and realized measures [Hansen et al., 2012], and multiple-indicator intraday structures [Engle and Gallo, 2006]. We keep the backbone deliberately plain — per-bar least squares on the lag ladder — because the paper's questions concern what can be added to it, and its strength makes every addition a stringent test: on this panel the HAR backbone's coefficients are 94–97% directionally stable over two decades, so a predictor that survives next to it carries information the ladder does not.

Machine learning for volatility and return prediction. Tree ensembles are strong off-the-shelf estimators in the recent machine-learning literature on asset pricing and volatility [Gu et al., 2020, Christensen et al., 2023], typically in gradient-boosted form [Ke et al., 2017, Chen and Guestrin, 2016]. Interpretability machinery — functional ANOVA, additive and pairwise additive models [Lou et al., 2013], and Shapley-based attribution [Lundberg and Lee, 2017, Lundberg et al., 2018] — is usually applied *post hoc*. This paper uses the trees as a specification search. The structure they recover — sparse pairwise products gated on the session clock — is written back into the linear design as explicit columns, and the distilled model is estimated in the same rolling least-squares path as every other arm.

Sparse selection versus dense shrinkage. The linear estimators are standard families — ridge [Hoerl and Kennard, 1970], lasso [Tibshirani, 1996], elastic net [Zou and Hastie, 2005], with the least-angle-regression path [Efron et al., 2004] behind our exact online lasso solve — but the question they are put to has a now-substantial literature: whether economic signal is concentrated in a few predictors or spread thinly across many. Giannone et al. [2021] compare sparse and dense representations across macro, micro, and finance prediction problems and find that the posterior rarely concentrates on a sparse model — sparsity is largely an illusion, and dense methods (ridge, factor models) predict best. Kozak et al. [2020] reach the parallel conclusion for the cross-section of expected returns, where a dense, heavily shrunk stochastic discount factor dominates sparse characteristic selections. Shen and Xiu [2025] supply the estimation theory: in the weak-signal regime the zero predictor is asymptotically Bayes-optimal, ridge can improve on it while lasso cannot for any penalty, and the ranking extends to nonlinear methods (dense random forests over sparse boosted trees, ℓ_2- over ℓ_1-penalized networks); lasso's failure reflects signal *weakness*, not necessarily the absence of sparsity. The rare-and-weak detection literature [Donoho and Jin, 2004] formalizes the underlying phase structure, in which a dense regime of individually undetectable effects is separated from the sparse regime by a sharp boundary. Our head-to-head (Section 5.1) is the intraday-volatility instance of this phenomenon, measured under a stricter protocol than the literature's cross-validation standard: every penalty is selected causally on a short forward tail, and the comparison is scored on a consistent loss. That protocol isolates a mechanism the asymptotic results abstract from: on short validation tails the price of a hyperparameter mistake is asymmetric across families — benign for ridge, whose loss is flat in the penalty near the optimum, destructive for selectors, whose penalty is partly a selection dial — which is why the paper's final model carries fixed penalties rather than tuned ones.

Retransformation. A mean model estimated on a log or square-root scale and scored on the original scale is not inverted by the algebraic inverse: by Jensen's inequality the naïve back-transform is biased for the original-scale mean. The standard nonparametric correction is Duan [1983]'s smearing estimator. Manning [1998] and Manning and Mullahy [2001] place that correction inside the forecast: under heteroscedasticity a single smearing factor is itself biased, so the retransformation is a modeling choice rather than a reporting footnote. This paper fits on a diurnally-adjusted square-root scale and scores QLIKE on raw variance, so the same gap applies. The back-transform is treated as part of the evaluation protocol — a calibrated, strictly causal second-moment correction that replaces the retrospective

in-window Duan scalar — and the comparison is reported in Section 4.2.

The clock of the variance risk premium. Hizmeri and Bevilacqua [2025] identify *when* volatility uncertainty is incorporated into real-time option prices. Volatility of volatility measured at 10:00 EST, during the U.S.–European overlap, predicts next-day variance-asset returns; that content is gone by midday. Variance prices respond with a delay, consistent with underreaction and slow-moving beliefs about volatility. This paper's last-bar option reading (Section 4.5) is the 15:30 end of the same clock. If the morning is when the early bird gets the vol, 15:30 is when that bird has already flown: under the headline forecast of Section 5.4 quoted implied variance exceeds the forecast on 60% of expiration days, so a two-sided reading of the forecast is mostly a short. What the forecast adds is not size on that short — sizing on the gap does not improve on its sign (Appendix D.11) — but the direction of the other days: the rule differs from always short only on the days it buys, and those days are worth +103.7 [+10.7, +200.1] premium units to it over the sample. We do not restage their next-day VVIX regression; the object here is same-day remaining variance against the expiration-day chain.

3 Data

3.1 The Panel

The raw data are 30-minute bars for the S&P 500 complex, gridded to a regular 30-minute frequency. The panel's extent is not set by a chosen date: **a bar exists in the panel if and only if the realized-variance source printed on it** — the same observed-print fact the availability indicators of Section 3.2 are built from, applied as the grid's own existence criterion. Under this rule the panel begins at the source's first print, in January 1998, and ends at its last, in April 2024. No weekend rule, holiday calendar, or session table is supplied anywhere: what is absent — weekends, and sessions that do not yet exist — is absent for that one reason, that the source did not print. The converse deserves saying, because a reader will assume otherwise: the source is a near-24-hour *futures* tape, so exchange holidays and early closes are not absent from it. A cash-market holiday is a shortened futures session and prints regular-hours bars ending at 13:00 or 11:30 with no 16:00 bar at all; an early close prints post-close bars right through 16:00, on which realized variance collapses to a few percent of what that slot normally carries. Both are legitimate rows of a futures panel and are kept. They are not legitimate rows of a *cash-session* exercise, and the last-bar option reading of Section 4.5 excludes them by an explicit calendar.

The panel's session structure therefore evolves as the market's did. The source prints around the clock from its first day — the panel's first bar is an overnight bar — but the early sessions are sparser: roughly 7,500 bars per year in 1998 (median 32 bars per trading day before 2004), growing through roughly 10,600 bars in 2003 to a stable $\approx$ 12,400 bars per year from 2005 onward, when the near-24-hour cycle (Sunday 18:30 through Friday 20:00, 48 bars per full trading day, roughly 34 of them outside regular hours) is fully live. The Friday boundary in that cycle is a grid convention and not a market fact: the futures week ends at 17:00 ET, and the bars the 20:00 cut keeps after it carry a thin tape — median

thirty-minute volume about one and a quarter percent of the 17:00 bar's — which the panel treats like any other low-activity slot rather than dropping. A time-of-day slot that thickens or enters mid-sample behaves exactly like a feed going live: the rolling per-slot estimators of Section 3.2 have whatever history the slot has and accumulate it causally. Any statement about intraday structure in this panel refers to the full cycle as it stood at the time, not to the 09:30–16:00 window alone.

The existence rule replaced an earlier calendar-only filter (weekend trim plus forward fill) after that filter produced a measurable incident: bars following an early close existed on the calendar grid as forward-filled dead-session bars, and on one such bar — the only one in more than twenty-five thousand — two feeds' availability indicators disagreed for a single bar, which turned a nearly singular solve into forecasts orders of magnitude off target. On the overlapping 2005–2024 span the rule removes 4,925 such dead-session bars ($\approx 2\%$ of the calendar grid). The accounting is: 303,442 print-bars on the grid, 300,317 panel rows after the construction burn-in, and 273,554 scored evaluation rows after the 24,000-bar training window and the calibration burn-in of Section B.1.

The forecasting target is realized variance. For each 30-minute bar it is computed as the sum of squared one-minute log-returns within the bar:

$$RV_{d,j} = \sum_{i=1}^{M_{d,j}} r_{d,j,i}^2, \qquad M_{d,j} \le 30, \tag{1}$$

where $r_{d,j,i}$ is the i-th one-minute log-return inside the j-th 30-minute bar of day d, and $M_{d,j}$ is the number of returns the tape printed in that bar. The vendor supplies that count beside every bar; it reaches exactly 30 and never exceeds it, which is how the sampling frequency is read off the file rather than assumed. Seventy-one percent of the panel's bars are full, and short bars are overnight and early-sample bars, not regular-hours ones: 99% of bars in the 10:00–16:00 window carry all thirty returns. Bars are indexed sequentially as RV_t. The forecast horizon is one bar ahead. The arrays carry that alignment the other way round: row t holds the target RV_t beside features every one of which is shifted by one bar, so a row's regressors are known at the end of bar $t-1$ and the model standing there forecasts the bar the row is labelled by. Throughout, "realized variance" denotes the target RV_t, and "volatility" denotes its square root; models are estimated on the square-root scale (Section 3.2). The construction in Equation (1) is the standard realized-measure definition. The heterogeneous autoregressive model of Corsi [2009] uses the trading day as its base unit; here both the target and the lag ladder are defined at 30-minute resolution.

The evaluation panel. Feature construction consumes a fixed burn-in of 3,125 bars at the start of the sample (January to June 1998), a constant of the panel construction that exceeds the longest feature lookback. This reduces the grid's 303,442 print-bars to a usable panel of 300,317 bars. The walk-forward protocol consumes a further 24,000 bars as the initial training window, and the calibration burn-in of Section B.1 consumes the first evaluation window. Every arm reported in this paper is scored on the identical remaining rows; the paired bar-for-bar comparisons used throughout require this shared row index. The scored evaluation panel contains $n = 273{,}554$ bars.

Exogenous information. Beyond the target's own history, the design uses 41 exogenous channels. The channels are organised into thematic buckets that partition the columns exactly ($7 + 12 + 6 + 6 + 3 + 3 + 4 = 41$):

- **Moments** (7): own-asset higher-order return moments: bipower variation, realized semi-variance, autocovariance, absolute, cubic, and quartic returns.
- **Liquidity** (12): trading volume, turnover (total, buy-initiated, sell-initiated), effective and quoted spreads, number of trades.
- **Market, equal-weighted** (6) and **value-weighted** (6): cross-sectional stock-factor aggregates of return moments, bipower variation, semi-variance, and absolute returns.
- **Sentiment** (3): StockTwits attention, sentiment score, and sentiment count.
- **Implied volatility** (3): VIX, VVIX, VIX3M.
- **Volatility demand** (4): options order-flow indicators (SPX open/close, all options open/close, open only).

The buckets differ in column count; liquidity contributes twelve columns and sentiment three. A per-bucket comparison therefore mixes information content with dimensionality unless it is read per feature. Availability also differs across buckets: the cross-sectional aggregates are observed on 138,357 bars, the options order-flow bucket on 54,062 bars, and the VIX family does not exist for the earliest part of the sample. Row deletion would score each bucket on a different sub-sample. Section 3.2 describes the impute-and-indicate construction used instead; it scores every bucket on the same rows.

FOMC calendar channels. Four further columns enter as a scheduled public calendar overlay on top of the 41 exogenous channels. They are not part of the 7+12+6+6+3+3+4 = 41 partition above. The source is the `fomc release` flag: one 14:30 bar per scheduled FOMC statement (246 events; last live scheduled print 2023-11-01 14:30). `fomc_release` is 1 on that statement bar; `fomc_day` is 1 on every bar of a scheduled-statement day; `fomc_until_inv` and `fomc_since_inv` are the inverse bar-distance to the next and previous scheduled event. The unscheduled 2020-03-03 action is excluded from all four. After the last live scheduled print the four channels are missing (a dead tail), so the impute-and-indicate construction of Section 3.2 sees a stopped feed rather than a wall of zeros.

OLS on the HAR ladder plus this FOMC calendar bucket beats the baseline (HAR + calendar OLS) on the shared evaluation panel (Table 1): scheduled calendar information in the design, not a post-forecast filter. Joint OLS on all features, with the FOMC channels included among the rest, still loses to the baseline (HAR + calendar OLS).

Table 1: FOMC calendar channels on the shared evaluation panel of Section 3.1, $n = 273{,}554$. QLIKE and Diebold–Mariano t versus the baseline (HAR + calendar OLS); negative t favors the arm.

Arm	QLIKE	DM t
baseline (HAR + calendar OLS)	0.22521	+0.0
FOMC calendar bucket (OLS)	0.22400	−3.6
all features (OLS)	0.22819	+4.6
ridge (joint)	0.22551	+0.5
lasso	0.22231	−4.6
block-diag ridge	0.22206	−5.6

3.2 Preparation

This subsection specifies the preparation pipeline. The pipeline is presented in full because, measured on this panel, the representation of missing and stale data has a larger effect on QLIKE than any model-class, basis, or penalty decision reported in this paper; the magnitudes are given under "Rolling robust scaling" below. At any given bar, 36% of the exogenous design consists of forward-filled values rather than new measurements; on the sparsest channels the figure is 78%. Without availability information, an estimator cannot distinguish a filled value from a new measurement.

Overnight fills and availability indicators. The 41 exogenous channels are not all published on the near-24-hour grid. Most are US cash- or extended-session equity quantities; each prints only inside its own session. Averaged across channels, 0.638 of panel rows carry a print from the raw feed and the remaining 0.362 do not. Each channel is processed in the following order:

1. Compute a binary availability indicator from the raw feed: the indicator is 1 on bars where the feed printed and 0 otherwise.

2. Forward-fill the channel within its availability window.

3. Where the channel is structurally absent (no prior print exists), substitute a fixed in-distribution neutral value.

Every value column is accompanied by its indicator column. The indicator is computed before the fill. An earlier build of this pipeline computed the indicator after the fill, so every forward-filled bar was labelled observed. Under that build, the volatility-demand feed stopped publishing on 2023-08-31, the unlimited forward fill carried its final August print across the following eight months (8,453 bars, 3.44% of the sample), and the indicator read ≈ 1 throughout. Three further feeds (`vix`, `vix3m`, `vvix`) stopped publishing the same way on 2024-02-12. Every built cache is now gated on one check: for each channel, the mean of the constructed availability indicator must equal the fraction of rows on which the raw feed printed. The two quantities are estimates of the same number, so any gap indicates a

construction error of this kind. On the uncorrected panel the check fails for 31 of 41 channels. The check requires no model, no held-out data, and no threshold tuning. We recommend it for any mixed-frequency panel in which exogenous series are filled.

The indicator also enters estimation as a regressor. Its linear coefficient absorbs any level shift between the pre- and post-availability eras, and the value column contributes only where the feed exists. Every bucket, including implied volatility and volatility demand, is therefore scored on the same 218,934-row index. The exact paired significance tests of the later sections require this shared index.

Stage 1: diurnal adjustment. Intraday volatility has a time-of-day profile. Let $s(t) \in \{1, \ldots, 48\}$ index the time-of-day slot of bar t, and let $\mathcal{W}_t = \{j : j < t,\ s(j) = s(t)\}$ be the set of the most recent $W = 20$ observations sharing that slot, with at least 5 required before the baseline is valid. For non-negative series, the baseline is the same-slot rolling mean and the adjustment divides by it:

$$\bar{RV}_{t|s} = \frac{1}{|\mathcal{W}_t|} \sum_{j \in \mathcal{W}_t} RV_j, \qquad \widetilde{RV}_t = \frac{RV_t}{\bar{RV}_{t|s}}, \tag{2}$$

Signed series (e.g. autocovariance) are divided by the same-slot rolling standard deviation instead. The window condition is $j < t$, not $j \le t$, so the baseline at bar t uses no information from bar t; the adjustment is non-anticipative. Warm-up bars pass through with baseline 1. They are consumed by the burn-in and never enter the evaluation panel.

Stage 2: the winsorized square-root target. After the diurnal adjustment, each variable class receives a fixed variance-stabilising map. The map is chosen in advance from the class's support and tail behaviour; no transform parameters are fitted. The maps are: $x \mapsto \sqrt{x}$ for squared-return and volatility-like measures, including RV itself, bipower variation, turnover, and effective spread; $x \mapsto \operatorname{sign}(x)\sqrt{|x|}$ for autocovariance; $x \mapsto x^{1/3}$ and $x \mapsto x^{1/4}$ for third and fourth moments; and $x \mapsto \log x$ for the implied-volatility family. Models are estimated on the adjusted $\sqrt{RV}$ scale. After transformation, each series is clipped to rolling 1st and 99th percentile bounds estimated on the trailing $W_{\text{clip}} = 240$ observations (about five trading days) ending at $t-1$. Winsorization is not inverted at evaluation time; only the diurnal division and the square root are inverted. Because $\mathbb{E}[\sqrt{X}]^2 \neq \mathbb{E}[X]$, squaring a forecast of $\sqrt{RV}$ gives a downward-biased forecast of RV. The standard remedy in the literature is the classical retransformation estimator of Duan [1983] — add the in-sample residual variance to the squared forecast. We refine on that traditional construction with a calibrated, strictly causal *second-order adjustment*: a Mincer–Zarnowitz mean map re-estimated at every window boundary, and an adaptive second moment that tracks the model's own recent error level bar by bar. The refinement is specified and analyzed in Appendix B; its sensitivity against the na"ive inverse and the classical estimator is measured in Section 4.2. In raw form the back-transform is the traditional one:

$$\widehat{RV}_t^{\text{raw}} = \left(\hat{f}_t^{\,2} + \hat{\sigma}_\varepsilon^2\right) \cdot \bar{RV}_{t|s}^{\text{baseline}}, \qquad \hat{\sigma}_\varepsilon^2 = \frac{1}{N}\sum_i (y_i - \hat{y}_i)^2. \tag{3}$$

HAR features. The target's own history enters through trailing rolling means of the adjusted series on a base-2 geometric ladder,

$$\mathcal{L} = \{1,\ 2,\ 4,\ \ldots,\ 2^{k_{\max}}\}, \qquad \overline{RV}_t^{(k)} = \frac{1}{k}\sum_{i=0}^{k-1} \widetilde{RV}_{t-1-i}, \tag{4}$$

The rungs span roughly 30 minutes, 2.5 hours, half a trading day, 2.6 days, 13 days, and 65 trading days. The sum starts at $t-1$, so the feature at bar t contains no contemporaneous information. The ladder adapts the daily/weekly/monthly averages of Corsi [2009] to six intraday horizons. The same ladder is applied to each included exogenous channel, contributing $|\mathcal{L}| = 6$ rolling-mean columns per channel.

Rolling robust scaling. Input features to the linear estimators are standardised by subtracting a rolling median and dividing by a rolling interquartile range. Both statistics are estimated from the current training buffer and updated at every step. Two rules govern the estimation.

First, the median and IQR are estimated from observed rows only, i.e. rows on which the availability indicator is 1. If the statistics are instead estimated over forward-filled values, a series whose feed has stopped contributes identical repeated values, its rolling IQR shrinks toward zero, and the standardising division divides by a value near zero. On the uncorrected panel, the stale volatility-demand columns reached 2,260 rolling IQRs by this mechanism; the raw series did not move, and the denominator shrank.

Second, a scale is used only after 512 observed values have accumulated, and the IQR is floored at 1% of the largest scale the channel has shown. This rule exists because masked estimation fails at the start of a feed's life: with few observed rows, a low-dispersion opening window would otherwise set the scale for all subsequent bars.

The three elements — indicators computed before the fill, masked estimation, and neutral substitution — are applied together. Each element applied alone does not remove the failure. Applied together, they reduce the worst channel's maximum standardised value from 2,260.4 to 96.2 rolling IQRs. On the October 2023 window affected by the stale feed, the uncorrected panel scored QLIKE 6.37170 and the repaired panel scored QLIKE 0.12391. Every modelling effect reported in this paper lies between 0.0005 and 0.004 QLIKE. This difference in magnitude is the reason the preparation pipeline is presented before the modelling sections.

All stages are causal by construction, and the two places where the panel as shipped falls short of that are named here rather than left implicit. Every rolling statistic, fill, winsorization and scaling above uses only information dated strictly before t; the HAR ladder, the target, and the benchmark of Section 4.1 were re-derived under prefix truncation and reproduce bit for bit. Two stages of the exogenous build do not. The neutral value substituted where a channel is structurally absent (step 3 above) is the full-sample median of that column rather than a trailing one; and the availability indicators enter the design after the forward fill rather than before it, so on a filled bar they assert a live state the feed did not print. Neither stage touches the HAR block or the benchmark, which read no exogenous column. Their footprint on the design matrix is that 456 of 1,264 columns move under prefix truncation, by up to 0.8 standard deviations on the bars the last-bar reading of Section 5.4 uses and by much more at isolated points early in the sample. Both are one-line

pipeline changes; both require the panel to be rebuilt and every exogenous arm re-scored, and that rebuild is pending.

3.3 Descriptive Analysis

4 Methods

The methods come in two halves. The first half fixes the evaluation framework before any comparison is made: the comparator each arm is measured against and the testing machinery that accompanies it (Section 4.1); and the inheritance rule that makes the whole stack binding on every downstream comparison (Section 4.2). The back-transformation from the fitting scale to the variance scale, and the calibrated, strictly causal second-order adjustment it carries, are defined in Appendix B and summarized in Section 3.2. The second half defines what is compared: the estimators — one unpenalized member, the penalized linear family, and two tree ensembles — with the walk-forward protocol and the causal tuning rule every tuned arm obeys (Section 4.3); and ridge with a block-diagonal Tikhonov penalty (Section 4.4). Every estimator is fit on the shared panel of Section 3 and subject to the hyperparameter discipline of Section 4.1: no quantity is selected on evaluation loss.

4.1 The OLS–HAR Benchmark

Every result in the remainder of this paper is referenced to a single fixed comparator: ordinary least squares on the six HAR features of Equation (4) plus calendar dummies, with no exogenous variables, refit at every bar. The forecasting protocol is:

1. Walk-forward with a rolling origin. The training window is the most recent $T_{\text{train}} = 24{,}000$ bars, which is the window's definition. Measured on the panel that is about 480 trading sessions, not the 500 a flat 48 bars per session would imply, because early-sample and shortened sessions print fewer. The window slides forward by one bar per prediction.

2. The model is refit at every bar. The refit is exact and computationally feasible because the sliding-window normal equations admit rank-1 up-dating (add the entering bar) and down-dating (remove the exiting bar).

3. The benchmark is read from the ridge path at $\alpha = 0$; it is therefore exact OLS.

4. Each forecast is mapped back to raw variance through Equation (3).

5. Forecasts are scored by QLIKE [Patton, 2011],

$$L_{\text{QLIKE}} = \frac{1}{N}\sum_{t=1}^{N}\left[\, r_t - \log r_t - 1 \,\right], \qquad r_t = \frac{RV_t^{\text{raw}}}{\widehat{RV}_t^{\text{raw}}}, \tag{5}$$

computed on the full 273,554-bar evaluation panel. QLIKE is used for three reasons: it is scale-invariant in the level of volatility; it is strictly consistent for the conditional variance; and it penalises under-prediction more heavily than over-prediction, which matches

the direction in which forecast errors are more costly for this target. QLIKE values are not comparable across preprocessing regimes: on the diurnally adjusted panel of the reference study, the same class of models scores several tenths of a QLIKE unit lower than on a raw, unadjusted panel. Every comparison in this paper is therefore made within one preprocessing regime, and every table is labelled with the panel it was computed on.

Table 2: The benchmark: per-bar OLS on the HAR ladder plus calendar dummies, scored on the frozen panel under the evaluation protocol of Sections B.1–4.2. QLIKE is computed on the raw variance scale; the secondary metrics are computed on the fitting (square-root) scale where the model is estimated, on the same scored pairs.

Scored bars	273,554 (positivity exclusion removes 0.2%)
QLIKE (raw scale)	0.22521
Out-of-sample R^2 (fit scale)	0.417
Mincer–Zarnowitz α (fit scale)	-0.034
Mincer–Zarnowitz β (fit scale)	1.039
MSE (fit scale)	0.1397
MAE (fit scale)	0.2208

Squared-error metrics computed on the raw variance scale are uninformative on this panel: the median absolute error is 3.3×10^{-7} against a maximum of 35.6, so a handful of volatility-spike bars dominates every squared-error aggregate — the asymmetry that motivates QLIKE as the primary loss (Section 4.1).

Differences in QLIKE at the fourth decimal place require a significance assessment. Every comparison against this benchmark is therefore accompanied by a Diebold–Mariano test [Diebold and Mariano, 1995] on the per-bar loss differential. The test procedure is:

1. Compute the per-bar QLIKE loss for each of the two arms on the shared row index, and take the per-bar difference.

2. Estimate the long-run variance of the loss differential with a Newey–West estimator.

3. Apply the small-sample correction of Harvey et al. [1997] to the test statistic.

Where several arms are compared simultaneously, we report the Model Confidence Set of Hansen et al. [2011]. Every arm is scored on the identical row index, so all of these tests are paired bar-for-bar comparisons.

Hyperparameter discipline. No quantity anywhere in this paper is selected using evaluation loss. Every hyperparameter that is tuned is tuned *causally*: the penalty (or penalty vector) is re-selected periodically from a grid fixed in advance, by minimizing validation error on a forward split of the current training window — fit block, embargo, validation tail — so that no forecast's configuration uses information from that forecast's future. Every hyperparameter that is not tuned is a fixed constant, set before the campaign was scored and reported with its arm: the fixed-penalty arms exist precisely as controls for what tuning contributes, and the structural constants of the design (training-window length, re-selection

cadence, winsorization quantiles, block widths, estimation-window lengths) are conventions of the panel and estimator construction, disclosed where they are defined and never optimized against any loss.

4.2 A Common Evaluation Protocol

Every forecast in this paper passes through Equation (18) before any QLIKE number is computed. Every downstream comparison — the linear-class comparisons of Section 5.1, and the tree and final-model results of Sections 5.2 and 5.3 — scores its arms on raw-scale pairs produced by exactly this map, with exactly these window and residual choices.

Because the calibration pair and the second-moment coefficients are model-specific, the correction does not shift all arms by a common constant. It inflates the forecasts of an arm with larger residual variance by more, and it straightens the forecasts of a miscalibrated arm by more. Under QLIKE's asymmetric penalty, this interacts with each arm's calibration in a way that a shared correction would not. A comparison run under a different convention — an uncalibrated squared forecast, a a scalar classical estimator, or no correction at all — would be a comparison of different forecast objects, and it would not necessarily induce the same ranking.

That comparison is run, restricted to exactly three conventions on the identical persisted per-bar arrays and the identical exclusion rule. Convention *none* is the naïve inverse of Section B.1: the squared forecast with the diurnal adjustment reversed, $\widehat{RV}_t^{\text{raw}} = \hat{y}_t^2 B_t$, with no second-moment term. Convention *Duan* is the classical Duan [1983] estimator — the in-window scalar special case identified after Equation (18) — with one scalar per evaluation window, the mean of that window's own squared fit-scale residuals, added to the squared forecast before the baseline reversal. Convention *second-order adjustment* is the calibrated back-transformation of Equations (14)–(18), the convention every headline number in this paper inherits. Table 3 reports each completed arm's pooled QLIKE under the three conventions, with within-convention ranks. The rank-stability summary is the Kendall rank correlation between the second-order-adjustment ranking and each alternative's: $\tau = 0.682$ against *none* and $\tau = 0.859$ against *Duan*.

Table 3: The benchmark under the three back-transform conventions. The full per-arm table is Appendix Table 10; Kendall τ of each alternative ranking against the second-order-adjustment ranking over all arms: none 0.682, traditional Duan 0.859.

	No correction	Classical Duan	Second-order adjustment
OLS on HAR + calendar (benchmark)	0.24097	0.23165	0.22521

The correction is treated as part of the evaluation protocol: it is frozen here, before the first comparison, and no downstream section revisits it. Any future change to the second-moment model — including the learned correction contemplated for this slot — would constitute a change of protocol, and would require re-scoring every comparison that inherited the old one.

4.3 Estimators and Causal Tuning

Minimum-norm least squares. The unpenalized estimator is *minimum-norm least squares*: among all coefficient vectors that minimize the training-window sum of squares, the unique one of minimal ℓ_2 norm, computed through the pseudoinverse of the window gram at the standard machine-precision cutoff. This is the $\alpha \to 0^+$ limit of ridge regression, it is defined at any design rank, and it carries no penalty and no tuned quantity of any kind — the strongest attribution claim an unpenalized estimator supports. The rank cutoff is not a tuned quantity either: the centered gram's spectrum is gapped by more than five orders of magnitude around the retention threshold (on the widest design, largest discarded eigenvalue 1.8×10^{-11} relative, weakest retained 6.2×10^{-6}, with no eigenvalue between them at any bar), so every tolerance across four decades retains the identical subspace and reproduces the reported forecasts bitwise. Each arm is refit at every bar over a rolling window of 24,000 bars (about 480 trading sessions). On a full-rank design (the benchmark's 52 columns) minimum-norm least squares coincides exactly with ordinary least squares. An earlier pass of this campaign used the plain normal-equations solve, whose solution is arbitrary in near-null directions of the design; two single-window incidents in the campaign record trace to that path, and its results are archived. Because the pseudoinverse is defined at any rank, the estimator requires no rank guards; the only design intervention retained is the go-live participation rule of Section 3.2, which is a statement about data existence, not numerics.

Estimators. The design matrix X has one row per bar in the current training window and one column per feature: the HAR ladder of the target, the calendar features, and the full exogenous block of 523 columns. Let y be the vector of targets over the same window, and let x_s denote row s of X. The two estimators solve, at each refit,

$$\hat{\beta}^{\text{ridge}} = \arg\min_{\beta} \sum_{s} \left(y_s - x_s^\top \beta\right)^2 + \alpha \left\|\beta\right\|_2^2,$$

$$\hat{\beta}^{\text{lasso}} = \arg\min_{\beta} \sum_{s} \left(y_s - x_s^\top \beta\right)^2 + \lambda \left\|\beta\right\|_1,$$

where the sums run over the bars of the training window and the penalties α and λ are held constant between the periodic re-selections described below. The ℓ_1 penalty produces a coefficient vector supported on a subset of columns; the ℓ_2 penalty produces a coefficient vector with all entries nonzero. The elastic net [Zou and Hastie, 2005] interpolates the two,

$$\hat{\beta}^{\text{enet}} = \arg\min_{\beta} \sum_{s} \left(y_s - x_s^\top \beta\right)^2 + \lambda \left(\tfrac{1-\rho}{2} \left\|\beta\right\|_2^2 + \rho \left\|\beta\right\|_1\right),$$

recovering ridge at $\rho = 0$ and the lasso at $\rho = 1$. Its role in this paper is diagnostic as much as predictive: in the causally tuned arms the mixing share ρ is *free* — re-selected from $\{0.25, 0.5, 0.75, 1.0\}$ jointly with the penalty strength at every retune — so the estimator, not the author, chooses how much selection to apply, and the selected trajectory is part of the evidence in Section 5.1. All three penalized families are parameterized in the units of `scikit-learn`'s solvers, against which every custom solver below was validated.

Walk-forward protocol. The panel is computed as 100 walk-forward chunks over a rolling 500-trading-day (24,000-bar) training window. Every arm is refit *every bar*: at each bar the oldest bar leaves the window, the newest enters, and the coefficients are re-estimated before the next forecast is issued. Each arm re-selects its own penalty periodically and causally: every 250 solves, the current training window is split forward into a fit block, a 25-bar embargo (covering the longest moving average entering the features), and a 125-bar validation tail, and the penalty value minimizing validation MSE on that tail is adopted. No forecast's hyperparameters ever see its own future. The lasso and elastic-net arms are solved as exact every-bar online homotopies [Garrigues and El Ghaoui, 2008]: the solution and active set at each bar are obtained by exactly updating the previous bar's solution for the one-row change in the window, not by a warm-started batch solver terminated at a tolerance, so the head-to-head is not confounded with optimization error. The implementation is a re-authored recursive elastic net (Gram-form active-set homotopy with rank-1 Cholesky maintenance, validated coefficientwise against `scikit-learn`). The ridge arm runs on an exact rank-one update path.

Tree ensembles. The nonlinear members of the estimator set are two implementations of gradient-boosted regression trees. Gradient boosting fits an additive ensemble forward-stagewise: starting from a constant, at stage m it adds a shallow regression tree f_m fit to the negative gradient of the loss at the current ensemble's predictions,

$$F_m(x) = F_{m-1}(x) + \eta\, f_m(x), \qquad f_m \approx \arg\min_{f \in \mathcal{T}} \sum_i \Big(-\frac{\partial \ell(y_i, F_{m-1}(x_i))}{\partial F_{m-1}(x_i)} - f(x_i) \Big)^2,$$

where $\mathcal{T}$ is the space of depth-limited regression trees, $\eta \in (0, 1]$ is the shrinkage rate, and for the squared loss used throughout this paper the negative gradient is exactly the current residual, so each tree is a regression tree on the residuals of the ensemble so far.

XGBoost [Chen and Guestrin, 2016] fits each tree to a second-order Taylor expansion of the loss rather than the raw gradient. Writing $g_i = \partial \ell / \partial F_{m-1}(x_i)$ and $h_i = \partial^2 \ell / \partial F_{m-1}(x_i)^2$, the per-tree objective

$$\mathrm{Obj}_m \;\approx\; \sum_i \Big[g_i f_m(x_i) + \tfrac{1}{2} h_i f_m(x_i)^2 \Big] + \gamma T + \tfrac{1}{2} \lambda \sum_{j=1}^{T} w_j^2,$$

with T leaves and leaf weights w_j, has closed-form optimal weights $w_j^* = -G_j/(H_j + \lambda)$, where G_j and H_j are the gradient and Hessian sums over the rows in leaf j, and a split is accepted only if it improves $\frac{1}{2}\big[G_L^2/(H_L + \lambda) + G_R^2/(H_R + \lambda) - G^2/(H + \lambda)\big]$ by more than γ. The curvature term makes the leaf weights loss-aware (for squared error $h_i = 2$ and the scheme reduces to a regularized mean), and γ prices complexity per added leaf explicitly; row and column subsampling decorrelate successive trees. Our configurations use its histogram split finder (`tree_method="hist"`).

LightGBM [Ke et al., 2017] uses the first-order scheme with three engineering choices that matter at this design's width. First, feature values are binned into histograms, so split search costs depend on the bin count rather than the 24,000-row window length. Second, gradient-based one-side sampling keeps all large-$|g_i|$ rows and samples small-$|g_i|$ rows with

compensating weights, concentrating the fit on the hard bars. Third, exclusive feature bundling merges mutually exclusive columns — a property this design has in abundance, since an availability indicator and its siblings are zero whenever another feed's indicator is one — shrinking the effective width without changing the fitted function class. Trees grow leaf-wise (best-first) under a leaf-count cap rather than level-wise: deeper trees for the same budget, with the cap standing in for depth control.

Roles in the campaign. The two families enter twice, under different tuning disciplines. In the causal-tuning battery they are tuned arms like any other: their hyperparameters are re-selected on the same periodic forward-split schedule as every tuned arm, and the battery fully crosses the five estimators of this subsection with the nine feature buckets on the battery's shared panel (n = 218,934 thirty-minute bars, identical rows in every cell). In the frozen *expert bank* the configurations are fixed permanently before any scored bar: a dev-span study (rows before 2003, screen refits every 250 bars) searched each family's box — LightGBM: `num_leaves` 15–255 and `min_child_samples` 20–500 and `lambda_l1`/`l2` 10^{-8}–10 on log scales, `learning_rate` 0.01–0.3, `n_estimators` 50–500, feature and bagging fractions 0.5–1.0; XGBoost: `max_depth` 3–10, `min_child_weight` 1–100, `reg_alpha`/`reg_lambda` 10^{-8}–10, `gamma` 10^{-8}–1 on log scales, the same learning-rate and estimator ranges, subsample and column-subsample 0.5–1.0 — and the menu froze the top eight unique configurations plus two seeded Sobol points per family (2×10 experts). Each expert is then refit on the trailing 24,000-bar window at the disclosed refit cadence, on the identical wide design the linear arms read, with a fixed seed and threads following the slot count, so the tree class and the linear class are compared on the same information set with no per-arm tuning freedom. The shared comparator in both roles is the per-bar OLS–HAR incumbent (Section 4.1).

4.4 Block-Diagonal Tikhonov

A single ridge penalty cannot serve the HAR backbone and the wide exogenous design at once. Penalizing the HAR-plus-calendar design alone raises QLIKE relative to plain OLS (ridge $t = +5.0$ against the incumbent): those columns are few, well-conditioned, and individually strong, so any shrinkage on them is pure bias. The wide exogenous design requires heavy shrinkage — that is the head-to-head's result. A scalar α large enough for the weak exogenous columns over-shrinks the HAR columns; a value small enough for the HAR columns under-shrinks the exogenous columns.

The resolution is a **block-diagonal Tikhonov** penalty [Hoerl and Kennard, 1970, van Wieringen, 2023, van de Wiel et al., 2021]: one jointly solved least-squares problem, blocks ordered from strong and low-dimensional to weak and wide, each shrunk harder than the last. The B-block problem

$$\hat{\beta} = \arg\min_{\beta_1,\ldots,\beta_B} \Big\| y - \sum_{b=1}^{B} X_b \beta_b \Big\|_2^2 + \sum_{b=1}^{B} \alpha_b \|\beta_b\|_2^2, \qquad \alpha_1 < \alpha_2 < \cdots < \alpha_B,$$

reduces to ordinary unit-penalty ridge by column scaling. With

$$\tilde{X} = \big[\, X_1/\sqrt{\alpha_1} \;\cdots\; X_B/\sqrt{\alpha_B} \,\big] \quad \text{and} \quad \gamma_b = \sqrt{\alpha_b}\,\beta_b,$$

$$\Big\| y - \sum_b X_b \beta_b \Big\|_2^2 + \sum_b \alpha_b \|\beta_b\|_2^2 = \|y - \tilde{X}\gamma\|_2^2 + \|\gamma\|_2^2,$$

so $\hat{\beta}_b = \hat{\gamma}_b/\sqrt{\alpha_b}$. After scaling, every nested design runs in the same exact rank-one rolling least-squares path as every other linear model in this paper, refit at every bar. Setting a trailing block's coefficients to zero recovers the previous design exactly. There is no selection step and no tuned dial: the penalties are fixed constants.

Two blocks. The two-block design uses $B = 2$ and $(\alpha_1, \alpha_2) = (1, 100)$. Block 1 ($X_1$) is the HAR ladder of the target. Block 2 (X_2) is the same HAR averaging construction applied to every exogenous feature — HAR-of-`all_features`, the full wide basis. The two blocks are not fit separately and stacked. Setting $\beta_2 = 0$ reduces the model to ridge at $\alpha = 1$ on the backbone, which differs from the OLS–HAR incumbent of Section 3 only by that nominal penalty. The estimator matches the measured coefficient structure: block 2 keeps every column and shrinks them two orders of magnitude harder than block 1.

4.5 A Last-Bar Same-Day Option Reading

The one-bar forecast can be read against a listed price on the last thirty minutes of an expiration day. What follows fixes the instrument, the two measurements, the position rules, and the control; constants were set before the portfolio was scored.

Instrument. S&P 500 weekly (SPXW) options with zero days to expiry, cash-settled on the day's official close. The panel is the expiration-day chain at 15:30 ET (bid, ask, mid, underlying, quoted implied volatility), 2020 through 2025. At 15:30 the trade is the *straddle*: the nearest out-of-the-money call and the nearest out-of-the-money put of the same-day expiry, held as one position — the call at the smallest listed strike $K_c \geq S$ with a live mid and the put at the largest listed strike $K_p \leq S$ with a live mid, where S is the 15:30 underlying print. When S sits on a listed strike the two strikes coincide; otherwise they are one strike apart. Holding the straddle from 15:30 to the close is the *last-30-min trade*. Entry $P_{15:30}$ is the sum of the two mids. The position is held to the close and settled at intrinsic against the official S&P 500 close S_{cl},

$$\Pi_{\text{long}} \;=\; \big(S_{\text{cl}} - K_c\big)^+ + \big(K_p - S_{\text{cl}}\big)^+ \;-\; P_{15:30}, \tag{6}$$

with no delta hedge and no share unwinding. The long return is $R = \Pi_{\text{long}}/P_{15:30}$; a short earns $-R$. By construction $R \geq -1$, with equality on the days the close settles between K_p and K_c, either endpoint included, and both legs expire worthless. The 16:00 chain underlying is retained only as a check on the settlement print. Early-close sessions are dropped. The session ends at 13:00 on those days, so the 15:30 chain row is stamped after the close and carries stale quotes: the settlement is already known there and the short is a certain $+1$. The test is a non-positive time to expiry on the 15:30 row — the vendor's `early_close` column is true on every row of this file and is not used. Twelve such sessions appear in the chain. Six of them fall inside the forecast window, and the five of those that carry a 16:00 forecast row are the days the filter takes out of the scored frame; the remaining one never had a forecast row to lose.

Forecast remaining variance. Eight per-bar forecast series enter the same rules on the same contracts. Seven are estimators the body of the paper defines: the baseline (HAR + calendar OLS) forecast (Section 4.1), the block-diagonal ridge (HAR versus exogenous block, separate penalties; Section 4.4), LightGBM and XGBoost on the wide all-features design (Section 5.2, frozen expert menu), a causally tuned lasso on that same design under the selection protocol of the paper's tuned ridge (Section 5.1), a lasso at a fixed $\alpha = 10^{-4}$, and a causally tuned elastic net — the interpolating ℓ_1/ℓ_2 family — on the same wide all-features design (battery grid `logspace(-6,-2,5)` at `l1_ratio=0.5`, arm `b3_enet_tuned`). The eighth is not a competing model: it is the block-diagonal ridge again, refit on a design that carries no FOMC calendar columns, and it is carried as a diagnostic row — "block-diagonal ridge, without the FOMC columns" — because the gap between it and the ridge of record is what those calendar channels are worth on this trade, and nothing else in the table isolates them. All eight are stored on the estimation scale $y = \sqrt{RV/B}$. The 15:30 row is mapped to raw thirty-minute variance by a causal second-order back-transform *adapted from* the evaluation protocol of Section 4.2: a Mincer–Zarnowitz map $m = a + b\widehat{y}$ and a residual second moment $\hat{\sigma}^2$, both fit on the previous 250 trading sessions only, then $\widehat{RV} = (m^2 + \hat{\sigma}^2)B$, an estimate of $E[RV]$ for the 15:30–16:00 bar. Adapted, not inherited, and the three differences are stated rather than buried. The protocol's map is an ordinary least-squares fit over the preceding *evaluation window*, and its second moment opens at that window's mean squared residual and then decays exponentially within the window at $\lambda = 0.6$ (Equations (14)–(16)). The map used here is a *weighted* fit, over a flat lookback of 250 trading *sessions*, and its second moment is that same window's residual variance held fixed across it. The differences are what a daily exercise can refit — this section scores 866 daily observations, not a quarter million bars — but they mean no number in this section is produced by the map that produced the QLIKE tables, and none is carried between the two. This *session-bar recalibration* is the one of the diagnostics parked in Appendix D.15; every table of Section 5.4 uses the 16:00-bar recalibration defined below (Equation (7)). Forecast files end on 30 April 2024. The chain itself runs to 31 December 2025, so the 413 expiration days from 1 May 2024 onward — the most recent twenty months of contiguous tape — carry no forecast row and are not scored in Section 5.4; the trade's live form on them is reported in Appendix D.5.

Two conventions matter for honesty. *Alignment*: forecast rows are labelled by the end of the bar they predict, so the row read at 15:30 is the one stamped 16:00 — issued at 15:30 for the thirty minutes that remain — and not the stamp-15:30 row, which was issued at 15:00 for a bar already finished; every input to the signal below is therefore known at 15:30. *Weighting*: the map and the residual moment solve weighted least squares with weights $1/\max(\widehat{y}_u,\, q_{10})^2$, where q_{10} is the tenth percentile of $\widehat{y}$ within the same trailing window, and the fit window is restricted to regular-session bars. *Regular session* is an explicit mask, not an assumption, because the panel underneath is a futures panel (Section 3.1): a bar enters the fit only if its own stamp lies in 10:30–16:00, if its date carries a 16:00 stamp at all — which drops cash-market holidays, whose futures session ends at 13:00 or 11:30 — and if its date is not one of the NYSE 13:00 early closes of 2001–2025, which drops the post-close bars those days print through 16:00. The mask is not cosmetic. Of the 69,298 bars the stamp window alone would admit it removes 1,083, which is 1.56% of them: 510 rows on 48 of the 55 early-close dates in the 2001–2025 calendar, and 573 rows on a further 125 dates that

carry no 16:00 stamp at all — exchange holidays, whose rows here are futures-only bars, and a few dates whose tape stops early. Those bars carry far less variance than a regular one, so in a weighted fit they enter at a low y and drag the calibrated forecast down on every scored row.

One related contamination is disclosed rather than repaired. The per-slot factor B that scales the target is a trailing twenty-session mean of the same slot, and it is built upstream of this section, before any of the masking above: on an early close it absorbs that day's post-close bars, so for the following twenty sessions the 16:00 slot's B sits a few percent low. Ninety-four of the 866 scored days fall inside such a window. The effect on the trade is second order — $\widehat{RV} = (m^2 + \hat{\sigma}^2)B$ moves with B, but so does the y the map is calibrated against, and the two largely cancel — and a separate audit bounded what survives the cancellation at under a tenth of a Sharpe ratio and a handful of flipped positions. Repairing it at the source means regenerating the forecast tables, and it is part of the pending campaign.

The weighting addresses a level effect — an unweighted fit is dominated by the largest bars — and was adopted on a forecast-accuracy criterion, not on any portfolio outcome.

Which panel produced which row. The eight columns were not all produced on one design, and the tables say which each came from. The *FOMC panel* — the 1,264-column design that carries the FOMC calendar block — is the panel of record here: the block-diagonal ridge and the fixed lasso are fitted on it. The causally tuned lasso, the elastic net, LightGBM and XGBoost are still on the 1,144-column *earlier* panel, which has no such block; a run of those four on the panel of record is pending, and the tables mark them so. The baseline is invariant to the swap — it reads no exogenous column, and re-derived on both panels it agrees to 2×10^{-11} — so its row is unaffected. For an arm that does read exogenous columns the swap is not cosmetic: on the ridge it moves the 16:00 forecast by a median of 1.1%, and by more than 1% on 53% of trading days. That is why the diagnostic row exists, and why a ranking that mixes rows from the two panels would not be a like-for-like ranking. Each table's file, panel, chunk tree, code fingerprint and refit contract is recorded in `results/spxw_pnl/MANIFEST.md`.

Refit contract, hyperparameters, and what is not recoverable. Every linear arm is refit at each bar on a rolling window of 24,000 bars (about 480 trading sessions); the tuned lasso and elastic net re-select their penalty from the fixed grid every 250 solves, on the embargoed forward tail of the current window. The two tree columns are reported as shipped and are less clean than that. Their settings are indices 00 and 16 into a frozen twenty-arm menu whose file is absent both from the working tree and from the repository's history, so the hyperparameters behind those two columns cannot be recovered; the two arms were hand-picked from that bank and the basis of the pick is not recorded. Their refit cadence is mixed as well: one cluster refit at every bar and another every tenth bar, both contributed chunks to the same harvest, and neither the chunk metadata nor the shipped table records which cadence produced which stretch. A cadence coarser than one bar is stale but still causal — the fit still sees only strictly prior data, and an instrumented replay found no look-ahead at either cadence — so this is a provenance defect and not a leak. One caveat attaches to the ridge of record too: its table came from a cluster run identified by code fingerprint

rather than by a commit. The same estimator specification reproduces the corresponding earlier-panel chunks to 2×10^{-13}, so the estimator is confirmed; what cannot be re-derived on a workstation is the panel build under that exact code. A campaign that regenerates all eight columns under one committed revision — tree menu re-frozen, and cadence, thread count, library versions and code hash written into every chunk — is the stated remedy, and it has not been run.

Two facts about the FOMC block itself belong with that. The release feed behind it ends on 1 November 2023, so 122 of the 866 scored expiration days — 14.1% — fall after the feed is dead, and the meetings of 13 December 2023, 31 January 2024 and 20 March 2024 carry no FOMC signal at all. And the release-*surprise* channel is a dead constant over the whole panel: it is logged and then winsorized on a rolling 240-bar window, which clips a single-bar event to the running bound every time. So "FOMC in the design" means, operationally, the meeting-day indicator and the bars-until and bars-since counters, and it means them only through 1 November 2023. The diagnostic row above measures the calendar channels as they actually are in the shipped panel, not as they would be with a feed that reaches the end of the sample.

What the 16:00 target is. The forecast series predict RV on the paper's own panel, which is built from a near-24-hour futures tape (Section 3.1). The row read at 15:30 is therefore the realized variance of the futures 15:30–16:00 bar; it is not a measurement of the cash market's settlement procedure, and it is nonzero on days the cash market is shut. The option, by contrast, settles at the official S&P 500 close. On a regular session the two describe the same half hour of the same index and the comparison is the intended one; the fit mask above exists precisely to keep the sessions on which they are not aligned out of both the calibration and the trade.

Forecasts fitted for the 16:00 bar alone. The eight columns above fit one coefficient vector to all 48 half-hour bars of the day. The trade reads one bar, so a second family of forecasts is fitted for that bar alone — the *per-bar* forecasts. Their rows are the 15:30–16:00 bars (stamp 16:00) of the previous 2,000 sessions, about eight years; the target is the paper's $y = \sqrt{RV/B}$, winsorized as for every other arm; and the regression is re-solved every session. Every input is known at 15:30: for the target and for each exogenous series, its mean over the last $k \in \{1, 5, 25, 125, 625, 3125\}$ half-hour bars of the near-24-hour series, overnight bars included; an is-present flag per series; and calendar dummies (day of the week, option-expiry day and week, month and quarter end). In the design these means are the columns `har_ma_`k (the target's own ladder) and `adj_`x`_ma_`k (the ladder of series x after the diurnal adjustment of Section 3.2). The design of all 48 bars also interacts the target's ladder with open- and close-of-session indicators; in a series with one bar per session those 12 columns are either zero or exact copies of `har_ma_`k, so the per-bar design leaves them out (the pooled twins below keep them). Three input sets are used, each named in the tables by its design name:

- `baseline` (HAR + calendar): no exogenous series;
- `all_features`: every exogenous series of the paper's design, the FOMC calendar channels included;

- `live_feasible`: the 16 series a 15:30 forecaster can rebuild in real time from feeds a retail account has — from one-minute index-futures bars the seven return moments `sumret`, `sumabsret`, `sumret3`, `sumret4`, `sumpret2`, `sumbipow`, `sumautocov` and the activity counts `sumvolume`, `numobs`; the Cboe indices `vix`, `vvix`, `vix3m`; and the four FOMC calendar channels `fomc_release`, `fomc_day`, `fomc_until_inv`, `fomc_since_inv`. It leaves out the constituent cross-section, the message-board sentiment and the option-volume series, none of which a retail feed delivers during the session.

The linear estimators are OLS, ridge, lasso, and an elastic net with equal ℓ_1 and ℓ_2 weight. Each penalized one re-selects its penalty every 250 sessions from a fixed log-scale grid (ridge 10^{-2} to 10^{3}, lasso and elastic net 10^{-6} to 10^{-2}, one value per decade) by squared error on the last 125 sessions of the current window after a 25-session embargo — the causal tuning rule of Section 4.3, counted in sessions. The per-bar trees — LightGBM, XGBoost and a random forest — are fitted on the same rows, inputs and window and refit either every ten sessions or every session; the two cadences are separate rungs. *Untuned*, LightGBM and XGBoost take the settings of the paper's two tree columns (read from their stored chunk metadata, leaf minimums scaled to the 2,000-row window) and the random forest the library defaults (the *shipped settings*). They are tuned in two ways, each on the same validation tail and by validation squared error. The *random search* re-selects the settings every 250 sessions from 32 fixed candidates of a grid centred on the shipped settings. *Optuna* runs 50 trials of a tree-structured Parzen estimator at each tuning point, the first trial being the shipped settings, and re-tunes every 1, 5, 25 or 250 sessions; the forecast, refit every session, uses the best of the first 10, 25 or all 50 trials of the latest tuning point. Each tuned path is also run with the settings chosen by validation QLIKE. Two LSTMs complete the per-bar family. The first reads as a sequence the design rows of the last $L \in \{5, 20\}$ sessions at the 16:00 bar. The second reads the last $N \in \{13, 48, 96\}$ half-hour bars of the near-24-hour series, ending with the 15:30 bar, one step per bar, with bar-level inputs in place of the ladders (the bar's adjusted target and adjusted values of the input set's series, their availability and activity indicators, the calendar dummies and the half-hour clock), so that the network learns the aggregation the ladders impose. Both choose L or N, a hidden size of 16 or 64, dropout of 0 or 0.2 and a learning rate of 10^{-3} or 10^{-2} every 250 sessions on the same validation tail by squared error, average five seeds, and are refit every session (the first also every ten sessions, as a rung). Every per-bar fit — linear, tree or LSTM — first drops the columns that are constant on its own training window and every exact copy of an earlier column, keeping first occurrences: the *per-window mask*. The linear estimators apply it at every penalty tune, as their identifiability rule; the trees and LSTMs at every refit and tuning point. Tree and LSTM forecasts depend in their last bits on the processor's vector instruction class, and a tuning choice can flip on such a difference, so every tree and LSTM table is computed on one processor model. Each per-bar linear forecast has a *pooled twin*: the identical specification fitted on all 48 bars of the same 2,000 sessions with one coefficient vector, so that the difference between the two is the per-bar fit alone.

Two recalibrations. A forecast f_t on the fitting scale becomes a variance forecast in one of two ways. The *session-bar recalibration* is the map of the paragraph "Forecast remaining

variance" above: a weighted Mincer–Zarnowitz fit of level and slope, with its residual second moment, over the regular-session bars of the previous 250 sessions. The *16:00-bar recalibration* keeps the forecast's own level and adds only a second-moment term, estimated on the bar the trade reads,

$$\widehat{RV}_t \;=\; \big(f_t^2 + \bar{e}_t\big)\, B_t, \qquad \bar{e}_t \;=\; \frac{1}{|D_t|}\sum_{u \in D_t} \big(y_u - f_u\big)^2, \tag{7}$$

where D_t holds the 16:00 bars of the previous 250 sessions (at least 63), ending the session before t. It is the second-moment correction of the evaluation protocol (Section 4.2) made causal and specific to the traded bar. Because it has no level map, a forecast fitted on all 48 bars keeps whatever level bias it carries at 16:00, which the session-bar map would partly remove; the two recalibrations therefore give one forecast two different trades, and no number computed under one is set beside a number computed under the other. Under the 16:00-bar recalibration QLIKE is scored against the per-bar specification's 16:00 target, the same target for every forecast, and two forecasts are compared on the same days: the Diebold–Mariano statistic of the daily QLIKE difference [Diebold and Mariano, 1995], with a Newey–West variance (Bartlett kernel; lag 6 on 866 days) and the correction of Harvey et al. [1997], and, for QLIKE and Sharpe-ratio differences, a 95% interval from a circular block bootstrap of the days (blocks of 21 days, 2,000 draws, the same resampled days for every interval).

Quoted implied volatility. Implied volatility is the chain's quoted value on each of the two legs at 15:30; the straddle's quote is the mean of those two. The vendor quote is an *hourly* volatility while the window that remains is thirty minutes, so the quote is rescaled before it is compared with anything,

$$\mathrm{IV}_{30,t} \;=\; \mathrm{IV}_t^{\mathrm{hourly}} \big/ \sqrt{2}, \tag{8}$$

and $\mathrm{IV}_{30,t}^2$ is the implied variance of the remaining bar — the same window, and the same units, as $\widehat{RV}_t$. The vendor field is not always a quote: it is the output of a bisection on [0.0005, 0.025] that returns a bracket node when it fails to converge, so the two bounds and the first interior nodes of that bisection are censored values rather than prices, and they are masked here. On the five scored days where a nearest leg carries one, the implied volatility is recovered instead by inverting the Black–76 price of the straddle — call plus put at zero rate on the 15:30 index level, the vendor's own convention, over the remaining half hour — for the total volatility that reproduces the 15:30 midpoint. On uncensored legs that same inversion reproduces the quoted midpoint to quote precision (median model-to-quote price ratio 1.0000, 5th to 95th percentile 0.9969 to 1.0062), so "quoted" and "inverted" implied volatility coincide there. Dropping the censored days outright instead of recovering them moves no sign(s) Sharpe ratio in Section 5.4 by more than 0.06.

Signal. The two measurements are compared in variance space,

$$s_t \;=\; \widehat{RV}_t - \mathrm{IV}_{30,t}^2, \qquad \mathrm{VRP}_t \;=\; \mathrm{IV}_{30,t}^2 - \widehat{RV}_t \;=\; -\,s_t, \tag{9}$$

so $s_t > 0$ says the forecast is above the price of the remaining variance and $s_t < 0$ says the premium is positive. The daily portfolio return is $R'_t = q_t R_t$ for a position q_t fixed at 15:30 from information dated t or earlier.

Position rules. Two rules are scored on the same R_t and the same days; only q_t changes.

- *Always short*: $q_t = -1$ every day. The rule reads no forecast — it is the variance-risk-premium portfolio with nothing in it — and it is the control for the other. It is called the always-short control throughout, and it is one row of every table because it is the same portfolio whichever forecast sits beside it.
- sign(s): $q_t = +1$ when $s_t > 0$ and $q_t = -1$ otherwise — long the straddle when the forecast exceeds implied variance, short when it does not, with the tie at $s_t = 0$ resolved short so that the position is never flat. Direction only, at unit size. This is the headline rule.

Compounding at a fixed fraction of wealth. The tables above price one unit of premium per day and add P&L up. A reinvesting trader bets a fraction f of current wealth instead, so wealth compounds as $W_T = \prod_t (1 + f\, q_t R_t)$ and the natural objective is the annualized mean of $\log(1 + f R'_t)$. Ruin is one bad day: any $f \geq 1/|\min_t R'_t|$ puts wealth at or below zero on the worst day, so the unhedged short portfolio, with a worst day near -10 premium units, can only ever bet a small fraction of wealth no matter how good its mean. The fraction is fixed at $f = 0.03$ — three percent of wealth per day, a chosen round number, the same on every day and for both rules; nothing is estimated — and every realized wealth factor is asserted positive. Here f is a share of wealth deployed as body premium, the unit of the tables above. A defined-risk variant, which would bound the worst day by construction, is parked and not reported here. The exercise was run on the block-diagonal ridge under the session-bar recalibration and is reported in Appendix D.15.

Scoring and control. The unit of account is the expiration day. For a portfolio of n daily returns, $t = \sqrt{n}\,\overline{R'}/\hat{s}(R')$, and the annualized Sharpe ratio is $\overline{R'}/\hat{s}(R') \times \sqrt{252}$; the two carry the same information at fixed n, and both are reported so the tables read either way. The $\sqrt{252}$ is a per-*trade*-day convention: one row of the daily series is one trading period, and 252 of them make a year. This frame does not trade 252 days a year. SPXW listed Monday, Wednesday and Friday expirations before June 2022 and every session after, so over 2020–2024 it trades 200.3 days a year and reaches 252 only from mid-2022 on. A calendar-time annualization — fill the non-expiration sessions with a zero return and keep $\sqrt{252}$ — multiplies every annualized figure in this section by 0.892 and leaves every comparison between rules and columns unchanged. The convention is stated rather than switched, because the per-trade-day figure is the one a reader can recompute from a table row's own mean and standard deviation. Fills are at the 15:30 midpoint throughout, and no borrow or market-impact model is applied. The one alternative fill that is reported is the *crossed spread*: both legs bought at the ask when the rule is long and both sold at the bid when it is short, at entry only, since the position is held into a cash settlement and pays no

exit spread. It is quoted beside the midpoint figure in Section 5.4 rather than replacing it, because the midpoint figure is the one every other number in this section is computed at.

The always-short control is the 15:30 end of the clock in Section 2: Hizmeri and Bevilacqua [2025] locate the predictive content of volatility uncertainty at 10:00 EST and gone by midday; by the last bar the quoted implied already embeds a premium over remaining realized variance. On the scaled comparison of Equation (8) and under the 16:00-bar recalibration, implied variance exceeds the forecast on 62 to 73% of days for the paper's eight forecasts and on 60% for the headline forecast of Section 5.4, so the portfolio leans short more often than long. What the forecast contributes at this hour is the *sign* of the day — which side of the quoted premium to stand on. Sizing beyond the sign was tested rather than assumed, under the session-bar recalibration, and it is not retained: scaling the position by the size or the rank of $|s_t|$ raised no portfolio's Sharpe ratio with an interval above zero (Appendix D.11). The compounding exercise above bets a fixed fraction of wealth and reads nothing from the signal.

4.6 Implied Volatility by Root Finding

Where an option price is read as a volatility, the map is the Black–Scholes–Merton formula [Black and Scholes, 1973, Merton, 1973] at zero rate on a forward-like underlying — the futures form, and the form the option vendor's own quoted field uses. Only the product $\sigma\sqrt{\tau}$ enters that form, so it is carried here as a single *total volatility* $s = \sigma\sqrt{\tau}$ over the window that remains. For a strike K on an underlying level F,

$$V^{\mathrm{BSM}}(s; F, K) \;=\; \phi\big[\, F\,\Phi(\phi d_1) - K\,\Phi(\phi d_2)\,\big], \qquad d_1 \;=\; \frac{\log(F/K) + \tfrac{1}{2}s^2}{s}, \quad d_2 \;=\; d_1 - s, \tag{10}$$

with $\phi = +1$ for a call and $\phi = -1$ for a put and Φ the standard normal distribution function.

What is inverted is the *package*, not a leg. The recovery of Section 4.5 needs the one volatility that reprices the nearest out-of-the-money call and put together, so the root is taken of

$$V^{\mathrm{pkg}}(s) \;=\; V^{\mathrm{BSM}}(s; F, K_c) + V^{\mathrm{BSM}}(s; F, K_p) \;-\; P^{*}, \tag{11}$$

where P^{*} is the quoted package midpoint. The map $s \mapsto V^{\mathrm{pkg}}(s) + P^{*}$ is continuous and strictly increasing on $s > 0$ — its derivative is the sum of two positive vegas — and rises from the intrinsic value $(F - K_c)^{+} + (K_p - F)^{+}$ at $s \to 0$ to $F + K_p$ as $s \to \infty$, so a root exists and is unique exactly when P^{*} lies inside that interval, and there is no closed form for it.

It is found by bisection alone, on the bracket $[10^{-8}, 1]$ in total volatility: up to one hundred halvings, stopped when the bracket is narrower than 10^{-12}, which a unit bracket reaches in the first forty. Monotonicity makes the bracket sufficient and no Newton polish is applied; a guaranteed bracket is worth more at this precision than the last digits. Package midpoints at or below intrinsic — and any quote at $\tau = 0$ — have no root, and are returned as missing rather than clipped to a bound.

Two conventions are inherited rather than chosen. First, the underlying passed in is the 15:30 index level itself and not a put–call-parity forward: on legs where the vendor's field is a genuine quote, an inversion on the index level reproduces that field to 10^{-4} and a

parity-forward inversion does not, so a re-inverted leg stays on the same footing as the legs around it. Second, the vendor publishes an *hourly* standard deviation, so the total volatility returned here is converted as $\mathrm{IV}^{\mathrm{hourly}} = s/\sqrt{h}$ over the h hours that remain — at 15:30, $s\sqrt{2}$ — before it is compared with anything.

5 Results

The results proceed from estimator comparisons to nonlinear structure. Section 5.1 establishes the coefficient structure of the exogenous signal — dense but weak. Section 5.2 locates the recoverable nonlinearity. Section 5.3 reports the block-diagonal Tikhonov.

5.1 Lasso versus Ridge

The exogenous panel carries information beyond the HAR lag structure of Corsi [2009]; whether any of it is recovered is decided by the estimator. This subsection holds the information set fixed and varies only the penalty of the linear estimator. Two estimators are compared: the lasso [Tibshirani, 1996], which applies an ℓ_1 penalty and sets a subset of coefficients to exactly zero, and ridge [Hoerl and Kennard, 1970], which applies an ℓ_2 penalty and shrinks all coefficients toward zero without setting any to zero. The result of the comparison is a property of how the predictive coefficients are distributed across the columns of the design. In this section that property is measured directly: ridge attains lower out-of-sample QLIKE than the lasso on the full design (Section 5.1); the lasso zeroes 304 of 523 exogenous columns on the full basis, of which 235 are dead or constant availability indicators (Section 5.1); and five principal components retain 72% of the feature variance while losing essentially all of the predictive signal (Section 5.1). The coefficient structure is *dense but weak*, in the sense of the sparse-versus-dense prediction literature [Giannone et al., 2021, Kozak et al., 2020, Shen and Xiu, 2025]: many nonzero coefficients, none individually large. Section 4.4 defines the estimator built for this structure: ridge with a block-diagonal Tikhonov penalty whose first design uses $\alpha = 1$ on the HAR-of-target block and $\alpha = 100$ on the HAR-of-all-features block, with no selection step.

The head-to-head.

Design. The two penalized linear estimators are run on the full feature design — every exogenous column at once — on one shared out-of-sample panel of n = 218,934 thirty-minute bars, with identical rows in both cells; nothing varies between the two arms except the penalty.

Comparison protocol. Both arms are compared against the shared incumbent of Section 3: per-bar OLS on HAR plus calendar features with no exogenous inputs, at the benchmark QLIKE of 0.22521. For each arm, let ℓ_t^{arm} and ℓ_t^{inc} be the per-bar QLIKE losses [Patton, 2011] of the arm and the incumbent at bar t, and let $d_t = \ell_t^{\mathrm{arm}} - \ell_t^{\mathrm{inc}}$. The Diebold–Mariano statistic [Diebold and Mariano, 1995] is the sample mean $\bar{d}$ divided by its Newey–West HAC

standard error, with the small-sample correction of Harvey et al. [1997] applied; a negative t favors the arm.

Result. The comparison is run on the full design — the HAR ladder, the calendar features, and the full exogenous block — with both estimators under the same causal tuning protocol, so that neither family carries an informed penalty choice the other lacks (Table 4). Under the evaluation protocol, causally tuned ridge (0.22452, DM -1.1) beats the incumbent; causally tuned lasso (0.22566, DM +0.5) does not. Keeping every column and shrinking produces a lower loss than selecting a subset, when both families choose their penalty by the same causal rule. The elastic net (causally tuned) on the same full design scores 0.22464 (DM -0.6), interpolating ridge and lasso, closer to ridge, and not distinguishable from the incumbent at conventional size.

Table 4: Ridge versus lasso versus elastic net on the full design under the evaluation protocol, out-of-sample QLIKE with the Diebold–Mariano t-statistic against the OLS–HAR incumbent in parentheses. Panel: the frozen panel of Section 3, $n = 273{,}554$ identical rows. Ridge is the fine-grid causal tune already harvested on this root (`b1_ridge_tuned_fine`); lasso is `b2_lasso_tuned`; elastic net is `b3_enet_tuned`. Negative t favors the arm.

	Ridge	Lasso	Elastic net
Causally tuned	0.22452 (-1.1)	0.22566 (+0.5)	0.22464 (-0.6)
Incumbent (per-bar OLS–HAR)		0.22521	

A penalty-level sweep locates where the tuned arms operate. Holding the ridge family fixed and varying only the constant, QLIKE falls monotonically in α across the tested grid: 0.23219 at $\alpha = 0.1$, 0.23159 at 0.3, 0.23083 at 1, 0.23003 at 3, and 0.22909 at 10, still decreasing at the grid's edge; the tuned-ridge arm, which re-selects from a grid reaching $\alpha = 1000$, lands below every fixed point. The penalty *level* moves the loss by more than the penalty *family* does at any fixed level, and the wide design rewards heavier shrinkage than the $\alpha = 1$ default — the direction the block penalties of Section 4.4 take further. One caveat: all DM statistics are computed against the incumbent, a third model, so the ridge-minus-lasso differences themselves carry no significance statement yet.

The diagnosis: dense, not sparse. The lasso's own solution path shows what ℓ_1 selection does on this design. On the full basis the lasso zeroes 304 of 523 exogenous columns. Of those 304, 235 — 77% — are dead or constant availability indicators. Most of the lasso's selection therefore removes columns that carry no signal. Removing such columns does not lower QLIKE relative to ridge, because ℓ_2 shrinkage makes the same columns small without making them zero. Where the ℓ_1 penalty zeroes live columns, QLIKE rises: the margin by which ridge beats the lasso is carried by the live columns the lasso zeroes, not by the dead ones.

A principal-component check measures whether the spread of coefficients across columns is a low-dimensional structure in a rotated basis. Truncating the design to five principal components retains 72% of the feature variance and loses essentially all of the predictive

signal. The signal therefore does not lie in the high-variance directions of the feature covariance. The measured structure is: (i) no small subset of columns concentrates the signal — the only columns the lasso can zero without raising QLIKE are dead or constant ones; and (ii) no small set of principal directions concentrates it. The coefficient structure has many small nonzero entries in the coordinate basis and is not low-rank in the data's covariance geometry. This is the dense-but-weak structure of Giannone et al. [2021], verified on the solution path rather than inferred from the loss ordering.

The block-diagonal Tikhonov of Section 4.4 is the estimator built for this structure; its scores are in Section 5.3.

5.2 Gradient-Boosted Trees

The linear results are now in place: the exogenous signal is dense and weak (Section 5.1). The remaining question is what nonlinear structure adds. First, gradient-boosted trees beat the best linear arm on every feature set, by a small margin. Second, decompositions of the tree fits attribute the bulk of that margin to additive-plus-pairwise structure, and the pairwise part to products of columns the panel already contains, gated on the session clock.

The expert bank. The twenty frozen configurations of Section 4.3 were walked per bar on the full panel and scored under the evaluation protocol. Seventeen of the nineteen completed experts beat the benchmark, at QLIKE 0.21796–0.22108 against the benchmark's 0.22521 (DM -11.3 to -7.6); the two seeded Sobol points — included deliberately as unoptimized controls — lose to it (0.23064 and 0.23853), so the Optuna selection is worth 1–2 $\times 10^{-2}$ over a random draw. One expert's harvest is still completing; the bank aggregates below are computed on the nineteen complete experts over their full common row set (263,782 bars, 2003 onward), and the twentieth's inclusion is expected to move them only marginally.

The bank beats its members. The bank is aggregated two ways on the same trailing-125-bar tail with a 25-bar embargo, refreshed every 250 bars (the identical schedule to the linear tuners): a *selection* arm, which fills the next block from the lowest-tail-loss expert, and a *hedged* arm, which applies exponential weights $\propto \exp(-\eta \sum \ell)$ with $\eta = \sqrt{8 \ln K / L}$ — the standard weighted-average forecaster rate, no free constant. All tree rows are restricted to 2003 onward, since the menu was tuned on the pre-2003 dev span. The selection arm scores 0.21710 (DM -4.5 against the benchmark); the hedged arm scores 0.21296 (DM -15.3) — **better than every individual expert, including the best of them** (0.21796), a paired improvement of -0.0046 over the strongest member. The ordering (selection $<$ best expert $<$ hedge, with hedge -0.00413 better than selection at DM -5.3) is the paper's recurring tuning lesson in its purest form: causal selection on a short tail is expensive, and weighting is cheaper than selecting. The selection trajectory splits nearly evenly across families (296 XGBoost to 262 LightGBM boundaries) with era-specific leans (LightGBM in 2009 and 2013, XGBoost in 2018), and no expert dominates: the most-selected fills 22% of boundaries.

Table 5: The tree expert bank under the evaluation protocol (Section 4.2). Individual experts are scored on the full panel; the bank aggregates are computed on their common row set (263,782 bars, 2003 onward). DM t is against the OLS–HAR benchmark (0.22521); negative favours the arm.

	QLIKE	DM t
Best individual expert	0.21796	−11.3
Median completed expert	0.21969	—
Weakest Optuna-selected expert	0.22108	—
Sobol controls (2, unoptimized)	0.23064, 0.23853	—
`tree_tuned` (causal best-expert selection)	0.21710	-4.5
`tree_hedge` (exponential weights)	0.21296	-15.3

Comparison discipline against the linear ladder. The tree rows above are 2003+ by construction (263,782 bars); the linear arms are scored on 248,686–273,554 bars under their own row rules. Pooled levels across these row sets are not comparable (the campaign's row-set discipline of Section 4.2) — the hedged bank's 0.21296 and the linear champion's 0.21336 are separated by 2×10^{-5} *on different row sets*, which is exactly why the trees-versus-products verdict is reserved for the scorer's paired increments on common rows, registered as the block-ladder harvest completes (Appendix B.1). What is comparable as stated: the hedged bank's -0.00790 improvement over the benchmark at DM -15.3, and the bank-internal ordering above.

Where the nonlinearity lives. Three decompositions of the tree fits were computed in an earlier phase of this project.

Interaction order. A functional-ANOVA decomposition of a fitted depth-eight boosted tree attributes approximately 55% of its explainable variance to additive terms, 9% to pairwise interactions, and 36% to a higher-order remainder concentrated in no identifiable interaction. Within the additive part, the HAR block accounts for 96%.

Saturation at pairwise order. On the deployed residual-stack panel of the earlier build (n = 194,934; levels not comparable to the battery), an expressivity ladder holds the base model fixed and varies only the correction stage. A purely additive correction scores 0.12757. Adding bilinear pairwise interactions scores 0.12108. Nonlinear pairwise shapes score 0.12080. The deployed bagged additive-plus-pairwise stage (a GA2M in the sense of Lou et al. 2013) scores 0.12033. An Optuna-searched unrestricted-depth XGBoost correction stage scores 0.12094, which is worse than fitting no correction at all (0.12081). The ladder improves monotonically up to pairwise order; the unrestricted stage does not improve on the pairwise stages.

Distillation into products. On an earlier build of the same panel family, a penalized linear base scores 0.12516 and a residualized boosted-tree correction improves it to 0.12414. Adding to the base a single hand-written interaction — the HAR lags multiplied by a late-day session dummy, six columns — scores 0.12457, which recovers 58% of the tree correction's improvement. The full set of HAR-by-six-regime interactions scores 0.12453, or 62%. The

session dummy in the recovered term marks the late-day bars at which the change of sign in HAR's persistence coefficient is documented in Section 3.

The tree experiments therefore yield a specification statement: **the panel's recoverable nonlinearity is a sparse set of pairwise products of columns the design already contains**.

5.3 Block-Diagonal Tikhonov

The block-diagonal Tikhonov of Section 4.4 is one solver with nested block penalties at a 24,000-bar window. The design is two blocks, $(\alpha_1, \alpha_2) = (1, 100)$: HAR barely penalized, the wide exogenous block shrunk hard. It scores QLIKE 0.22350, Diebold–Mariano -3.1 against the OLS–HAR incumbent.

Table 6: Block-diagonal Tikhonov under the evaluation protocol (Section 4.2), 24,000-bar window. DM t is against the OLS–HAR benchmark (0.22521); negative DM favours the arm.

	QLIKE	DM t	n
OLS on HAR + calendar (benchmark)	0.22521	—	273,554
Two-block $(1, 100)$	0.22350	-3.1	272,971

What the block-diagonal ridge carries runs in a linear estimator — one solver, two penalties, refit every bar. The next subsection reads that one-bar forecast against the last thirty minutes of the expiration-day option chain.

5.4 A Last-Bar Same-Day Option Reading

The forecasts of Table 6 are one-bar forecasts of realized variance. This subsection reads them against the last thirty minutes of the expiration-day chain, under the rules of Section 4.5: the straddle bought or sold at 15:30 ET and cash-settled at the close, with the position fixed by the sign of $s_t = \widehat{RV}_t - \mathrm{IV}^2_{30,t}$. Joining the chain to the forecasts leaves 866 expiration days common to every forecast (3 January 2020 to 30 April 2024). The 413 expiration days after that date — the most recent twenty months of chain — carry no forecast row and are not scored here; the trade's live form on them is reported in Appendix D.5. Every forecast is turned into a variance by one rule, the 16:00-bar recalibration of Equation (7), so every row of Table 7 is scored the same way on the same days: the paper's eight forecasts, fitted on all 48 bars of the day (panel A); the pooled twins of the per-bar forecasts (panel B); and the per-bar forecasts fitted for the 16:00 bar alone, linear (panel C), and trees and LSTMs (panel D, one row for each of their 7 rungs).

Table 7: The last-30-min trade under one scorer: sign(s) on the straddle, 15:30 ET to the close, on the 866 expiration days common to every row (3 January 2020 to 30 April 2024). Every forecast is turned into a variance by the 16:00-bar recalibration (Equation (7)). QLIKE: the recalibrated forecast's mean QLIKE at the 16:00 bar; vs R: its percent difference from the reference R, the block-diagonal ridge (negative: more accurate), and the Diebold–Mariano statistic of that difference. $\text{Sharpe}_{\text{ann}}$: the portfolio's Sharpe ratio annualized by $\sqrt{252}$ per trade day, at the 15:30 midpoint and at the crossed spread (bought at the ask, sold at the bid, at entry only; the position settles in cash). ΔSharpe: the midpoint Sharpe ratio minus that of R and of always short on the same days, with a 95% paired interval (circular block bootstrap of the days, blocks of 21 days). Buys: the share of days the rule is long. Inputs: the input set of Section 4.5 by its design name; panel A reads the paper's full design. Panel D holds one row per rung of the per-bar trees and LSTMs of Section 4.5, each LightGBM or the LSTM on `live_feasible`, so that consecutive rows differ in one thing; the other trees and input sets of every rung are counted in the text. †: the four columns still fitted on the earlier design panel (Section 4.5); H: the headline forecast (bold). The rows are selected from the closing-strategy master table, which scores 220 forecasts the same way on the same days.

			vs R		$\text{Sharpe}_{\text{ann}}$		ΔSharpe, mid [95%]		
forecast	inputs	QLIKE	%	DM	mid	crossed	vs R	vs always short	buys, %
always short (no forecast)					0.20	−0.27	−0.80 [−2.01, +0.59]		0.0
A. The paper's forecasts: one regression for all 48 bars of the day									
baseline (HAR + calendar OLS)		0.1172	+10.8	+2.33	1.17	0.69	+0.17 [−0.74, +1.13]	+0.97 [−0.20, +1.96]	27.5
block-diagonal ridge^{R}		0.1058			1.00	0.53		+0.80 [−0.59, +2.01]	36.0
block-diagonal ridge, without the FOMC columns		0.1050	−0.8	−1.63	1.12	0.65	+0.12 [−0.35, +0.63]	+0.92 [−0.45, +2.11]	36.1
LightGBM†		0.1006	−4.9	−1.41	1.48	0.99	+0.47 [−0.44, +1.38]	+1.27 [+0.05, +2.41]	27.6
XGBoost†		0.1030	−2.7	−0.77	1.34	0.85	+0.33 [−0.76, +1.48]	+1.13 [−0.06, +2.25]	27.9
lasso (causally tuned)†		0.1024	−3.2	−1.36	1.24	0.76	+0.24 [−0.55, +0.99]	+1.04 [−0.15, +2.09]	30.8
lasso ($\alpha = 10^{-4}$)		0.1067	+0.8	+0.85	1.54	1.07	+0.54 [+0.07, +1.06]	+1.33 [−0.18, +2.68]	37.9
elastic net (causally tuned)†		0.1035	−2.2	−0.85	1.11	0.63	+0.11 [−0.75, +1.01]	+0.91 [−0.38, +2.03]	29.4
B. Pooled twins: the per-bar specification fitted on all 48 bars									
ridge	`baseline`	0.1106	+4.5	+1.29	1.24	0.77	+0.24 [−0.64, +1.16]	+1.04 [−0.36, +2.26]	37.0
ridge	`all_features`	0.1042	−1.5	−0.75	0.99	0.51	−0.01 [−0.73, +0.76]	+0.78 [−0.57, +2.00]	35.2
ridge	`live_feasible`	0.1034	−2.2	−1.01	1.12	0.64	+0.12 [−0.63, +0.90]	+0.92 [−0.31, +2.08]	32.9
C. Per-bar linear: one regression for the 16:00 bar alone									
OLS	`baseline`	0.0975	−7.8	−1.93	1.33	0.85	+0.32 [−0.66, +1.31]	+1.12 [−0.12, +2.16]	36.8
ridge	`baseline`	0.0982	−7.2	−1.80	1.22	0.74	+0.21 [−0.79, +1.20]	+1.01 [−0.30, +2.11]	37.9
lasso	`baseline`	0.0972	−8.1	−1.97	1.51	1.04	+0.50 [−0.49, +1.53]	+1.30 [+0.04, +2.35]	38.0
elastic net	`baseline`	0.0969	−8.4	−2.02	1.29	0.82	+0.29 [−0.73, +1.27]	+1.09 [−0.20, +2.14]	37.6
ridge	`all_features`	0.1004	−5.1	−1.15	1.66	1.20	+0.66 [−0.15, +1.54]	+1.46 [+0.11, +2.74]	36.1
lasso	`all_features`	0.0998	−5.7	−1.97	1.45	0.98	+0.44 [−0.45, +1.32]	+1.24 [−0.08, +2.52]	39.6
elastic net	`all_features`	0.0994	−6.0	−1.75	1.33	0.86	+0.33 [−0.56, +1.24]	+1.13 [−0.27, +2.48]	39.1
ridge^{H}	`live_feasible`	0.1005	−5.0	−1.42	1.90	1.44	+0.90 [+0.19, +1.66]	+1.70 [+0.09, +3.11]	40.2
lasso	`live_feasible`	0.0964	−8.8	−3.04	1.83	1.36	+0.82 [−0.18, +1.86]	+1.62 [+0.20, +2.91]	38.9
elastic net	`live_feasible`	0.0968	−8.5	−3.24	1.41	0.94	+0.41 [−0.33, +1.26]	+1.21 [−0.20, +2.48]	37.8
D. Per-bar trees and LSTMs: one row per rung, the same rows, target and window									
LightGBM, shipped settings, refit every 10	`live_feasible`	0.0991	−6.3	−1.93	1.59	1.12	+0.59 [−0.25, +1.39]	+1.39 [−0.03, +2.63]	39.8
LightGBM, shipped settings, refit daily	`live_feasible`	0.0955	−9.7	−2.88	1.42	0.95	+0.42 [−0.58, +1.48]	+1.22 [−0.03, +2.45]	37.3
LightGBM, random search, refit every 10	`live_feasible`	0.0992	−6.2	−1.82	1.47	1.00	+0.47 [−0.33, +1.33]	+1.27 [−0.10, +2.51]	38.0
LightGBM, random search, refit daily	`live_feasible`	0.0994	−6.1	−1.78	1.63	1.17	+0.63 [−0.22, +1.58]	+1.43 [+0.08, +2.63]	39.0
LightGBM, Optuna tuned daily, refit daily	`live_feasible`	0.1026	−3.0	−0.93	0.87	0.40	−0.13 [−1.34, +1.09]	+0.67 [−0.43, +1.83]	38.7
LSTM over sessions, refit daily	`live_feasible`	0.1193	+12.7	+2.15	0.91	0.43	−0.10 [−1.13, +1.04]	+0.70 [−0.53, +1.90]	37.8
LSTM over half-hour bars, refit daily	`live_feasible`	0.1103	+4.3	+0.83	1.53	1.06	+0.53 [−0.40, +1.52]	+1.33 [+0.14, +2.47]	37.1

Read the control first. Selling the straddle every day earns a mean of 0.014 premium units per day, an annualized Sharpe ratio of 0.20 [−0.63, 1.14] at the midpoint and −0.27 [−1.10, 0.63] at the crossed spread: a premium that 866 days cannot separate from zero at the midpoint and that the spread consumes.

The paper's forecasts under one scorer. Putting the paper's forecasts in charge of *direction* lifts every one of them above the control: the eight sign(s) portfolios trade at Sharpe 1.00 to 1.54 at the midpoint and 0.53 to 1.07 at the crossed spread. The trade turns on the paired comparison — both rules trade the same contracts on the same days — and paired against always short the eight improvements run from +0.80 to +1.33, with one of the eight intervals above zero (LightGBM, +1.27 [+0.05, +2.41]). The paper's forecast of record, the block-diagonal ridge, trades at 1.00 (0.53 crossed), +0.80 [−0.59, +2.01] above always short. Under this recalibration a forecast fitted on all 48 bars keeps its own level at the 16:00 bar (Section 4.5), and the same estimator without the FOMC calendar columns trades higher by +0.12 [−0.35, +0.63], an interval that includes zero: the calendar channels are not resolved on this trade. The eight columns take the same side as the block-diagonal ridge on 79 to 96 percent of days. One of them separates from it: the fixed lasso, on the same design panel, trades at 1.54, above the block-diagonal ridge by +0.54 [+0.07, +1.06], while taking its side on 96 percent of days.

The per-bar tables, rebuilt. Every per-bar row of Table 7 comes from one re-run of the 16:00 bar that changed four things, each defined in Section 4.5 and reported in full in Appendix D.19. The per-bar design drops the 12 session-edge columns that, in a series with one bar per session, are either zero or exact copies of `har_ma_`k (34, 244 and 640 columns become 22, 232 and 628 on `baseline`, `live_feasible` and `all_features`). Every tree and LSTM fit now drops, as every linear fit already did, the columns that are constant on its own training window or copy an earlier one; it keeps a median of 17 of 22, 138 of 232 and 369 of 628 columns. Trees and LSTMs are refit every session as well as every ten, and the trees are tuned by Optuna as well as by the random search. Tree and LSTM forecasts depend on the processor's instruction class in their last bits and, through the tuning, beyond them: re-run on a second class, LightGBM's best-of-50 choice changed on 18 of 20 tuning points and a LightGBM forecast moved by up to 5.5% of its value, while re-run on the same class every one of those points was bit-identical for all three trees. Every one of the 541 chunks behind the tree and per-bar LSTM tables was therefore computed on one processor class. The linear forecasts barely move: their identifiability mask had already removed the duplicated columns, so the headline keeps every position and its QLIKE moves only at rounding, and over the master table's 55 per-bar linear forecasts QLIKE changes by at most 0.09% and at most two positions, with no Sharpe difference resolved.

The forecasts built for the traded bar. Panels B to D hold the forecasts of Section 4.5 that read the 16:00 bar on their own design: the pooled twins trade at 0.99 to 1.24, the per-bar linear forecasts at 1.22 to 1.90, and the rungs of the trees and LSTMs at 0.87 to 1.63. Against the block-diagonal ridge, 2 of the 27 other forecasts in the table have a Sharpe-difference interval wholly above zero and none has one wholly below; over all 219 other forecasts of the master table the count above zero is 10. A table of this many forecasts on 866 days is a search, and its intervals are read as such.

The headline forecast, and why this one. The headline forecast of this section is the per-bar ridge on the `live_feasible` inputs (H in Table 7). It is chosen for two reasons,

neither of them its Sharpe ratio. It reads only the 16 series a forecaster can rebuild at 15:30 from a retail feed, so the trade it scores is one that can be run; and it is the 15:30 study's forecast of record, the per-bar estimator that study defined first, not the maximum of the table: over the master table's 220 forecasts the highest Sharpe ratio belongs to the per-bar lasso on `live_vix_only` (1.91), and taking the maximum of that many Sharpe ratios would select on the trade's own noise. The headline ranks 2 of 220 on the Sharpe ratio and 63 of 220 on QLIKE.

Its forecast is more accurate than the paper's, but not significantly: QLIKE 0.1005, −5.0% against the block-diagonal ridge, Diebold–Mariano −1.42 ($p = 0.156$), with a day-block interval on the daily loss difference of [−0.0113, +0.0005]. Its trade is resolved against both references. The sign(s) portfolio has a Sharpe ratio of 1.90 at the midpoint and 1.44 at the crossed spread, earns 0.134 premium units a day, is right on 54.6% of days and long on 40.2%. Paired on the same days it trades above the block-diagonal ridge by +0.90 [+0.19, +1.66] at the midpoint (+0.91 [+0.19, +1.67] crossed) and above always short by +1.70 [+0.09, +3.11] (+1.71 [+0.12, +3.12] crossed); the second interval clears zero narrowly.

What it is *not* separated from matters as much. The lasso on the same inputs — the same rows, target and window, only the penalty changed — trades at 1.83, −0.08 [−0.89, +0.79] against the headline, takes its side on 89 percent of days, and forecasts better on QLIKE (−4.1%, DM −1.20). The elastic net on the same inputs is at −0.49 [−1.08, +0.14], the ridge on all features at −0.24 [−1.07, +0.54], and the table's top Sharpe ratio at +0.01 [−0.76, +0.81]. Over the master table's 219 other forecasts, none trades above the headline with an interval above zero and 43 trade below it with an interval below zero. The headline is a forecast of record, not a winner: Table 7 resolves that it trades above the paper's forecast and above the control, and it does not resolve an order among the per-bar forecasts.

One change at a time. Table 8 walks from the paper's forecast to the headline one change at a time, every step paired on the same days. Replacing the block-diagonal ridge by the per-bar specification's ridge fitted on all 48 bars — design, window and penalty change together — moves nothing on the trade (−0.01 [−0.73, +0.76]). Fitting the 16:00 bar alone is the largest step: +0.68 [−0.12, +1.45] on `all_features` and +0.78 [−0.08, +1.52] on `live_feasible`. On HAR + calendar the same step changes QLIKE by −11.2 [−15.7, −6.9] percent, a resolved gain (DM −4.59), and leaves the trade where it was (−0.03 [−0.66, +0.64]): a better forecast of the bar's variance is not by itself a better trade. Restricting the per-bar ridge from all features to the `live_feasible` inputs moves the trade by +0.24 [−0.54, +1.07], and adding those inputs to HAR + calendar by +0.69 [−0.23, +1.70]. From the headline, changing only the estimator to another penalty or to a tree with the shipped settings moves the Sharpe ratio by between −0.08 (the lasso) and −0.63 (XGBoost). The headline's resolved lead over the block-diagonal ridge is exactly the sum of three steps of the table — the specification step, fitting the 16:00 bar alone on `all_features`, and restricting those inputs to `live_feasible` — none of which is resolved alone; the largest is fitting the traded bar on its own.

The last five blocks of Table 8 walk the rungs of the 16:00 re-run the same way, for LightGBM and the LSTM on `live_feasible`. Dropping the duplicated columns leaves the headline where it was. The per-window mask moves no rung's QLIKE or Sharpe ratio with

an interval that excludes zero (LightGBM with the shipped settings refit every ten sessions: +1.0 [−0.4, +2.3]% on QLIKE, −0.17 [−0.80, +0.39] on the trade). Refitting every session instead of every ten makes LightGBM more accurate, −3.6 [−7.4, −0.8]% (DM −2.73), without moving its trade (−0.17 [−1.11, +0.59]); the LSTM's QLIKE changes by −2.5 [−7.8, +3.2]%, not resolved. Tuning does not help: against the shipped settings, both refit daily, the random search changes QLIKE by +4.1 [−0.2, +8.9]% and Optuna tuned every session by +7.4 [+1.4, +16.3]%, a resolved loss of accuracy, and tuning every session rather than every 250 moves Optuna's trade by −0.55 [−1.42, +0.30]. Reading the last half-hour bars instead of the last sessions moves the LSTM's QLIKE by −7.5 [−20.9, +4.7]% and its trade by +0.62 [−0.26, +1.54]. Of all 26 steps of the table, none has a Sharpe interval that excludes zero.

Table 8: One change at a time: paired differences between two forecasts of Table 7 that differ in one thing (the step's "to" forecast minus its "from" forecast), on the same 866 days under the 16:00-bar recalibration. QLIKE: percent difference in mean QLIKE at the 16:00 bar with a 95% day-block interval (negative: the "to" forecast is more accurate), and its Diebold–Mariano statistic. ΔSharpe: difference in the sign(s) portfolio's annualized Sharpe ratio at the midpoint and at the crossed spread, with 95% paired intervals on the resampled days of Table 7. Same side: the share of days the two forecasts take the same position. The first step changes the design, the training window and the penalty together; every other step changes one thing. Steps the master table also stores are reproduced by it exactly. The last five blocks are the rungs of the 16:00 re-run (Section 4.5) for LightGBM and the LSTM on `live_feasible`: the design row pairs the headline with its table before the re-run; the mask rows and the LSTM refit every ten sessions, whose unmasked or ten-session tables are not rows of Table 7, are read from the re-run's own paired comparisons (same days, same scorer and resampling); every other campaign step is recomputed here and reproduces the re-run's report. DM —: not stored for that pair.

step (to minus from)	QLIKE, % [95%]	DM	ΔSharpe, mid [95%]	ΔSharpe, crossed [95%]	same side, %
From the paper's forecast to the per-bar specification					
block-diagonal ridge → pooled ridge, `all_features`	−1.5 [−5.8, +2.3]	−0.75	−0.01 [−0.73, +0.76]	−0.02 [−0.73, +0.76]	87
Fit the 16:00 bar alone: pooled twin → per-bar ridge					
`baseline`	−11.2 [−15.7, −6.9]	−4.59	−0.03 [−0.66, +0.64]	−0.03 [−0.66, +0.63]	88
`all_features`	−3.6 [−10.7, +3.1]	−0.92	+0.68 [−0.12, +1.45]	+0.69 [−0.11, +1.46]	83
`live_feasible`	−2.8 [−7.9, +2.6]	−0.87	+0.78 [−0.08, +1.52]	+0.80 [−0.07, +1.54]	88
Inputs of the per-bar ridge: other set → `live_feasible`					
`all_features` → `live_feasible`	+0.2 [−4.9, +5.5]	+0.06	+0.24 [−0.54, +1.07]	+0.24 [−0.54, +1.08]	86
`baseline` → `live_feasible`	+2.4 [−5.8, +12.3]	+0.53	+0.69 [−0.23, +1.70]	+0.70 [−0.23, +1.72]	82
Estimator on `live_feasible`: headline ridge → alternative					
lasso	−4.1 [−11.2, +1.8]	−1.20	−0.08 [−0.89, +0.79]	−0.08 [−0.90, +0.79]	89
elastic net	−3.7 [−9.8, +0.8]	−1.38	−0.49 [−1.08, +0.14]	−0.50 [−1.09, +0.14]	91
LightGBM, shipped settings, refit every 10	−1.4 [−6.8, +3.7]	−0.46	−0.31 [−1.04, +0.41]	−0.32 [−1.04, +0.41]	85
XGBoost, shipped settings, refit every 10	−1.1 [−6.9, +4.8]	−0.32	−0.63 [−1.51, +0.16]	−0.64 [−1.54, +0.16]	86
random forest, shipped settings, refit every 10	+1.3 [−6.4, +8.1]	+0.35	−0.21 [−0.84, +0.42]	−0.22 [−0.85, +0.41]	85
LightGBM, random search, refit every 10	−1.3 [−6.6, +4.1]	−0.35	−0.43 [−1.22, +0.35]	−0.44 [−1.23, +0.35]	85
XGBoost, random search, refit every 10	+6.5 [−0.3, +15.2]	+1.52	−0.44 [−1.14, +0.25]	−0.44 [−1.14, +0.25]	84
random forest, random search, refit every 10	+18.4 [+4.8, +39.1]	+2.41	−0.48 [−1.17, +0.24]	−0.48 [−1.18, +0.23]	85
Campaign, design: the 12 session-edge duplicates dropped					
per-bar ridge (the headline)	+0.0 [+0.0, +0.0]	—	+0.00 [+0.00, +0.00]	+0.00 [+0.00, +0.00]	100
Campaign, per-window mask: masked minus unmasked					
LightGBM, shipped settings, refit every 10	+1.0 [−0.4, +2.3]	+1.31	−0.17 [−0.80, +0.39]	−0.17 [−0.80, +0.39]	95
LightGBM, shipped settings, refit daily	−1.0 [−2.4, +0.3]	−1.47	−0.06 [−0.49, +0.37]	−0.06 [−0.49, +0.37]	95
LightGBM, random search, refit every 10	−0.9 [−2.7, +0.8]	−0.93	−0.12 [−0.67, +0.43]	−0.12 [−0.68, +0.43]	94
LightGBM, random search, refit daily	−0.1 [−2.1, +1.8]	−0.10	+0.01 [−0.55, +0.58]	+0.01 [−0.55, +0.58]	94
LSTM over sessions, refit every 10	+0.3 [−3.0, +3.4]	+0.15	−0.05 [−0.69, +0.57]	−0.06 [−0.71, +0.57]	92
Campaign, refit cadence: every 10 sessions → every session					
LightGBM, shipped settings	−3.6 [−7.4, −0.8]	−2.73	−0.17 [−1.11, +0.59]	−0.18 [−1.11, +0.59]	93
LSTM over sessions	−2.5 [−7.8, +3.2]	−0.77	−0.17 [−0.80, +0.48]	−0.17 [−0.80, +0.48]	89
Campaign, tuning (LightGBM, refit daily)					
shipped settings → random search	+4.1 [−0.2, +8.9]	+1.70	+0.21 [−0.61, +1.07]	+0.22 [−0.60, +1.08]	90
shipped settings → Optuna, tuned daily	+7.4 [+1.4, +16.3]	+2.32	−0.55 [−1.37, +0.24]	−0.55 [−1.37, +0.25]	89
Optuna: tuned every 250 sessions → daily	−2.8 [−9.2, +2.9]	−0.98	−0.55 [−1.42, +0.30]	−0.56 [−1.44, +0.29]	90
Campaign, LSTM sequence: sessions → half-hour bars					
LSTM, refit daily	−7.5 [−20.9, +4.7]	−1.25	+0.62 [−0.26, +1.54]	+0.63 [−0.25, +1.54]	80

Trees and LSTMs, every rung. Over every tree, input set and rung of the re-run the reading of Table 8 holds. No tree is more accurate than the per-bar ridge on the same inputs:

of 72 QLIKE intervals (three trees, three input sets, the four rungs with the shipped settings or the random search and Optuna at four tuning cadences) none lies below zero and 32 lie above, and none trades above the ridge with an interval above zero (two below). Refitting every session lowers the QLIKE of all nine tree forecasts (three trees, three input sets) with the shipped settings, with three intervals below zero, and moves one trade with an interval that excludes zero, XGBoost on `all_features` (+0.52 [+0.10, +1.02]). Tuning never makes a tree refit daily more accurate than the shipped settings refit daily: the random search is less accurate with an interval above zero in 6 of 9 cases and Optuna in 19 of 36, and no tuned tree's trade differs from the shipped one's with an interval that excludes zero. Tuning Optuna every session, every 5 or every 25 sessions rather than every 250 is more accurate with an interval below zero in 8 of 27 cases and never resolved on the trade. On the trial budget, the best validation score falls in the last ten of the 50 trials at 27 to 40% of tuning points, against 20% if the order did not matter, yet out of sample the best of 50 is more accurate than the best of the first 10 or 25 in none of 36 cases, and less accurate in one. The mask itself changes the trees little: a median of 0.7% in QLIKE, five of 36 intervals above zero and none below, and one Sharpe interval below zero. The LSTM over sessions is more accurate with daily refits (−1.9, −2.5 and −10.1% on `baseline`, `live_feasible` and `all_features`, two intervals below zero) but stays +6.8 to +30.0% above the ridge in QLIKE; the LSTM over half-hour bars is resolved more accurate than it only on `all_features` (−14.3 [−26.9, −2.0]%) and is +9.2 to +11.5% above the ridge, every interval above zero; neither LSTM's trade differs from the ridge's with an interval that excludes zero. On the trade the best of the master table's 147 tree and LSTM forecasts, the random forest on `live_feasible` with the random search's settings chosen by QLIKE and refit every ten sessions (1.90), is −0.00 [−0.56, +0.59] against the headline, and none of the 147 trades above the headline with an interval above zero.

Where the P&L comes from: a tail trade. Summed without compounding, the headline's portfolio makes 116.2 premium units at the midpoint over the 866 days (87.4 crossed). Its 20 best days carry 68% of that (86% crossed); all 20 are buys, all 20 worst days are sells, and 40 days add up to the whole total. What the rule contributes is which of the tail days it owns: of the 20 days on which the straddle paid most it bought 14, against 8.0 expected at its buy rate ($p = 0.006$). Month-ends are not the source: on the 52 last sessions of a month, where the long straddle pays on average, the rule makes −6.0 [−26.1, +12.7], against +116.9 [+59.8, +174.1] on the 735 days more than one session away. Against always short the difference is +103.7 [+10.7, +200.1] premium units, all of it earned on the days the rule buys — an interval that excludes zero, and that of the Sharpe difference in Table 7 does too — but without the difference's 10 best days it is +6.0 [−59.8, +72.7]. The edge is a tail trade: a handful of long days carries it, and 866 days hold few of them. Appendix D.14 gives the decomposition by cell.

At the crossed spread. Every Sharpe ratio of Table 7 is given at both fills. Charging the spread at entry — both legs bought at the ask when the rule is long, both sold at the bid when it is short, with nothing charged at the exit because the position settles in cash — costs every forecast's portfolio between 0.46 and 0.48 of annualized Sharpe and takes always

short from 0.20 to −0.27. It is one crossing per day at the quoted touch, with no size, no queue and no model of impact — a bound on the fill rather than a fill model — but it is the crossing the venue charges.

Other structures. The straddle is one choice of instrument for sign(s). Appendix D.13 writes down 31 alternatives — strangles at several distances from the index, iron butterflies and iron condors at three wing widths, put and call verticals, call and put butterflies, 1×2 ratio spreads, and the straddle with a wing hedge on the days the rule sells — and scores them under the headline forecast on the same days, per unit of straddle premium. None beats the straddle: none of their paired Sharpe-difference intervals lies above zero at either fill. The closest at the midpoint, the iron butterfly with the widest wings (1.94), is +0.04 [−0.04, +0.14] above it, and at the crossed spread the straddle ranks first. Wings cost at every width once the spread is paid: 6 of the 6 iron-butterfly and wing-hedge rows lie below the straddle with an interval that excludes zero at crossed fills, against 2 at the midpoint; the median entry cost of crossing is 2.9% of the straddle's own premium and 4.0% to 6.1% of an iron butterfly's.

Diagnostics parked in Appendix D. The previous version of this subsection read the paper's forecast under the session-bar recalibration and qualified it by two sample splits, the share of settlement pins, the correlation of the eight forecast columns, a decomposition of QLIKE into threshold decisions with its Murphy diagram, a same-day regression of the settlement return on the signal, a dating (shift) test of the forecast row, and compounding at a fixed fraction of wealth. Those paragraphs are parked, verbatim, in Appendix D.15: they concern the paper's forecast under the other recalibration and are not set beside Table 7. Sizing the position by the size of the gap does not improve on its sign either; that test, too, was run under the session-bar recalibration (Appendix D.11).

The trade as a whole is a demonstration under the stated constants of Section 4.5, not a trading claim: there is no delta hedge, no borrow and no market-impact model, and the crossed spread is a bound on the fill rather than a fill model. What survives is worth stating plainly. Under one recalibration and on the same 866 days, the per-bar ridge on the `live_feasible` inputs trades the last-30-min straddle at a Sharpe ratio of 1.90 (1.44 crossed), above the paper's forecast by +0.90 [+0.19, +1.66] and above always short by +1.70 [+0.09, +3.11]. Its forecast is not significantly more accurate than the paper's (DM −1.42, $p = 0.156$); no single change on the way from the paper's forecast is resolved on its own, the largest being to fit the traded bar alone; and it is not separated from the lasso on the same inputs (−0.08 [−0.89, +0.79]). None of the 147 tree and LSTM forecasts of the 16:00 re-run — refit every session or every ten, with the shipped, searched or Optuna-tuned settings — trades above it with an interval above zero. The edge is a tail trade: the 20 best days carry 68% of its P&L, all of them buys, and without its 10 best days the advantage over always short is +6.0 [−59.8, +72.7]. The sample ends with the forecast panel in April 2024; after it the trade's live form has shown no edge (Appendix D.5).

6 Conclusion

On a quarter-million-bar intraday panel, we asked what exogenous information adds to a HAR backbone, what estimator can extract it, and what the nonlinearity is made of. The answers form a single chain. The exogenous panel carries real, statistically significant information — but the information is spread densely and weakly across hundreds of columns, so it is invisible to selection and to unpenalized joint fits, and recoverable only by shrinkage. The appropriate shrinkage is block-structured: almost none on the strong low-dimensional backbone, heavy and uniform on the wide exogenous block. The recoverable nonlinearity is small beside that information effect and is concentrated at pairwise order — products of columns the design already contains, gated on the session clock — a specification statement the tree fits deliver. The final model is ridge with a block-diagonal Tikhonov penalty, linear in its parameters throughout: two nested blocks, two fixed penalties, one exact rolling solve, and it outperforms the benchmark by a margin an order of magnitude beyond the fourth decimal place at strong paired significance. The paper closes by reading the forecasts against the last thirty minutes of the expiration-day chain (Section 5.4). Under one recalibration and on the same days, the sign of the gap between forecast and quoted implied variance carries a direction edge on the straddle: every forecast improves on always short in point estimate, and a ridge fitted for the 15:30–16:00 bar alone, on inputs a forecaster can rebuild live, is resolved above both always short and the paper's own forecast at both fills. The forecast built for the whole panel is therefore not the one to trade at its last bar — but the accuracy gain behind the difference is not significant, no single step from one forecast to the other is resolved alone (the largest is fitting the traded bar on its own), and the edge is a tail trade: a few long days carry it, the sample holds few of them, and the trade's live form has shown no edge after the sample ends. Sizing on the size of the gap does not improve on its sign.

Two findings sit beside the model rather than inside it, and may matter more broadly. First, the evaluation protocol is a modeling choice: the causal calibrated back-transform beats the retrospective Duan convention on every arm, and its adaptive second moment — not any conditional-mean device — is among the largest measured effects in the study. Second, causal hyperparameter tuning on short validation tails is repeatedly worse than well-chosen constants, because the price of a selection mistake is asymmetric across estimator families. Both findings are about the discipline surrounding the model, and both are measured, not assumed.

The limitations are stated where they bind. Results are computed on one asset complex and one loss; levels are comparable only within one preprocessing regime and one protocol, and arms that survive only under the legacy protocol are labeled as such in the appendix. The conditional-variance panel remains on the legacy scalar back-transform and has not been re-scored under the protocol used in the body. The natural extensions are equally concrete: promoting the mean-reverting second-moment refinement to the protocol (a re-score of every comparison); and model confidence sets over the feature sets once per-bar losses are persisted. None of these changes the chain above; each tightens a link in it.

A Master Table of Campaign Arms

Table 9 lists every arm of the unification campaign: one frozen panel (Section 3), one scoring protocol (Section B.1). QLIKE is the pooled loss under the evaluation protocol; Δ is the *paired* mean per-bar loss differential against the benchmark, and DM its Diebold–Mariano t-statistic, both computed on the row intersection of the two arms. **Protocol note.** Two row sets appear in this table: arms run under the original scoring protocol and arms run under the `oos_mult=2` protocol (chunks 9–99, 248,686 rows), which excludes the earliest out-of-sample period. Pooled QLIKE *levels* are not comparable across protocols — the excluded rows carry above-average loss, so `oos_mult=2` arms read mechanically lower. The Δ and DM columns are paired on common rows and are comparable throughout; level comparisons should be made only within a protocol. The gated two-block ridge and the *trail*-family arms are the `oos_mult=2` panel.

B The Back-Transformation

B.1 The Back-Transformation

Every model in this paper is fit on the preprocessed scale. Realized variance RV_t is diurnally adjusted and then passed through the square-root transform, so the regression target is $y_t = \sqrt{\widetilde{RV}_t}$, where $\widetilde{RV}_t$ denotes the diurnally adjusted variance at bar t. Evaluation is performed on the raw variance scale. This is because QLIKE — the primary loss — is a strictly consistent scoring rule for the conditional variance, not for any transform of it. A forecast issued in square-root space must therefore be mapped back to raw space before it is scored.

The naïve inverse squares the forecast and reverses the diurnal adjustment. Squaring does not commute with conditional expectation: by Jensen's inequality, $\mathbb{E}[\sqrt{X}]^2 \leq \mathbb{E}[X]$, with equality only in the degenerate case. Decompose the target as $y_t = \hat{y}_t + \varepsilon_t$, where $\hat{y}_t$ is the model's forecast of y_t and ε_t is a conditionally mean-zero innovation. Then

$$\mathbb{E}\big[y_t^2 \,\big|\, \mathcal{F}\big] = \hat{y}_t^2 + \mathbb{E}\big[\varepsilon_t^2 \,\big|\, \mathcal{F}\big]\,, \tag{12}$$

where $\mathcal{F}$ is the conditioning information set. The conditional mean of the squared target exceeds the squared forecast by the conditional second moment of the square-root-scale innovation. A forecast that is well calibrated in $\sqrt{RV}$-space is, once squared, biased low in RV-space, and the size of the bias equals the residual variance of the model being evaluated.

Two facts about this bias determine the design. First, QLIKE penalizes under-prediction more heavily than over-prediction, and the bias of the uncorrected squared forecast is downward. Second, the bias equals each model's own residual variance, so it differs across models. The back-transform therefore includes an explicit second-moment term, and the estimator of that term is specified as part of the evaluation procedure.

The correction is a two-component causal calibration: a mean map re-estimated at every window boundary on the window before it, and a second moment seeded there and adapted bar by bar. Fix the chunked walk-forward grid: the backtest is split into contiguous evaluation windows indexed $k = 0, 1, \ldots$, tiling the panel in row order. The inputs for window k are

three per-bar arrays over its bars: the forecasts $\{\hat{y}_t\}$ on the transformed (square-root, diurnally adjusted) scale, the diurnal baseline factors $\{B_t\}$, where B_t is the factor that reverses the Stage-1 adjustment at bar t, and the raw realizations $\{RV_t^{\text{raw}}\}$. The evaluation-scale target is the *unwinsorized* transformed realization, reconstructed exactly from the persisted raw target,

$$y_t^{\text{raw}} = \sqrt{RV_t^{\text{raw}}/B_t}. \tag{13}$$

Mean calibration. The forecast is passed through one affine map per window,

$$m_t = \hat{a}_k + \hat{b}_k\,\hat{y}_t, \qquad (\hat{a}_k, \hat{b}_k) = \arg\min_{a,\,b} \sum_{s \in k-1} \left(y_s^{\text{raw}} - a - b\,\hat{y}_s\right)^2, \tag{14}$$

the Mincer–Zarnowitz regression of the evaluation-scale target on the forecast, estimated by ordinary least squares over the valid bars of window $k-1$ and applied to every bar of window k. The map absorbs the level and slope miscalibration the fit-scale construction leaves behind, including the attenuation induced by training on the winsorized target. For a window whose predecessor is missing from the harvest — a mid-panel gap, which a complete harvest does not contain — the map falls back to the identity $(\hat{a}_k, \hat{b}_k) = (0, 1)$; every such window is flagged in the harvest accounting.

Conditional second moment. Let $e_s = y_s^{\text{raw}} - m_s$ denote the residuals against the calibrated mean as applied at their own bar. The second-moment term is adaptive. Window k opens at the seed

$$\hat{\sigma}^2_{(k)} = \frac{1}{N_{k-1}} \sum_{s \in k-1} e_s^2, \tag{15}$$

the mean of window $k-1$'s squared residuals, and within the window the term tracks the model's own recent error level causally,

$$\hat{\sigma}^2_t = \lambda\,\hat{\sigma}^2_{t-1} + (1-\lambda)\,e^2_{t-1}, \qquad \lambda = 0.6, \tag{16}$$

so the variance applied at bar t uses residuals through bar $t-1$ only. The decay sits on a broad, flat optimum: every value of λ between 0.4 and 0.6 scores within 10^{-4} of the best on the benchmark arm, and $\lambda = 0.6$ is the consensus choice across the block ladder's rungs. Positivity is trivial — a convex combination of squares — and the estimator does not extrapolate: no value of window k's forecasts can move it. The scalar of Equation (15) held fixed across the window — the protocol's first iteration — loses to the retrospective in-window Duan baseline on the benchmark (0.23353 against 0.23165) precisely because it cannot track the level; the adaptive form beats that baseline on every arm measured, while remaining strictly causal (benchmark 0.22519, paired ΔQLIKE -0.0065 at DM -5.7 against the Duan convention). For a window whose predecessor is missing from the harvest, the seed falls back to the in-window scalar mean of $(y_t^{\text{raw}} - m_t)^2$; every such window is flagged in the harvest accounting.

Relation to the literature. Standard practice corrects the back-transform with a *static* scalar: the analytic log-normal factor $\exp(\hat{\sigma}^2_\varepsilon/2)$ of the log-RV literature, deployed since Andersen et al. [2003]'s lognormal–normal mixture and extended by Bekierman and Manner [2018] with the state variance of their state-space HAR, or Duan [1983]'s nonparametric mean of the exponentiated residuals. Whether the correction is worth applying at all is contested:

Demetrescu et al. [2020] compare bias corrections for exponentially transformed forecasts, find none uniformly best, and recommend skipping the correction under high persistence; Halbleib and Voev [2011] reach a parallel conclusion in the multivariate case, finding parsimonious uncorrected forecasts best. The origin of the second moment's predictability is the realized estimator's own measurement error [Asai et al., 2012], whose variance scales with the volatility level.

Time-varying second moments are nonetheless standard *inside* transform-scale models, and each ingredient of the adaptive correction has a precedent. Volatility clustering in the residuals of realized-variance models motivates GARCH$(1,1)$ innovation variances in the HAR-GARCH of Corsi et al. [2008] — under a log-scale specification the retransformation factor becomes $\exp(m_t + h_t/2)$ in place of $\exp(m_t + \hat{\sigma}^2/2)$ — with Bollerslev et al. [2009] modeling log bipower variation and Ciarreta and Zarraga [2016] the Spanish intraday electricity market with the same device. Inside log-variance GARCH models the Jensen gap is handled by construction, at a price: converting $\mathbb{E}[\log \sigma^2_{t+h} \mid \mathcal{F}_t]$ forecasts to $\mathbb{E}[\sigma^2_{t+h} \mid \mathcal{F}_t]$ requires the full h-step predictive distribution, approximated by Monte Carlo convolution of the one-step predictives [Andersen et al., 2006]. The multivariate analogue of the second-moment correction is exact: Chiriac and Voev [2011] forecast the Cholesky factors of realized covariance matrices and derive the analytic second-moment correction from the forecast-error covariance. Outside finance, Manning [1998] shows that under heteroscedastic log-scale errors the scalar Duan [1983] factor is biased and must be replaced by a retransformation conditional on an estimated variance function; Manning and Mullahy [2001] find the conditional correction consistent but cumbersome enough to recommend sidestepping the back-transform entirely.

What we have not found is the correction used as a post-hoc *evaluation protocol*: a causal second-moment model of an arbitrary forecaster's own residuals, applied uniformly across heterogeneous model classes at evaluation time. In every precedent above the time-varying second moment is estimated jointly within one model's likelihood and serves that model's own forecasts. Equation (16) is the standard scalar correction with its level allowed to adapt causally — the static scalar is recovered at $\lambda = 1$. Equation (17) below is the covariance-stationary GARCH$(1,1)$ of Bollerslev [1986], with the intercept variance-targeted as in Engle and Mezrich [1996], applied as an evaluation-layer second moment rather than estimated inside a forecast model's likelihood.

A mean-reverting refinement of the recursion. The volatility clustering that motivates Corsi et al. [2008] is present in this panel's residuals: the autocorrelation of e_t^2 is 0.15 at lag one and decays to zero within roughly one hundred bars. A covariance-stationary GARCH$(1,1)$ second moment [Bollerslev, 1986],

$$\hat{\sigma}_t^2 = \hat{\omega}_k + \hat{A}_k\, e_{t-1}^2 + \hat{B}_k\, \hat{\sigma}_{t-1}^2, \qquad \hat{\omega}_k = (1 - \hat{A}_k - \hat{B}_k)\, \bar{e}_{k-1}^2, \tag{17}$$

targets the intercept to the previous year's mean squared residual [Engle and Mezrich, 1996], so the process reverts to a causally estimated level instead of integrating shocks permanently. The EWMA of Equation (16) is the integrated special case ($\hat{\omega} = 0$, $\hat{A} + \hat{B} = 1$) of Engle and Bollerslev [1986]. The pair $(\hat{A}_k, \hat{B}_k)$ is re-selected at each year boundary by scored QLIKE on the previous year over a coarse grid ($\hat{A} \in \{0.02, 0.05, 0.1, 0.15, 0.2, 0.3\}$, $\hat{B} \in \{0.2, \ldots, 0.9\}$), and the recursion is seeded from the previous year's terminal state; every quantity applied in year k is a function of data through year $k-1$. Selections concentrate at persistent

mean-reverting dynamics: $\hat{A} = 0.3$, the top of the grid, in most years, with $\hat{B}$ between 0.2 and 0.6, so $\hat{A} + \hat{B}$ lies between 0.5 and 0.9. Paired against the EWMA on identical bars, the refinement improves the benchmark's pooled QLIKE from 0.22072 to 0.21842 — ΔQLIKE -0.00230 at DM -9.10, the paired differential negative in 21 of 23 years — and improves the paper's best mean model from 0.21386 to 0.21044 — ΔQLIKE -0.00342 at DM -6.00, negative in 21 of 23 years.[1] That the gain is at least as large on the stronger mean model argues that the harvestable structure is volatility-of-volatility in the target, in the sense of Corsi et al. [2008], rather than an artifact of a weak conditional mean. The deployed protocol nevertheless retains the EWMA of Equation (16): it has a single parameter on a flat optimum and no yearly selection step, and promoting the refinement would be a change of protocol in the sense of Section 4.2, requiring a re-score of every comparison that inherits it. Equation (17) is reported as an available upgrade, not adopted.

Calibration burn-in. The first evaluation window of each arm is the calibration burn-in, parallel to the feature construction's burn-in at the start of the panel: it estimates the two maps for its successor and is not scored — none of its bars enters any loss, any join, or any calibration diagnostic. Scoring begins at the second window. Every scored forecast is therefore computable at its own bar, with no exceptions.

The conditional-variance record. Two bar-conditional parameterizations of the second moment were implemented and rejected on measurement. An affine form, $\hat{\sigma}_t^2 = \max(\hat{c}_0 + \hat{c}_1 \hat{y}_t^2,\, 0)$, fit on the previous window's squared residuals, admits zero variance on quiet bars and lets the raw forecast collapse: under it, twenty bars carried 99% of one arm's pooled QLIKE. A log-linear form, $\hat{\sigma}_t^2 = \exp(\hat{d}_0 + \hat{d}_1 \hat{y}_t^2)\, \hat{S}$, with $\hat{S}$ the Duan [1983] retransformation factor for the log-back-transform, is structurally positive but extrapolates exponentially at bars whose $\hat{y}_t^2$ lies outside the previous window's fitted range: the second moment reached 10^{44}, inflating every arm's loss and collapsing the Diebold–Mariano differentials toward zero. A bar-conditional second moment therefore requires a model with controlled extrapolation, which is the reserved learned-correction slot for this section; the protocol uses the scalar of Equation (15).

A shrunk conditional second moment. The two rejected forms fail for the same reason: neither is anchored to the estimator that works. A third form removes that failure mode by construction. Let

$$z_t \;=\; \log\!\big((y_t^{\mathrm{raw}} - m_t)^2\big),$$

and predict z_t by ridge regression on a standardized design, fit on the rolling window and refit at every bar. The penalty is selected causally on the same embargoed forward split used for every other hyperparameter in this paper, from a grid that includes $\alpha = \infty$; at that grid point the fitted value is the training-window mean of z and the assembled variance is *exactly* the window-level second moment of Equation (15) — the seed of the adaptive recursion (16). The window-level protocol is therefore nested as the infinite-shrinkage limit, and conditioning is adopted only where the forward split prefers it. At finite α the variance is

$$\hat{v}_t \;=\; \exp(\hat{z}_t)\, c_k, \qquad c_k \;=\; \frac{\text{mean of } (y^{\mathrm{raw}} - m)^2 \text{ on window } k-1}{\text{mean of } \exp(\hat{z}) \text{ on window } k-1},$$

[1] The shootout's EWMA comparator runs one continuous recursion seeded at the first scored year for a like-for-like race, so its pooled levels differ slightly from the frozen protocol's headline levels, which re-seed at every window boundary; the paired differentials are unaffected.

so each window's variance *level* is pinned to the scalar estimator's level and conditioning can only redistribute variance across bars within a window. No floor, clip, or additive constant appears anywhere in the construction.

Measured against the benchmark on the shared panel, the conditional protocol is the largest single effect in this study: applied to the benchmark's own residuals it improves QLIKE by +0.00365 at DM +3.5, and applied to the paper's best mean model by +0.00246 at +2.1. Both are an order of magnitude larger than any increment produced by changing the conditional mean. These conditional-variance runs were scored under the campaign's first protocol — the window-level scalar of Equation (15) held fixed within each window; under the adaptive recursion of Equation (16) part of this gain is absorbed by construction, since the conditional model and the recursion trade on the same level channel, and re-scoring the sidecar under the new protocol is reserved.

Which state carries the second moment. That result might be taken to show that the exogenous panel predicts forecast uncertainty; a control arm rejects that interpretation. Conditioning the variance on the backbone alone — the autoregressive ladder and calendar dummies, with no exogenous column — improves the benchmark by +0.00283 at DM +2.8, and beats the exogenous-state version head to head (-0.00082 at -2.9). The predictable part of the squared forecast error is therefore carried by the volatility level itself, which is what the realized-variance estimator's own measurement error scales with, and not by the exogenous state. The conditional second moment is a large and robust effect about heteroskedasticity in the target; it is not evidence that the exogenous buckets carry distributional information beyond the level, and this paper does not claim that they do.

Back-transform. Both members of the scored pair are mapped to the raw scale as

$$\widehat{RV}_t^{\text{raw}} = \left(m_t^2 + \hat{\sigma}_t^2\right) B_t, \qquad RV_t^{\text{raw}} = \left(y_t^{\text{raw}}\right)^2 B_t, \tag{18}$$

where the right-hand identity is Equation (13) inverted: the realization is the persisted raw target itself, never reconstructed from the winsorized fit target. The traditional scalar estimator of Duan [1983] is the same form with the identity mean map, a window-constant scalar, and in-window fit-scale residuals: the calibrated protocol differs from the classical Duan estimator in exactly three ways — the calibrated mean, the adaptive recursion of Equation (16) in place of the window-constant scalar, and causal residuals against that mean on the evaluation scale in place of the in-window fit-scale residuals (which see the scored window's own errors).

The full scoring procedure for one evaluation window is:

1. *Inputs.* The per-bar arrays $\{\hat{y}_t\}$, $\{B_t\}$, and $\{RV_t^{\text{raw}}\}$ over windows $k-1$ and k; reconstruct y^{raw} by Equation (13).

2. *Calibration.* Estimate $(\hat{a}_k, \hat{b}_k)$ by Equation (14) and the seed $\hat{\sigma}^2_{(k)}$ by Equation (15) on window $k-1$'s valid bars, with the stated fallbacks.

3. *Back-transform.* Map every bar of window k to the raw scale with Equation (18), producing the per-bar pairs $\left(\widehat{RV}_t^{\text{raw}}, RV_t^{\text{raw}}\right)$.

4. *Loss.* Evaluate QLIKE on the raw-scale pairs,

$$\mathrm{QLIKE} = \frac{1}{N}\sum_{t=1}^{N}\left(\frac{RV_t^{\mathrm{raw}}}{\widehat{RV}_t^{\mathrm{raw}}} - \log\frac{RV_t^{\mathrm{raw}}}{\widehat{RV}_t^{\mathrm{raw}}} - 1\right), \tag{19}$$

where the mean is taken over the bars at which both members of the pair are positive; bars failing that condition are excluded from the mean.

Because $\hat{\sigma}_t^2$ is a convex combination of squares and $m_t^2 \geq 0$, the calibrated forecast $\widehat{RV}_t^{\mathrm{raw}}$ is positive whenever B_t is and window $k-1$'s residuals are not identically zero; the positivity condition of the exclusion rule then binds only through the realization.

Three implementation choices define the estimator.

Window-level calibration, bar-level variance. The mean map $(\hat{a}_k, \hat{b}_k)$ is re-estimated at every window boundary from the window before it; the second moment opens at the previous window's scalar and then adapts bar by bar by Equation (16). On the raw scale the correction also varies through the baseline factor B_t. Relative to the squared calibrated forecast, the correction is largest at the bars where m_t^2 is smallest — the quiet bars — and largest in level immediately after the model's own recent errors have been large.

Causal, hence deployable. Every estimated quantity applied in window $k \geq 1$ is a function of data through window $k-1$ only, and windows are processed in strictly ascending order. The map is a quantity a deployed forecaster could have computed at t: the scored object is a real-time forecast, not a retrospectively corrected one. Scoring begins after the calibration burn-in, so there are no exceptions.

Residuals of the model itself, against the unwinsorized target. The residuals entering Equation (15) are the evaluated model's own out-of-sample forecast errors from the preceding window, measured against the unwinsorized evaluation-scale target of Equation (13) — not training-set residuals, not the residuals of any reference model, and not fit-scale residuals. The correction is therefore model-specific: an arm with larger residual variance receives a larger upward correction, so the correction is part of each arm's forecast, not a shared post-processing constant. With this choice the decomposition of Equation (12) holds for the evaluation target itself, tails included. One bias remains and is stated explicitly: the *coefficients* of each arm are estimated on the winsorized fit target, so the forecast $\hat{y}_t$ itself carries the fit-scale clipping. That is a modeling choice rather than a scoring defect; the mean calibration of Equation (14) absorbs its first-order attenuation, it is shared by every arm, and it cancels to first order in comparisons.

Protocol note. The scoring protocol's second-moment layer was upgraded on 2026-08-11 from the window-level scalar to the causal adaptive recursion of Section B.1 (Equation (16)). Rows harvested on Hoffman2 — the benchmark, the `oos_mult=2` panel, and every increment between them — are scored under the adaptive protocol. Rows whose chunks survive only on the decommissioned cluster (the original-protocol block ridges and the conditional second-moment panel) retain their *legacy* scalar-protocol levels; the two protocols shift QLIKE levels by roughly $5\text{–}8 \times 10^{-3}$, so legacy levels are not comparable to adaptive-protocol levels, while paired Δ and DM within each protocol remain valid. A mean-reverting GARCH$(1,1)$ refinement of the recursion (Equation (17)) is measured in Section B.1 but not adopted; promoting it would be a further change of protocol. The bucket and tuned grids completed their

harvest under the current protocol on 2026-08-13 (the per-bucket attribution flipped accordingly: no bucket beats the benchmark unpenalized). The head-to-head table (Table 4) still carries a first-iteration scalar-protocol incumbent (0.23353) pending the lasso arm, launched 2026-08-13.

Table 9: All campaign arms under the frozen panel and scoring protocol.

Model	QLIKE	Δ	DM t
Benchmark			
OLS on HAR + calendar (benchmark)	0.22521	—	—
Single-bucket attribution (minimum-norm least squares)			
HAR + moments bucket	0.22761	+0.00240	+0.7
HAR + liquidity bucket	0.22934	+0.00413	+5.9
HAR + market bucket (equal-weighted)	0.22687	+0.00166	+4.6
HAR + market bucket (value-weighted)	0.22642	+0.00121	+3.2
HAR + sentiment bucket	0.22563	+0.00042	+1.4
HAR + implied-volatility bucket	0.22624	+0.00103	+3.9
HAR + volatility-demand bucket	0.22620	+0.00099	+3.4
HAR + all exogenous columns (joint)	0.22939	+0.00418	+6.3
Bucket designs under causally tuned ridge			
HAR + moments bucket	0.22407	-0.00114	-2.1
HAR + liquidity bucket	0.22817	+0.00296	+3.2
HAR + market bucket (equal-weighted)	0.22645	+0.00124	+2.7
HAR + market bucket (value-weighted)	0.22600	+0.00080	+1.7
HAR + sentiment bucket	0.22593	+0.00072	+1.7
HAR + implied-volatility bucket	0.22620	+0.00099	+2.9
HAR + volatility-demand bucket	0.22628	+0.00108	+2.4
HAR + all exogenous columns (joint)	0.22450	-0.00071	-1.1
Bucket designs under causally tuned elastic net (free mixing)			
HAR + moments bucket	0.22496	-0.00025	-0.4
HAR + liquidity bucket	0.22806	+0.00286	+3.2
HAR + market bucket (equal-weighted)	0.22647	+0.00126	+2.6
HAR + market bucket (value-weighted)	0.22634	+0.00114	+2.2
HAR + sentiment bucket	0.22624	+0.00103	+2.2
HAR + implied-volatility bucket	0.22646	+0.00125	+2.8
HAR + volatility-demand bucket	0.22632	+0.00111	+2.4
HAR + all exogenous columns (joint)	0.22570	+0.00049	+0.4
Penalized linear, full design			
Ridge, fixed $\alpha = 0.1$	0.22820	+0.00300	+4.5
Ridge, fixed $\alpha = 0.3$	0.22761	+0.00240	+3.7
Ridge, fixed $\alpha = 1$	0.22679	+0.00159	+2.6
Ridge, fixed $\alpha = 3$	0.22594	+0.00074	+1.2
Ridge, fixed $\alpha = 10$	0.22506	-0.00015	-0.3
Lasso, fixed $\alpha = 10^{-4}$	0.22382	-0.00139	-2.2

continued

Table 9 (continued)

Model	QLIKE	Δ	DM t
Ridge, causally tuned	0.22450	-0.00071	-1.1
Elastic net, causally tuned	0.22464	-0.00057	-0.6
Lasso, causally tuned	0.22566	+0.00045	+0.5
Block ridges			
Two-block ridge (stated penalties)	0.22350	-0.00171	-3.1
Two-block ridge (documented penalties)	0.22200	-0.00321	-4.7
Two-block ridge (per-block causal tuning)	0.22224	-0.00297	-5.6
Block ridges, `oos_mult=2` protocol (248,686 rows)			
Two-block ridge (gated rows, per-block causal tuning)	0.21431	-0.00330	-5.8
Conditional second-moment arms			
Benchmark, exogenous-state variance conditioning	0.22125	+0.00365	+3.5
Benchmark, backbone-only variance conditioning	0.22043	+0.00283	+2.8
Best model, exogenous-state variance conditioning	0.21613	-0.00229	-2.1

C Back-Transform Convention Sensitivity, All Arms

Table 10 reports every completed arm's pooled QLIKE under the three back-transform conventions of Section B.1, with within-convention ranks. The rank agreement statistics quoted there (τ against the second-order-adjustment ranking: none 0.682, traditional Duan 0.859) are computed over this table.

Table 10: Pooled QLIKE per arm under the three back-transform conventions (within-convention rank in parentheses).

Arm	QLIKE (none)	QLIKE (Duan)	QLIKE (contract)
OLS on HAR + calendar (benchmark)	0.24097 (96)	0.23165 (132)	0.22521 (106)
HAR + moments bucket (OLS)	0.24041 (87)	0.22729 (102)	0.22761 (131)
HAR + liquidity bucket (OLS)	0.28789 (134)	0.23152 (131)	0.22934 (140)
HAR + market EW bucket (OLS)	0.24110 (98)	0.23043 (120)	0.22687 (128)
HAR + market VW bucket (OLS)	0.24061 (91)	0.22976 (116)	0.22642 (121)
HAR + sentiment bucket (OLS)	0.24175 (102)	0.22969 (114)	0.22563 (108)
HAR + implied vol bucket (OLS)	0.24187 (104)	0.22993 (117)	0.22624 (116)
HAR + vol demand bucket (OLS)	0.24746 (126)	0.23008 (118)	0.22620 (114)
HAR + all features bucket (OLS)	0.92974 (148)	0.23051 (121)	0.22939 (141)
Ridge, fixed $\alpha = 1$	0.25403 (128)	0.22816 (109)	0.22679 (127)
Lasso, fixed $\alpha = 10^{-4}$	0.24017 (85)	0.22512 (93)	0.22382 (93)
Ridge, causally tuned	0.24304 (117)	0.22665 (100)	0.22450 (100)
Lasso, causally tuned	0.24238 (109)	0.22800 (108)	0.22566 (109)
Elastic net, causally tuned	0.24008 (83)	0.22730 (103)	0.22464 (102)
HAR + moments bucket (ridge, causally tuned)	0.23868 (79)	0.22797 (107)	0.22407 (94)
HAR + liquidity bucket (ridge, causally tuned)	0.24417 (122)	0.23166 (133)	0.22817 (134)
HAR + market EW bucket (ridge, causally tuned)	0.24076 (92)	0.23084 (122)	0.22645 (122)

Table 10: (continued)

Arm	QLIKE (none)	QLIKE (Duan)	QLIKE (contract)
HAR + market VW bucket (ridge, causally tuned)	0.24017 (86)	0.23018 (119)	0.22600 (113)
HAR + sentiment bucket (ridge, causally tuned)	0.24271 (114)	0.23117 (127)	0.22593 (111)
HAR + implied vol bucket (ridge, causally tuned)	0.24136 (100)	0.23088 (123)	0.22620 (115)
HAR + vol demand bucket (ridge, causally tuned)	0.24091 (95)	0.23100 (125)	0.22628 (118)
HAR + all features bucket (ridge, causally tuned)	0.24304 (118)	0.22665 (99)	0.22450 (99)
HAR + moments bucket (elastic net, causally tuned, free mixing)	1.49280 (151)	0.22887 (110)	0.22496 (104)
HAR + liquidity bucket (elastic net, causally tuned, free mixing)	0.24390 (120)	0.23166 (134)	0.22806 (132)
HAR + market EW bucket (elastic net, causally tuned, free mixing)	0.24053 (90)	0.23145 (130)	0.22647 (124)
HAR + market VW bucket (elastic net, causally tuned, free mixing)	0.24041 (88)	0.23093 (124)	0.22634 (120)
HAR + sentiment bucket (elastic net, causally tuned, free mixing)	0.24116 (99)	0.23142 (129)	0.22624 (117)
HAR + implied vol bucket (elastic net, causally tuned, free mixing)	0.24249 (111)	0.23116 (126)	0.22646 (123)
HAR + vol demand bucket (elastic net, causally tuned, free mixing)	0.24087 (94)	0.23123 (128)	0.22632 (119)
HAR + all features bucket (elastic net, causally tuned, free mixing)	0.24206 (106)	0.22781 (105)	0.22570 (110)
HAR + moments bucket (elastic net, causally tuned, free mixing, extended penalty grid)	2.46967 (152)	0.23208 (135)	0.22655 (125)
HAR + liquidity bucket (elastic net, causally tuned, free mixing, extended penalty grid)	0.24565 (123)	0.23452 (143)	0.23037 (143)
HAR + market EW bucket (elastic net, causally tuned, free mixing, extended penalty grid)	0.24221 (107)	0.23426 (142)	0.22877 (139)
HAR + market VW bucket (elastic net, causally tuned, free mixing, extended penalty grid)	0.24411 (121)	0.23377 (139)	0.22829 (136)
HAR + sentiment bucket (elastic net, causally tuned, free mixing, extended penalty grid)	0.24221 (108)	0.23387 (140)	0.22809 (133)
HAR + implied vol bucket (elastic net, causally tuned, free mixing, extended penalty grid)	0.24267 (113)	0.23370 (138)	0.22832 (137)
HAR + vol demand bucket (elastic net, causally tuned, free mixing, extended penalty grid)	0.46802 (144)	0.62449 (151)	0.25886 (152)
HAR + all-features bucket (elastic net, causally tuned, free mixing, extended penalty grid)	0.24251 (112)	0.22974 (115)	0.22665 (126)
Ridge, fixed $\alpha = 0.1$	0.28959 (135)	0.22962 (113)	0.22820 (135)
Ridge, fixed $\alpha = 0.3$	0.26750 (131)	0.22901 (111)	0.22761 (130)
Ridge, fixed $\alpha = 3$	0.24698 (125)	0.22725 (101)	0.22594 (112)
Ridge, fixed $\alpha = 10$	0.24303 (116)	0.22630 (97)	0.22506 (105)
Ridge, fixed $\alpha = 30$	0.24101 (97)	0.22570 (95)	0.22445 (98)
Ridge, fixed $\alpha = 100$	0.24046 (89)	0.22555 (94)	0.22414 (95)
Ridge, fixed $\alpha = 300$	0.24204 (105)	0.22586 (96)	0.22424 (96)
Lasso, fixed $\alpha = 10^{-6}$	0.73215 (147)	0.71933 (154)	0.25016 (151)
Lasso, fixed $\alpha = 10^{-5}$	0.29023 (136)	0.69309 (153)	0.24527 (150)

Table 10: (continued)

Arm	QLIKE (none)	QLIKE (Duan)	QLIKE (contract)
Lasso, fixed $\alpha = 10^{-3}$	0.24009 (84)	0.22797 (106)	0.22540 (107)
Lasso, fixed $\alpha = 10^{-2}$	0.24615 (124)	0.23357 (137)	0.22758 (129)
Two-block ridge (stated penalties)	0.23885 (80)	0.22489 (92)	0.22350 (92)
Three-block ridge (stated penalties)	0.23129 (68)	0.21630 (42)	0.21486 (47)
Four-block ridge (stated penalties)	0.23128 (67)	0.21621 (35)	0.21485 (46)
Two-block ridge (documented penalties)	0.24085 (93)	0.22383 (88)	0.22200 (89)
Three-block ridge (documented penalties)	0.25568 (129)	0.22282 (87)	0.22085 (87)
Four-block ridge (documented penalties)	0.24153 (101)	0.22278 (86)	0.22069 (86)
Two-block ridge (per-block causal tuning)	0.23633 (75)	0.22458 (91)	0.22224 (90)
Three-block ridge (per-block causal tuning)	0.22590 (14)	0.21566 (17)	0.21309 (14)
Four-block ridge (per-block causal tuning)	0.22573 (11)	0.21542 (13)	0.21292 (12)
Four-block ridge (trailing-standardized transmission)	0.22937 (55)	0.21601 (24)	0.21480 (42)
Four-block ridge (trailing, factor levels only)	0.22945 (56)	0.21605 (25)	0.21473 (38)
Four-block ridge (trailing, lead-lag flow only)	0.23062 (64)	0.21620 (34)	0.21488 (48)
Four-block ridge (trailing transmission, per-block causal tuning)	0.22598 (15)	0.21518 (12)	0.21297 (13)
Four-block ridge (trailing factor levels only, per-block causal tuning)	0.22553 (6)	0.21515 (11)	0.21282 (8)
Four-block ridge with per-bucket exogenous penalties (cyclic causal tuning)	0.22627 (17)	0.21504 (8)	0.21316 (15)
Four-block ridge (trailing factor levels, $K = 40$, rank-shaped transmission penalty, causally tuned)	0.22556 (7)	0.21496 (6)	0.21281 (6)
Four-block ridge (trailing factor levels, $K = 40$, rank-shaped transmission penalty over an extended exponent grid, causally tuned)	0.22557 (8)	0.21494 (4)	0.21281 (7)
Four-block ridge (causal trailing-demeaned factor levels, $K = 40$, rank-shaped transmission penalty over the extended exponent grid, causally tuned)	0.22572 (10)	0.21508 (9)	0.21287 (10)
Four-block ridge (factor levels divided by their causal trailing SD, $K = 40$, rank-shaped transmission penalty over the extended exponent grid, causally tuned)	0.22551 (4)	0.21489 (2)	0.21279 (4)
Three-block ridge (backbone + wide exogenous block + causal trailing-SD factor levels, $K = 40$, no product block, causally tuned)	0.22649 (18)	0.21575 (18)	0.21415 (24)
Pure exogenous spiked-prior model: backbone plus broad exogenous block with a rank-40 frozen-factor covariance spike (causally tuned)	0.22697 (30)	0.21618 (32)	0.21426 (25)
Frame factorial: exogenous frame, top-40 trailing-SD scores, no product block (causally tuned)	0.22666 (22)	0.21597 (20)	0.21482 (44)
Frame factorial: product-base frame, top-40 frozen scores, fixed spiked prior, no product block (causally tuned)	0.22707 (35)	0.21629 (38)	0.21435 (29)

Table 10: (continued)

Arm	QLIKE (none)	QLIKE (Duan)	QLIKE (contract)
Mechanism test: product-base frame, top-40 frozen scores, cross-block covariance deleted (split backbone/exog spike), no product block (causally tuned)	0.22712 (37)	0.21630 (45)	0.21438 (30)
Mechanism test: product-base frame without HAR columns, top-40 frozen scores, no product block (causally tuned)	0.22710 (36)	0.21629 (39)	0.21434 (28)
Mechanism test: product-base frame without session columns, top-40 frozen scores, no product block (causally tuned)	0.22713 (39)	0.21630 (44)	0.21441 (33)
Mechanism test: product-base values-only frame, top-40 frozen scores, no product block (causally tuned)	0.22706 (33)	0.21630 (43)	0.21439 (31)
Mechanism test: product-base frame with within-family row permutation (cross-family correlation destroyed), top-40 frozen scores, no product block (causally tuned)	0.22706 (32)	0.21627 (37)	0.21439 (32)
Mechanism test: product-base frame, top-40 frozen spiked prior plus nonlinear product block (causally tuned)	0.22578 (13)	0.21551 (15)	0.21289 (11)
Two-block ridge (backbone + causal trailing-SD factor levels, $K = 40$, causally tuned)	0.22896 (50)	0.21892 (68)	0.21664 (78)
Two-block ridge with a FULL-RANK rotated backbone+exogenous block and causal trailing-SD factor scores (causally tuned)	0.94728 (149)	0.66677 (152)	0.23035 (142)
Two-block ridge with a full numerical-rank rotated backbone+exogenous block and causal trailing-SD factor scores (roundoff-null directions removed; causally tuned)	0.22650 (19)	0.21650 (51)	0.21449 (37)
Full-rank factorial: exogenous frame, raw orthogonal rotation (causally tuned)	0.22705 (31)	0.21635 (47)	0.21444 (35)
Full-rank factorial: exogenous frame, frozen mean/SD factor scores (causally tuned)	0.22656 (20)	0.21609 (27)	0.21412 (22)
Full-rank factorial: exogenous frame, causal trailing-SD factor scores (causally tuned)	0.22667 (23)	0.21643 (49)	0.21446 (36)
Power-law transfer test: exogenous frame, causal trailing-SD full-rank scores, rank-shaped penalty (causally tuned)	0.25186 (127)	0.21725 (56)	0.21562 (66)
Full-rank refinement: exogenous frame, frozen scores, rank-shaped penalty (causally tuned)	0.22963 (58)	0.21766 (57)	0.21497 (50)
Full-rank factorial: product-base frame, raw orthogonal rotation (causally tuned)	0.26889 (133)	0.21714 (55)	0.21674 (80)
Full-rank factorial: product-base frame, frozen mean/SD factor scores (causally tuned)	0.23726 (78)	0.21714 (54)	0.21690 (81)
Full-rank factorial: product-base frame, causal trailing-SD factor scores (causally tuned)	0.22678 (24)	0.21706 (53)	0.21498 (52)
Full-rank factorial: complete-panel frame, raw orthogonal rotation (causally tuned)	0.22707 (34)	0.21636 (48)	0.21442 (34)
Full-rank factorial: complete-panel frame, frozen mean/SD factor scores (causally tuned)	0.22657 (21)	0.21612 (29)	0.21414 (23)

Table 10: (continued)

Arm	QLIKE (none)	QLIKE (Duan)	QLIKE (contract)
Two-block ridge with a FULL-RANK rotated backbone+exogenous block, trailing-SD factor scores, and a widened flat penalty grid (causally tuned)	0.32342 (141)	0.58151 (150)	0.23176 (144)
Two-block ridge with a FULL-RANK rotated backbone+exogenous block, trailing-SD factor scores, and a smooth rank-shaped penalty (causally tuned)	0.40065 (143)	0.52878 (149)	0.22859 (138)
Three-block ridge with products restricted to (HAR/session) × exogenous-value interactions (causally tuned)	0.29253 (138)	0.21594 (19)	0.21368 (19)
Four-block ridge with (HAR/session) × exogenous-value products plus causal trailing-SD factor levels, $K = 40$ (causally tuned)	0.29169 (137)	0.21509 (10)	0.21336 (16)
Four-block ridge (trailing factor levels, $K = 40$, rank-shaped transmission penalty, causally tuned on HALF-DECADE per-block grids)	0.22552 (5)	0.21492 (3)	0.21276 (2)
Ridge, causally tuned on a half-decade grid	0.24306 (119)	0.22661 (98)	0.22452 (101)
Two-block PCR (full rank), variance ordering, James-Stein shrinkage	1.25820 (150)	0.23766 (144)	0.23780 (147)
Two-block PCR ($K = 30$), variance ordering, James-Stein shrinkage	0.53476 (145)	0.23409 (141)	0.23408 (146)
Two-block PCR ($K = 20$), variance ordering, James-Stein shrinkage	0.24245 (110)	0.22386 (89)	0.22259 (91)
Two-block PCR ($K = 30$), predictive ordering, James-Stein shrinkage	0.25642 (130)	0.23309 (136)	0.23277 (145)
Two-block PCR ($K = 20$), predictive ordering, James-Stein shrinkage	0.24289 (115)	0.22413 (90)	0.22188 (88)
Two-block PCR ($K = 10$), predictive ordering, James-Stein shrinkage	0.26847 (132)	0.22146 (83)	0.21984 (85)
Two-block PCR ($K = 5$), predictive ordering, James-Stein shrinkage	0.23901 (82)	0.22009 (81)	0.21793 (84)
Two-block ridge, unrotated exogenous set (frame-gated rows, causally tuned)	0.22695 (28)	0.21629 (40)	0.21431 (27)
Two-block PCR (full rank), variance ordering, causally tuned ridge	0.23256 (72)	0.21858 (66)	0.21585 (68)
Two-block PCR ($K = 30$), variance ordering, causally tuned ridge	0.23146 (70)	0.21856 (65)	0.21589 (69)
Two-block PCR ($K = 20$), variance ordering, causally tuned ridge	0.22892 (49)	0.21826 (63)	0.21545 (60)
Two-block PCR ($K = 30$), predictive ordering, causally tuned ridge	0.22978 (59)	0.21864 (67)	0.21595 (70)
Two-block PCR ($K = 20$), predictive ordering, causally tuned ridge	0.22811 (42)	0.21814 (59)	0.21560 (65)
Two-block PCR ($K = 10$), predictive ordering, causally tuned ridge	0.22829 (44)	0.21900 (70)	0.21629 (74)
Two-block PCR ($K = 5$), predictive ordering, causally tuned ridge	0.22868 (47)	0.21955 (77)	0.21673 (79)

Table 10: (continued)

Arm	QLIKE (none)	QLIKE (Duan)	QLIKE (contract)
Three-block ridge on an exponentially weighted gram (half-life selected causally with the block penalties)	0.30072 (139)	0.21646 (50)	0.21474 (39)
Three-block ridge on an exponentially weighted gram (fixed half-life 3,000 bars)	0.23652 (76)	0.21552 (16)	0.21391 (21)
Three-block ridge on an exponentially weighted gram (fixed half-life 12,000 bars)	0.23673 (77)	0.21503 (7)	0.21280 (5)
Three-block design, positive-part James-Stein shrinkage, no tuned hyperparameters	0.33246 (142)	0.24173 (145)	0.24115 (148)
Three-block design, positive-part James-Stein shrinkage with a serial-correlation-corrected noise scale	0.66819 (146)	0.24286 (146)	0.24240 (149)
Three-block design, nonparametric empirical-Bayes (Tweedie) per-coefficient shrinkage, no tuned hyperparameters	243990.30522 (155)	1.26414 (155)	0.33006 (155)
Three-block design in the PC basis, diagonal-whitened positive-part James-Stein, no tuned hyperparameters	68.74239 (154)	0.28576 (147)	0.26770 (153)
Three-block design, diagonal-whitened positive-part James-Stein (raw-basis control), no tuned hyperparameters	55.53465 (153)	0.31051 (148)	0.27935 (154)
Four-block ridge (trailing factor levels, $K = 40$, flat transmission penalty, causally tuned)	0.22538 (3)	0.21496 (5)	0.21278 (3)
Two-block ridge: HAR backbone + trailing factor levels at the full frame spectrum (causally tuned)	0.22824 (43)	0.21816 (61)	0.21600 (72)
Two-block ridge: HAR backbone + full-rank rotation of the production-scaled exogenous columns (causally tuned)	0.22695 (29)	0.21629 (41)	0.21431 (26)
Two-block ridge: HAR backbone + $K = 40$ frozen-frame directions over all twelve ladder rungs (causally tuned)	0.23320 (74)	0.21898 (69)	0.21600 (73)
As above plus unrotated availability indicators (causally tuned)	0.22897 (51)	0.21788 (58)	0.21518 (55)
HAR backbone + $K = 20$ PLS (supervised) directions over all twelve ladder rungs, plus availability indicators (causally tuned)	0.22903 (53)	0.21815 (60)	0.21556 (62)
HAR backbone + $K = 10$ PLS (supervised) directions over all twelve ladder rungs, plus availability indicators (causally tuned)	0.22716 (40)	0.21826 (62)	0.21519 (56)
Four-block ridge (FROZEN-standardized factor levels, $K = 40$, rank-shaped transmission penalty over the extended exponent grid, causally tuned)	0.22576 (12)	0.21548 (14)	0.21286 (9)
Ridge on ladder-expanded principal components with a BIPOLAR rank-tilted penalty (causally tuned)	0.22692 (26)	0.21922 (74)	0.21528 (58)
Three-block ridge with spectrum-tilted exogenous penalty, power and step families on WIDENED grids (causally tuned)	0.22600 (16)	0.21653 (52)	0.21368 (20)

Table 10: (continued)

Arm	QLIKE (none)	QLIKE (Duan)	QLIKE (contract)
Three-block ridge with spectrum-tilted exogenous penalty (generalized Tikhonov, causally tuned)	0.22498 (1)	0.21617 (31)	0.21340 (17)
Ridge on ladder-expanded principal components with rank-tilted penalty (causally tuned)	0.22682 (25)	0.21936 (75)	0.21522 (57)
Ridge on ladder-expanded principal components at $K = 40$ with rank-tilted penalty (causally tuned)	0.23031 (62)	0.21938 (76)	0.21630 (75)
Ridge on ladder-expanded principal components at $K = 80$ with rank-tilted penalty (causally tuned)	0.22712 (38)	0.21909 (72)	0.21545 (61)
Ridge on ladder-expanded principal components at the FULL frame rank K with rank-tilted penalty (causally tuned)	0.22745 (41)	0.21906 (71)	0.21556 (64)
Ridge on principal components computed SEPARATELY PER LADDER HORIZON ($K = 20$ per rung, rank-tilted penalty shared across rungs, causally tuned)	0.22692 (27)	0.21911 (73)	0.21498 (51)
Four-block ridge (trailing factor levels, $K = 40$, transmission penalty shape selected from power, exponential and step families by the causal tuner)	0.22566 (9)	0.21489 (1)	0.21273 (1)
Three-block ridge with spectrum-tilted exogenous penalty, power and step families (causally tuned)	0.22508 (2)	0.21625 (36)	0.21346 (18)
Two-block ridge: HAR backbone + $K = 20$ principal components (causally tuned)	0.23004 (60)	0.22029 (82)	0.21778 (83)
Two-block ridge: HAR backbone + $K = 40$ principal components (causally tuned)	0.30128 (140)	0.21852 (64)	0.21753 (82)
Four-block ridge (trailing, symmetric-part control)	0.22948 (57)	0.21601 (23)	0.21477 (40)
Four-block ridge (trailing, undecomposed cross-correlation)	0.23021 (61)	0.21600 (22)	0.21478 (41)
Four-block ridge (trailing, causally refreshed frame)	0.23265 (73)	0.21620 (33)	0.21492 (49)
Four-block ridge (trailing, $K = 5$ factors)	0.23088 (65)	0.21615 (30)	0.21481 (43)
Four-block ridge (trailing, $K = 10$ factors)	0.23118 (66)	0.21608 (26)	0.21484 (45)
Four-block ridge (trailing, $K = 40$ factors)	0.23134 (69)	0.21634 (46)	0.21535 (59)
Four-block ridge (trailing, lag-2 lead–lag)	0.22868 (46)	0.21598 (21)	0.21512 (54)
Four-block ridge (trailing, transmission base excluding the cadence-heterogeneous aggregate families)	0.23043 (63)	0.21609 (28)	0.21509 (53)
HAR + product block only (stated)	0.22910 (54)	0.22227 (85)	0.21575 (67)
HAR + product block only (documented)	0.24179 (103)	0.22776 (104)	0.22438 (97)
HAR + product block only (tuned)	0.22902 (52)	0.22219 (84)	0.21556 (63)
HAR + transmission block only (stated)	0.22874 (48)	0.21997 (79)	0.21639 (76)
HAR + transmission block only (documented)	0.23898 (81)	0.22929 (112)	0.22489 (103)
HAR + transmission block only (tuned)	0.22861 (45)	0.21997 (80)	0.21647 (77)
HAR + transmission block only (trailing)	0.23255 (71)	0.21996 (78)	0.21596 (71)

D Running Appendix: Side Analyses

This appendix collects analyses run alongside the paper that are not part of its argument: other entry clocks and holding periods for the same-day option trade, other forecast targets, the trade after the forecast panel ends, and several checks on the last-30-min trade itself. Each subsection states the question, gives the table, states the finding, and says why it is kept out of the main text.

Conventions. Unless a subsection says otherwise: the instrument is the straddle of Section 4.5 (the nearest out-of-the-money call and put of the same-day expiry, one position); R is the return of the long straddle per unit of midpoint premium and a short earns $-R$; Sharpe ratios are annualized with $\sqrt{252}$ per trade day, as in Table 7; a crossed fill buys at the ask and sells at the bid at entry, and a position that settles in cash pays no exit spread; the forecast is the block-diagonal ridge; and the scored frame is the 866 expiration days of Table 7. The last-30-min trade is the trade of Section 5.4: entry at 15:30, cash settlement at the close. Panel stamps are bar-end labelled, so the forecast used at a clock t is the one issued at t for the bar $[t, t+30]$.

D.1 Every Thirty Minutes: A Re-Picked Straddle

Question. Does the forecast trade at every thirty-minute bar of the session, not only the last? At each clock t from 10:00 to 15:00 the nearest out-of-the-money straddle is bought or sold at its midpoint and closed at the next stamp's midpoint of the same strikes, a thirty-minute hold; the 15:30 straddle is held to settlement, so that leg is the last-30-min trade. The strikes are picked afresh at every clock. The chain carries only the same-day expiry, so no thirty-minute implied variance is quoted. It is built from the quoted variance to the close, $\mathrm{IV}^2_{\mathrm{hr},t} h_t$ (the vendor's hourly implied volatility squared, times the hours left), multiplied by w_t, the bar's average share of the rest of the day's realized variance on earlier days. The signal is $s^{\mathrm{m}}_t = \widehat{RV}_t - \mathrm{IV}^2_{\mathrm{hr},t}\, h_t\, w_t$, and a day's return is the sum of its twelve legs. Table 11 scores four rules: always short; always short with the 15:30 leg left flat; $\mathrm{sign}(s^{\mathrm{m}})$ at every clock; and always short before 15:30 with $\mathrm{sign}(s^{\mathrm{m}})$ on the 15:30 leg.

Table 11: The re-picked straddle, pooled over the twelve clocks 10:00–15:30 on the 866 expiration days. mean/day and t belong to the daily sum of the twelve legs, and the Sharpe ratio is that sum's mean over its standard deviation times $\sqrt{252}$. Crossings per day count the trades that cross the spread (a re-pick onto the same strikes with the same sign is a hold); the break-even half-spread is the mean daily midpoint profit over the mean crossings per day, in percent of premium. Panel (a) leaves flat the 13 bars whose vendor implied volatility is a censored solver value; panel (b) re-inverts it from the straddle midpoint, as the main text does, which moves the block-diagonal ridge from 2.34 to 2.32.

	mean/day	t	Sharpe (mid)	Sharpe (crossed)	crossings per day	break-even half-spread (% prem.)	% bars bought
(a) four rules, block-diagonal ridge							
always short	0.160	3.53	1.90	−1.92	16.4	0.98	0.0
always short, flat at 15:30	0.146	6.36	3.43	−3.62	16.0	0.91	0.0
sign(s^{m})	0.200	4.35	2.34	−1.90	18.2	1.10	38.7
always short, sign(s^{m}) at 15:30	0.243	5.51	2.97	−1.10	16.7	1.46	3.3
(b) sign(s^{m}) *for each forecast*							
baseline (HAR + calendar OLS)	0.163	3.54	1.91	−2.35	—	—	—
block-diagonal ridge	0.198	4.30	2.32	−1.94	—	—	—
block-diagonal ridge, without the FOMC columns	0.192	4.19	2.26	−2.02	—	—	—
LightGBM	0.179	3.92	2.12	−2.16	—	—	—
XGBoost	0.173	3.75	2.02	−2.21	—	—	—
lasso (causally tuned)	0.180	3.96	2.13	−2.17	—	—	—
lasso ($\alpha = 10^{-4}$)	0.209	4.59	2.47	−1.82	—	—	—
elastic net (causally tuned)	0.151	3.26	1.76	−2.47	—	—	—

Finding. At the midpoint every rule earns (Sharpe 1.90 to 3.43) and sign(s^{m}) is ahead of always short (2.34 against 1.90; the eight forecasts 1.76 to 2.47). At the crossed spread every rule loses (−3.62 to −1.10; the eight forecasts −2.47 to −1.82). The book crosses the spread 16.0 to 18.2 times a day, and the median half-spread, 1.69% of premium, is above every rule's break-even half-spread (0.91% to 1.46%). The least negative rule at the crossed spread is the one that uses the forecast only on the settling 15:30 leg (−1.10).

Why it is not in the main text. No rule survives the spread, and the implied side of the daytime signal is a construction — a profile share of a quote for the rest of the day — not a price.

D.2 Time to the Close: Holding from Every Clock, Delta-Hedged

Question. Enter at any clock and hold to the close: does a forecast of the rest of the day, read against the quoted variance for the rest of the day, pick the side? At each clock t from 10:00 to 15:30 the nearest out-of-the-money straddle is bought or sold at its midpoint and held to cash settlement, twelve overlapping trades a day. Every position is delta-hedged in the index: at the entry stamp and at every later thirty-minute stamp to 15:30, the straddle's Black–Scholes delta (entry strikes, that stamp's spot, and that stamp's implied volatility over the hours left) is offset by index units held to the next stamp, the last from 15:30 to the official close. The hedge trades at the vendor spot at no cost, and R_t adds its profit to the settlement value. The signal stretches the forecast to the rest of the day, $s_t = \widehat{RV}_t/w_t - \mathrm{IV}^2_{\mathrm{hr},t} h_t$, which

has the sign of s_t^{m} above; at 15:30, $w_t = 1$ and $h_t = \frac{1}{2}$, and s_t is the signal of the last-30-min trade.

Table 12: Holding to the close from every clock, delta-hedged, pooled over the twelve entry clocks 10:00–15:30 (10,387 trades); columns as in Table 11. Each trade crosses the spread once, at entry. The 15:30 leg column scores that leg alone at the crossed spread.

	days	mean/day	t	Sharpe (mid)	Sharpe (crossed)	mean/day (crossed)	15:30 leg Sharpe (crossed)	% bars bought
always short	866	1.014	5.44	2.93	2.21	0.766	−0.23	0.0
sign(s)	866	0.563	4.46	2.40	1.35	0.315	0.86	38.7

Table 13: The position against the outcome, by entry clock (midpoint, hedged R). Accuracy is the share of days on which the position had the sign of R: a buy with $R > 0$ or a sale with $R \leq 0$. The accuracy of always short is $1 - P(R > 0)$. The bars with a censored implied volatility are left out.

entry	days	accuracy sign(s)	accuracy always short	buy share	$P(R > 0)$	$P(R > 0 \mid \text{buy})$	$P(R \leq 0 \mid \text{sell})$
10:00	864	0.573	0.660	0.351	0.340	0.376	0.679
10:30	865	0.566	0.686	0.348	0.314	0.329	0.693
11:00	865	0.525	0.679	0.422	0.321	0.318	0.676
11:30	864	0.565	0.675	0.388	0.325	0.358	0.696
12:00	866	0.539	0.630	0.424	0.370	0.392	0.647
12:30	866	0.540	0.648	0.426	0.352	0.374	0.664
13:00	866	0.545	0.642	0.483	0.358	0.400	0.681
13:30	866	0.585	0.648	0.312	0.352	0.400	0.669
14:00	864	0.517	0.634	0.492	0.366	0.381	0.649
14:30	865	0.518	0.609	0.394	0.391	0.384	0.605
15:00	862	0.566	0.618	0.215	0.382	0.378	0.617
15:30	861	0.551	0.627	0.400	0.373	0.404	0.648

A forecast fitted on the rest of the day. The stretch $\widehat{RV}_t/w_t$ reaches the rest of the day from a one-bar forecast. Three maps fitted on the rest-of-day variance itself replace it, one regression per entry clock for $\log RV_t^{\mathrm{rem}}$ on sessions strictly before the day: M1 on the log of the next-bar forecast, M2 adding the realized variance from the open to t, and M3 adding the previous session's rest-of-day variance (the constructions of Section D.10). The 15:30 clock is left out because there the rest of the day is one bar.

Table 14: Rest-of-day signals on the eleven entry clocks 10:00–15:00 (9,521 trades, the 866 days). QLIKE scores each forecast against the realized variance from t to the close. sign(s^{rem}) trades $F_t^{\text{rem}} - \text{IV}^2_{\text{hr},t} h_t$ with the fitted map in place of the stretch. ΔSharpe is the paired difference from the profile-stretch row at the midpoint, with a 95% moving-block bootstrap percentile interval.

	QLIKE, rest of day	Sharpe (mid)	Sharpe (crossed)	% bars bought	% bars sign differs	ΔSharpe (mid)	95% interval
always short (no forecast)	—	3.38	2.65	0.0	—	—	—
sign(s), profile stretch $\widehat{RV}_t/w_t$	0.2018	2.30	1.25	38.6	—	—	—
sign(s^{rem}), M1	0.2093	1.70	0.68	46.6	10.4	−0.60	[−1.11, −0.11]
sign(s^{rem}), M2	0.2132	1.15	0.22	48.3	15.9	−1.15	[−1.82, −0.59]
sign(s^{rem}), M3	0.2009	0.86	−0.07	48.4	17.3	−1.44	[−2.14, −0.85]

Finding. Held to the close and hedged, both rules pay their spread (Table 12): always short 2.93 at the midpoint and 2.21 crossed, sign(s) 2.40 and 1.35. The forecast-free control is ahead at both fills. The direction rule is right on 0.52 to 0.59 of days at each clock; always selling is right on 0.61 to 0.69, more often at 12 of the 12 clocks (Table 13). Where it buys, the straddle wins more often than on an average day at 9 of the 12 clocks. The fitted rest-of-day maps forecast the rest of the day no better than the stretch (pooled QLIKE 0.2093, 0.2132 and 0.2009 against 0.2018), buy more often (46.6% of bars for M1 against 38.6%), and trade worse: Sharpe 1.70, 1.15 and 0.86 against 2.30 at the midpoint, with all 6 of 6 paired intervals (two fills, three maps) below zero; M1's difference is −0.60 [−1.11, −0.11].

Why it is not in the main text. Before 15:30 the forecast does not improve on the forecast-free control, the daytime trades rest on a frictionless hedge the main text does not model, and the twelve trades of a day overlap.

D.3 The Sharpe Ratio Through the Day

Question. How does the Sharpe ratio of each rule change with the clock at which the position is taken? Table 15 keeps one clock at a time, one trade per expiration day, for the two books above: the thirty-minute re-pick (unhedged) and the delta-hedged hold to the close. The 09:35 row is the first stamp of the chain; it is scored with the forecast issued at 09:30 for the 09:30–10:00 bar, and only in the hold-to-close book.

Table 15: Annualized Sharpe ratio by entry clock, one trade per expiration day. Re-pick: the thirty-minute hold of Section D.1 (its 15:30 row is the last-30-min trade). To close: the delta-hedged hold of Section D.2. sign(s) is the block-diagonal ridge; the last column is the range over the eight forecasts of Table 7. Midpoint fills unless marked crossed. The censored implied volatility is re-inverted as in the main text; the hold-to-close notebook, which leaves those bars flat, differs from the sign(s) column by at most 0.06 (15:30: 1.33).

entry	days	re-pick always short	re-pick sign(s)	to close always short	to close always short, crossed	to close sign(s)	to close sign(s), crossed	to close, sign(s) eight forecasts
09:35	863	—	—	3.11	2.44	0.68	0.02	0.62 to 1.71
10:00	864	0.26	−0.11	3.32	2.29	1.66	0.65	1.32 to 2.04
10:30	865	0.64	0.39	4.18	3.51	1.92	1.27	1.52 to 2.16
11:00	865	0.46	0.56	4.20	3.58	1.33	0.73	1.13 to 1.75
11:30	865	2.19	1.61	3.95	3.33	1.97	1.36	1.48 to 1.97
12:00	866	1.12	1.13	3.17	2.58	1.20	0.62	0.82 to 1.44
12:30	866	1.78	0.64	3.30	2.72	0.91	0.34	0.71 to 0.91
13:00	866	0.23	0.96	2.66	2.09	1.37	0.81	0.80 to 1.37
13:30	866	0.79	1.02	2.51	1.97	1.63	1.09	1.22 to 1.63
14:00	866	0.50	0.49	2.04	1.36	1.04	0.38	0.50 to 1.16
14:30	866	1.00	0.57	2.00	1.50	0.67	0.17	0.47 to 0.98
15:00	866	2.51	1.36	1.93	1.40	1.33	0.80	0.60 to 1.44
15:30	866	0.20	1.34	0.24	−0.23	1.30	0.83	0.92 to 1.51

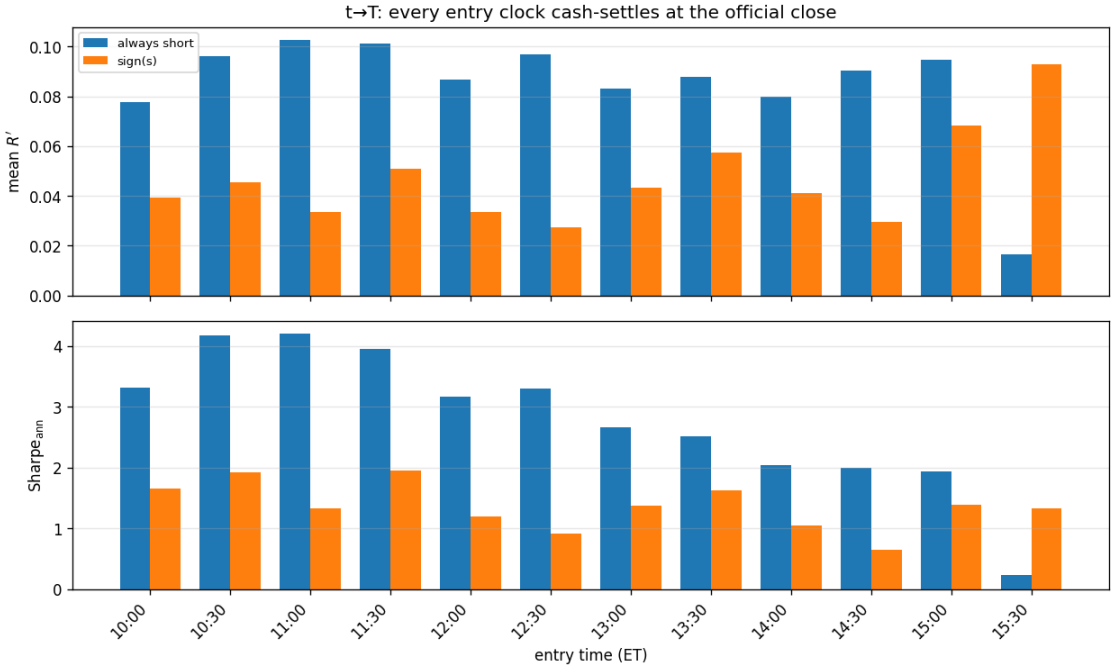


Figure 1: Delta-hedged hold to the close by entry clock, block-diagonal ridge: mean return (top) and annualized Sharpe ratio (bottom) of always short and sign(s) (holdclose notebook, where the bars with a censored implied volatility sit flat).

Finding. Hedged and held to the close, always short is strongest in the late morning (4.20 at 11:00, 4.18 at 10:30) and falls through the afternoon to 1.93 at 15:00 and 0.24 at 15:30

(−0.23 crossed). At 12 of the 12 clocks from 09:35 to 15:00 it is ahead of the sign(s) rule of every one of the eight forecasts at the midpoint; at the crossed spread it is ahead of the ridge's at 12 of them. The order reverses only at 15:30: sign(s) 1.30 (0.92 to 1.51 across the eight), 0.83 crossed. At the 09:35 open always short earns 3.11 (2.44 crossed) and the ridge's sign(s) 0.68 (0.02). In the unhedged re-pick the always-short Sharpe ratio moves between 0.20 and 2.51 from clock to clock, and sign(s) is ahead at 5 of 12 clocks, all before the spread.

Why it is not in the main text. What varies through the day is the premium earned by the forecast-free control, a different object from the one-bar forecast the paper studies; the forecast is ahead of that control only at the last bar, which is the main text's trade.

D.4 A Forecast Built for the Rest of the Day

Question. The rest-of-day signals above reach the close from a one-bar forecast. Does a model fitted directly on the rest of the day forecast it better? For every entry clock c from 09:30 to 15:30 the per-bar linear estimators (ridge, lasso, elastic net; 2000-session rolling window, causal penalty tuning) are fitted with the realized variance from c to the 16:00 close as the target, on the three column sets of the per-bar models. Each is compared on identical rows with the one-step construction of the same estimator and column set: the forecast of the bar starting at c divided by that bar's prior-days share of the rest of the day. The direct forecast is back-transformed plainly — the squared prediction times the arm's causal per-clock profile of the rest-of-day variance — with no second-moment correction.

Table 16: Direct rest-of-day forecasts against the one-step construction, 1,404–1,406 sessions from 2018-09-24 to 2024-04-30. ΔQLIKE is $100\,(\mathrm{QLIKE}_{\mathrm{direct}}/\mathrm{QLIKE}_{\text{one-step}} - 1)$ at one entry clock; the median is over the twelve clocks 09:30–15:00. Clocks better (worse) count the clocks at which the direct forecast's loss is lower (higher) with a Diebold–Mariano p below 0.05; clocks higher counts point estimates above zero. At 15:30 the two constructions are the same forecast.

column set	estimator	sessions	median ΔQLIKE (%)	clocks better	clocks worse	clocks higher (point)	15:30 |Δ| (%)
HAR + calendar	ridge	1,404–1,406	−18.8	11	0	0	0.000
HAR + calendar	lasso	1,404–1,406	−18.0	11	0	0	0.000
HAR + calendar	elastic net	1,404–1,406	−17.8	11	0	0	0.000
live-feasible (16 columns)	ridge	1,404–1,406	−30.6	6	0	2	0.000
live-feasible (16 columns)	lasso	1,404–1,406	−31.2	9	0	1	0.000
live-feasible (16 columns)	elastic net	1,404–1,406	−36.7	10	0	1	0.000
all features	ridge	1,404–1,406	−39.6	8	0	1	0.000
all features	lasso	1,404–1,406	−36.3	9	0	1	0.002
all features	elastic net	1,404–1,406	−44.4	8	0	1	0.018

Finding. The direct forecast has the lower loss at the median clock for every estimator and column set: −19% to −18% on HAR + calendar, −37% to −31% on the live-feasible set and −44% to −36% on all features. It is significantly better at 6 to 11 of the twelve clocks and

never significantly worse; at 15:30 the two agree to 0.018%. The plain back-transform has rare extreme values: 5 of the 108 clock cells carry a QLIKE above 1 on one side or the other, the largest the all features ridge at 14:00 (132.0 against 0.197 for the one-step construction), and the same kind of value gives the all-features ridge a plain QLIKE of 2.65 at 15:30. None of those cells is a significant comparison, which is why the table reports the median clock and per-clock tests rather than a pooled mean.

Trading test. The direct forecast replaces the stretch in the delta-hedged hold-to-close signal of Section D.2, $s_t = F_t^{\mathrm{rem}} - \mathrm{IV}_{\mathrm{hr},t}^2 h_t$, on the same days, bars, fills and hedge (Table 17; per-bar ridge, plain back-transform). Pooled over the entry clocks 10:00–15:30 it trades better by a point estimate: on the live-feasible set from 2.12 to 2.49 at the midpoint (+0.37 [−0.38, 1.05]) and from 1.00 to 1.48 crossed (+0.48 [−0.20, 1.20]); on all features +0.31 [−0.37, 0.97]; on HAR + calendar +0.12 [−0.63, 0.85]. Of these 6 intervals (three column sets, two fills), 6 include zero. Over all 18 estimator, column-set and back-transform combinations at the midpoint the direct forecast is ahead in 15, with 0 intervals excluding zero in its favour and 0 against. It buys less often (37.6% of bars against 43.6% on the live-feasible set), the mirror image of the fitted maps of Table 14, although it changes the QLIKE on these bars by −26.6%; both signals stay below always short (direct minus always short −0.44 [−2.02, 0.86], live-feasible, midpoint). The trading gain, where there is one, is inside its sampling error.

Table 17: Trading test of the direct rest-of-day forecast in the delta-hedged hold-to-close book, per-bar ridge, on the 866 days (another workstream's table, read as generated). Top: pooled over the entry clocks 10:00–15:30 (daily sums), Sharpe ratios of the current signal ($\widehat{RV}_t/w_t$) and the direct one (F_t^{rem}) and their paired difference, for the plain back-transform and a recalibration per entry clock. Bottom: by entry clock, plain, midpoint. Intervals: paired circular block bootstrap, 95% percentile; *: the percentile and basic intervals both exclude zero.

		midpoint			crossed spread			buy %	
bucket	F	current	direct	direct − current	current	direct	direct − current	cur.	dir.
No rest-of-day forecast: always short 2.93 / 2.21; block-diagonal ridge sign(s) *2.40 / 1.35 (midpoint / crossed)*									
plain back-transform									
live-feasible	plain	2.12	2.49	+0.37 [-0.38, +1.05]	1.00	1.48	+0.48 [-0.20, +1.20]	44	38
all features	plain	2.23	2.54	+0.31 [-0.37, +0.97]	1.13	1.58	+0.45 [-0.19, +1.16]	44	38
HAR + calendar	plain	1.33	1.45	+0.12 [-0.63, +0.85]	0.20	0.40	+0.20 [-0.51, +0.96]	46	45
recalibrated per entry time									
live-feasible	recal	1.40	1.82	+0.42 [-0.30, +1.18]	0.25	0.80	+0.55 [-0.14, +1.39]	52	46
all features	recal	1.33	1.81	+0.48 [-0.25, +1.28]	0.21	0.86	+0.65 [-0.06, +1.53]	52	46
HAR + calendar	recal	0.41	0.21	-0.20 [-1.00, +0.60]	-0.73	-0.83	-0.10 [-0.85, +0.72]	54	55

	always	live-feasible			all features			HAR + calendar		
entry	short	current	direct	direct − current	current	direct	direct − current	current	direct	direct − current
09:35	3.11	1.23	1.04	-0.19 [-1.29, +0.86]	1.35	1.06	-0.29 [-1.35, +0.84]	1.04	1.48	+0.44 [-0.77, +1.54]
10:00	3.32	0.53	1.79	+1.26 [+0.13, +2.51]*	0.64	1.60	+0.96 [-0.13, +2.00]	-0.07	0.85	+0.92 [-0.07, +2.02]
10:30	4.18	0.46	0.88	+0.42 [-0.64, +1.55]	0.53	1.49	+0.96 [-0.14, +2.02]	0.15	0.91	+0.76 [-0.31, +1.82]
11:00	4.20	0.85	1.84	+0.99 [-0.17, +2.20]	0.83	1.78	+0.95 [-0.13, +2.08]	0.76	0.93	+0.16 [-1.05, +1.32]
11:30	3.95	0.84	1.13	+0.28 [-0.89, +1.51]	0.11	1.52	+1.41 [+0.29, +2.68]*	0.25	0.35	+0.10 [-1.07, +1.26]
12:00	3.17	0.84	0.37	-0.47 [-1.60, +0.62]	0.85	1.04	+0.18 [-1.02, +1.40]	0.50	0.26	-0.24 [-1.31, +0.87]
12:30	3.30	1.20	1.29	+0.09 [-0.83, +1.10]	1.54	1.16	-0.39 [-1.33, +0.62]	0.84	0.76	-0.08 [-1.18, +1.00]
13:00	2.66	1.24	1.42	+0.18 [-0.94, +1.32]	1.48	1.19	-0.30 [-1.31, +0.78]	1.47	0.42	-1.04 [-2.01, -0.10]*
13:30	2.51	1.27	1.52	+0.24 [-0.89, +1.42]	1.42	1.76	+0.35 [-0.66, +1.39]	0.71	0.71	-0.00 [-1.27, +1.15]
14:00	2.04	2.09	1.85	-0.24 [-1.24, +0.76]	1.74	1.95	+0.21 [-0.78, +1.23]	1.15	1.10	-0.05 [-0.96, +0.84]
14:30	2.00	0.68	1.15	+0.47 [-0.15, +1.15]	1.06	1.22	+0.15 [-0.54, +0.85]	0.34	0.90	+0.57 [-0.29, +1.43]
15:00	1.93	0.35	1.43	+1.08 [+0.38, +1.85]*	0.74	1.53	+0.79 [-0.01, +1.60]	-0.00	0.45	+0.45 [-0.26, +1.22]
15:30	0.24	1.79	1.79	same forecast	1.85	1.85	same forecast	1.31	1.31	same forecast

Why it is not in the main text. The paper's target is the one-bar variance and its rest-of-session sections are parked. A better rest-of-day loss has also not been shown to be a better trade: the fitted maps of Table 14 forecast about as well as the stretch and trade worse.

D.5 After the Research Sample: May 2024 to September 2026

Question. The main text's scored frame ends with the forecast panel on 30 April 2024. What did the trade do afterwards? It has been run forward in the long-only form used live ("the card"): at 15:30 buy the straddle at the ask when the ask is at or below P^*, the Black–76 value of the straddle at the forecast's total volatility $\sqrt{\widehat{RV}_t}$, and hold it to the close; month-ends are bought at any price and are left to Section D.6. With the implied variance read at the ask, this is the buy side of sign(s). The forecast is the per-bar ridge on the free-feed column set of Section D.8, refitted on the panel extended with the post-2024

feeds (index-futures minute bars from a second vendor, hourly VIX prints). P&L per day counts R at the ask on the days bought and zero on the others. Studies 85 to 90 ask whether the loss after 2024 comes from the inputs (study 87), the buy threshold (88), the training window or the target (89), or the regime (90).

Table 18: The card after the research sample, month-ends excluded. Panel (a): research years 2021-09 to 2024-04 (564 days; the first 250 eligible days from 2020-01 are the walk-forward warm-up), after 2024-05 568 days; each walk-forward variant is set from the 250 eligible days before each decision. Panel (b): research years 2020-01 to 2024-04 (814 days; the target models start later, once their windows fill), after 2024-05 the same 568 days. Panel (c): the research years only (814 days) with the post-2024 input changes applied. The last column is the paired difference from the card on the same days — after 2024-05 in panels (a) and (b), on the research years in panel (c) — with a 95% circular block-bootstrap interval.

	% days bought research / after	research P&L/day (t)	after 2024-05 P&L/day (t)	minus the card [95%]
(a) the card and walk-forward variants (studies 88 and 90)				
card: buy iff ask $\le P^*$	38 / 51	0.0538 (1.60)	−0.0114 (−0.35)	—
88: trailing-optimal multiplier on P^*	36 / 36	0.0208 (0.77)	0.0225 (0.77)	+0.0339 [0.009, 0.061]
88: regression of R on $\log(P^*/\text{ask})$	14 / 24	0.0050 (0.32)	−0.0061 (−0.27)	+0.0052 [−0.046, 0.053]
88: premium tightened by trailing realized/implied	32 / 28	0.0572 (1.77)	0.0154 (0.66)	+0.0267 [−0.027, 0.075]
90: VIX tercile	32 / 26	0.0477 (1.53)	−0.0092 (−0.37)	+0.0021 [−0.047, 0.041]
90: realized variance so far, tercile	25 / 24	0.0126 (0.55)	−0.0132 (−0.59)	−0.0019 [−0.053, 0.042]
90: FOMC statement day	30 / 35	0.0434 (1.45)	−0.0381 (−1.45)	−0.0268 [−0.065, 0.002]
90: implied over trailing realized, tercile	24 / 27	0.0523 (1.83)	−0.0128 (−0.53)	−0.0015 [−0.041, 0.030]
90: the card's last 20 buys	26 / 23	0.0373 (1.27)	0.0073 (0.31)	+0.0187 [−0.021, 0.058]
(b) training windows and trade-aligned targets (study 89)				
training window 1000 sessions (card: 2000)	33 / 57	0.0518 (1.92)	−0.0190 (−0.54)	−0.0076 [−0.036, 0.020]
training window 500 sessions	38 / 58	0.0173 (0.64)	−0.0240 (−0.67)	−0.0126 [−0.049, 0.025]
training window 250 sessions	40 / 58	0.0223 (0.77)	−0.0124 (−0.35)	−0.0010 [−0.035, 0.034]
target $\log(RV/\text{IV}^2_{\text{ask}})$, ridge, 250 days	3 / 2	0.0058 (0.54)	0.0015 (0.22)	+0.0129 [−0.050, 0.071]
target $R > 0$, logistic, 250 days	18 / 18	−0.0010 (−0.06)	0.0015 (0.09)	+0.0129 [−0.057, 0.077]
target $\log(RV/\text{IV}^2_{\text{ask}})$, ridge, 500 days	5 / 5	0.0139 (0.73)	0.0017 (0.18)	+0.0131 [−0.050, 0.072]
(c) the research years rerun with the post-2024 inputs (study 87)				
research card (per-bar ridge, live-feasible columns)	34	0.0455 (1.72)	—	—
free-feed column set	34	0.0594 (2.16)	—	+0.0139 [−0.002, 0.035]
VIX family as hourly prints	34	0.0472 (1.71)	—	+0.0017 [−0.014, 0.022]
ES moments rescaled to the new vendor	35	0.0417 (1.58)	—	−0.0038 [−0.022, 0.012]
all three together	35	0.0360 (1.47)	—	−0.0095 [−0.045, 0.022]

Finding. The card has no edge after the research sample. On the SPX book from May 2024 to December 2025 it makes −0.0129 per day ($t = -0.31$, 212 buys on 395 days). In the walk-forward frame it makes 0.0538 per day ($t = 1.60$) in the research years, buying on 38% of days, and −0.0114 ($t = -0.35$, interval [−0.075, 0.057]) after April 2024, buying on 51%. The post-2024 inputs do not explain the loss: applied together to the research years they leave 0.0360 per day against 0.0455 (difference −0.0095, interval [−0.045, 0.022]), and each alone moves the P&L per day by at most 0.014. No walk-forward repair restores a material edge without giving up the research years: the trailing-optimal multiplier on P^* makes 0.0225 per day after 2024 ($t = 0.77$, interval [−0.033, 0.084]), above the card by 0.034 [0.009, 0.061] (Holm $p = 0.014$), but costs −0.033 [−0.078, −0.003] per day in the research years; every regime filter's post-2024 interval includes zero, the best being the card's own recent form (0.0073, difference interval [−0.021, 0.058]); shorter training windows lose −0.012 to −0.024

per day after 2024; and models of the trade-aligned targets buy on 2% to 18% of days and make −0.009 to 0.002. After 2024 the variance implied at the ask forecasts the 15:30–16:00 realized variance better than the model does (QLIKE 0.142 against 0.185 to 0.193 across the training windows); in the research years the order was the reverse (0.097 against 0.110).

Why it is not in the main text. It tests the live, long-only form of the trade on a rebuilt panel and on days the forecast panel of the main text does not cover. It is the trade's only record after April 2024.

D.6 The Month-End Long Straddle

Question. On the last session of each month the card buys the straddle at any price, with no forecast. Is 15:30 the right clock for that purchase? Study 83 buys, at each clock, the straddle nearest the spot at the ask and settles it at the close.

Table 19: The month-end long straddle by entry clock, 70 month-ends from 2020-01-31 to 2025-12-31, bought at the ask. The spread is the median relative spread at entry; minus 15:30 is the paired difference in mean R, with its paired t and the Holm-adjusted p over the twelve earlier clocks.

entry	median ask	median spread (%)	mean R	t	hit (%)	minus 15:30	paired t	Holm p
09:35	23.80	1.9	0.152	1.41	46	−0.281	−1.54	1.00
10:00	22.75	3.2	0.211	1.84	47	−0.222	−1.25	1.00
10:30	20.10	2.2	0.190	1.61	54	−0.242	−1.39	1.00
11:00	18.80	2.2	0.190	1.58	53	−0.242	−1.51	1.00
11:30	18.30	2.3	0.202	1.76	51	−0.230	−1.46	1.00
12:00	16.40	2.4	0.238	1.98	54	−0.194	−1.28	1.00
12:30	15.05	2.5	0.309	2.27	54	−0.124	−0.75	1.00
13:00	15.15	2.6	0.269	1.98	51	−0.163	−1.04	1.00
13:30	14.25	2.9	0.301	2.35	51	−0.132	−0.96	1.00
14:00	13.15	2.9	0.422	2.99	53	−0.011	−0.08	1.00
14:30	11.70	2.9	0.355	2.50	56	−0.077	−0.67	1.00
15:00	9.95	4.0	0.267	1.82	53	−0.166	−1.56	1.00
15:30	8.30	4.9	0.432	2.62	54	—	—	—

Finding. The purchase pays at every clock and most at 15:30 (mean R 0.432, $t = 2.62$); the earlier clocks earn 0.152 to 0.422, and no difference from 15:30 survives the Holm adjustment (smallest p 1.00). The 15:30 return is 0.189 over the first half of the month-ends and 0.676 over the second, from 2022-12-30. On the 40 SPX month-ends from December 2024 of Table 21 the 15:30 purchase earns 0.356 at the ask ($t = 1.45$), and entries from 15:40 to 15:50 are 0.11 to 0.13 higher, with Holm p of 1.00.

Why it is not in the main text. It is a calendar trade that reads no forecast.

D.7 Is 15:30 the Right Decision Time?

Question. Two studies move the decision. Study 80 reruns the per-bar ridge with every regressor lagged h bars, so that the forecast of the 15:30–16:00 bar uses information through 15:30 ($h = 1$, the forecast the card uses), 15:00 ($h = 2$) and so on, and trades the card rule at the 15:30 ask on the 866 days. Study 86 moves decision and entry together to τ between 15:00 and 15:50 on the SPX book of 403 sessions from 2024-12-06 to 2026-09-21: a session-by-session regression on the thirty-minute design moved to τ forecasts the variance from τ to the close, and the card buys at τ at the ask when the ask is at or below P^*.

Table 20: The decision clock on the 866 days (study 80): the card rule at the SPXW 15:30 ask with the forecast built from information through the last input. QLIKE is that forecast's loss on the 16:00 bar. The last column is the paired difference in P&L per day from $h = 1$, with a 95% block bootstrap interval (blocks of 20 days).

h	last input	QLIKE 16:00 bar	days bought	mean R bought	t	P&L/day	minus $h = 1$ [95%]
1	15:30	0.1054	286	0.152	1.97	0.0501	—
2	15:00	0.1209	341	0.010	0.17	0.0040	−0.046 [−0.082, −0.016]
3	14:30	0.1264	378	0.024	0.37	0.0105	−0.040 [−0.074, −0.004]
4	14:00	0.1467	403	0.015	0.24	0.0069	−0.043 [−0.077, −0.011]

Table 21: Decision and entry at τ on the SPX book (study 86), month-ends excluded, at the ask. The difference from 15:30 is paired, with a 95% block bootstrap interval (blocks of 20 sessions) and the Holm-adjusted one-sided p over the ten other clocks. The month-end columns buy every one of the 40 month-ends at τ.

τ	days bought	P&L/day	t	minus 15:30 [95%]	Holm p	month-end mean R	month-end t
15:00	208	0.0223	0.60	+0.033 [−0.042, 0.111]	1.00	0.320	1.52
15:05	207	−0.0075	−0.22	+0.003 [−0.082, 0.092]	1.00	0.173	0.76
15:10	208	−0.0251	−0.78	−0.015 [−0.078, 0.050]	1.00	0.282	1.26
15:15	214	0.0110	0.28	+0.021 [−0.045, 0.094]	1.00	0.310	1.36
15:20	221	−0.0318	−0.87	−0.022 [−0.072, 0.034]	1.00	0.249	1.06
15:25	223	−0.0034	−0.09	+0.007 [−0.037, 0.052]	1.00	0.250	1.06
15:30	227	−0.0103	−0.27	—	—	0.356	1.45
15:35	210	0.0057	0.15	+0.016 [−0.054, 0.087]	1.00	0.381	1.44
15:40	190	0.0068	0.18	+0.017 [−0.053, 0.087]	1.00	0.462	1.76
15:45	192	0.0425	1.03	+0.053 [−0.032, 0.140]	1.00	0.481	2.12
15:50	156	0.0059	0.15	+0.016 [−0.054, 0.087]	1.00	0.481	1.63

Finding. Only the 15:30 forecast trades. On the 866 days the card's buys return 0.152 ($t = 1.97$, 286 buys) with the 15:30 forecast and 0.010 ($t = 0.17$) with the 15:00 one; every earlier forecast earns less per day with the interval below zero ($h = 2$: −0.046 [−0.082, −0.016]). On SPX after 2024 no τ is distinguishable from 15:30 (smallest Holm p 0.773 over

both forecast designs and both fills) and no τ's P&L per day has $|t|$ above 1.48; the SPX spread is 2.7% to 5.7% of premium across the clocks, against 9.8% to 40.0% for the one-tenth-size XSP contract. Two further checks sit here. Adding one-minute index-futures features to the 15:30 forecast (study 82, 445 out-of-sample sessions) changes its QLIKE by -2.2% to 0.2% with $|t| \leq 0.66$. On 259 days with minute quotes, a limit order at P^* cancelled at once earns 0.0937 per day ($t = 1.70$), and leaving it working one minute longer costs -0.040 per day ($t = -2.85$), because the fills that come late are the ones that lose (study 84).

Why it is not in the main text. These confirm the timing the main text uses and settle execution details of the live form.

D.8 The VIX Alone

Question. A live forecaster must fetch every input each afternoon. Which columns does the forecast need? Four column sets are fitted the way the per-bar models are (one regression per bar, 2000-session rolling window; ridge, lasso and elastic net), each on the HAR lags and calendar columns plus: the live-feasible set, the sixteen design columns `sumret`, `sumabsret`, `sumret3`, `sumret4`, `sumpret2`, `sumbipow`, `sumautocov`, `sumvolume`, `numobs`, `vix`, `vvix`, `vix3m`, `fomc_release`, `fomc_day`, `fomc_until_inv`, `fomc_since_inv`; that set without `vvix` and `vix3m`; the free feed, the live-feasible set without `numobs`, `vvix`, `vix3m` (thirteen columns, all available from free sources); and `vix` alone.

Table 22: Column sets for the 15:30 forecast, on the 866 days. QLIKE is relative to the live-feasible set on the same rows, on the 16:00 bar and averaged over the 13 bars of the session; negative is better. The Sharpe ratios are those of sign(s) under the research scorer, which recalibrates the 16:00 bar on its own; the notebook's all-bar recalibration is quoted in the text.

column set	estimator	16:00 QLIKE vs live-feasible (%)	13-bar QLIKE vs live-feasible (%)	Sharpe (mid)	Sharpe (crossed)
HAR + calendar + VIX only	ridge	−5.92	+3.31	1.48	1.00
HAR + calendar + VIX only	lasso	−1.73	+3.11	1.42	0.94
HAR + calendar + VIX only	elastic net	−0.60	+3.34	1.34	0.86
live-feasible minus VVIX and VIX3M	ridge	−0.04	−0.44	1.86	1.39
live-feasible minus VVIX and VIX3M	lasso	+0.10	−0.10	1.85	1.38
live-feasible minus VVIX and VIX3M	elastic net	+0.14	−0.51	1.90	1.43
live-feasible (16 columns)	ridge	0.00	0.00	1.90	1.44
live-feasible (16 columns)	lasso	0.00	0.00	1.83	1.36
live-feasible (16 columns)	elastic net	0.00	0.00	1.41	0.94
free feed (ES 1-min bars + VIX + FOMC calendar)	ridge	−0.43	−0.01	1.83	1.36
free feed (ES 1-min bars + VIX + FOMC calendar)	lasso	−0.63	+0.76	1.91	1.45
free feed (ES 1-min bars + VIX + FOMC calendar)	elastic net	+0.11	+0.18	1.81	1.35

Finding. The VIX alone gives the best 16:00 forecast (QLIKE -5.9% for the ridge) but is $+3.3\%$ worse over the 13 bars, and it trades worse: 1.48 at the midpoint and 1.00 crossed against 1.90 and 1.44. Under the notebook's all-bar recalibration the order is the same (1.38 against 1.68). Dropping `vvix` and `vix3m` changes the 16:00 QLIKE by -0.04%, and the free

feed by −0.43% (Sharpe 1.83): the index-futures moments are what the trade uses, and of the Cboe indices only the VIX is needed.

Why it is not in the main text. It is an implementation question for the live forecast. It is also one more case of the loss and the trade disagreeing.

D.9 Per-Bar Against Pooled Forecasts

Question. A per-bar forecast fits one regression for each thirty-minute bar; its pooled twin fits one regression for all bars. Which is the better variance forecast? QLIKE is computed bar by bar for the 13 session bars on the 866 days, both tables through the same recalibration.

Table 23: Per-bar forecast against its pooled twin: the percent difference in mean QLIKE (negative: the per-bar forecast is better), with a 95% circular block-bootstrap interval (blocks of 21 days), over the session and on three bars labelled by bar end. DM t is the Diebold–Mariano statistic of the session mean. Bars better (worse) count the bars whose interval lies wholly below (above) zero.

per-bar forecast	session mean (%) [95%]	DM t	14:30 bar (%) [95%]	15:00 bar (%) [95%]	16:00 bar (%) [95%]	bars better	bars worse
ridge (HAR + calendar)	−8.5 [−11.0, −6.0]	−5.91	−30.5 [−34.4, −26.1]	−10.4 [−14.6, −5.8]	−6.3 [−10.4, −1.9]	3	2
ridge (all features)	−4.1 [−9.2, 0.8]	−1.59	−35.7 [−47.1, −23.5]	−7.9 [−15.3, 0.7]	−2.6 [−9.4, 4.4]	1	5
lasso (all features)	−8.8 [−14.1, −4.1]	−3.30	−36.1 [−49.7, −21.7]	−12.7 [−21.4, −2.7]	−2.5 [−6.7, 1.7]	2	1
elastic net (all features)	−9.0 [−14.2, −4.3]	−3.28	−38.0 [−51.2, −24.3]	−14.8 [−22.5, −6.6]	−0.7 [−6.4, 5.9]	3	1
ridge (live-feasible set)	−3.8 [−9.6, 2.2]	−1.28	−37.9 [−49.4, −26.1]	−8.9 [−16.1, −1.0]	+1.6 [−3.9, 7.9]	2	6
lasso (live-feasible set)	−3.3 [−11.0, 4.3]	−0.88	−38.9 [−51.0, −26.4]	−9.7 [−18.4, 0.5]	+2.7 [−3.7, 10.6]	1	4
elastic net (live-feasible set)	−4.4 [−11.3, 2.5]	−1.29	−39.0 [−50.9, −26.2]	−12.9 [−20.9, −3.9]	+0.9 [−3.8, 6.7]	2	6

Finding. Over the session every per-bar forecast has the lower loss (−9.0% to −3.3%), with the interval below zero for 3 of the 7. The gain sits mostly in one bar, 14:00–14:30 (−39.0% to −30.5%), the bar that holds the FOMC statement: on the 33 statement days the difference there is −78% to −43%, those days carry 0.45 to 0.54 of the pooled forecasts' loss at that bar, and on the other 833 days it is −16% to −5%. On the traded 16:00 bar only the ridge (HAR + calendar) resolves a gain (−6.3%, [−10.4, −1.9]).

Why it is not in the main text. The per-bar gain is mostly a calendar effect at a bar the trade does not use.

D.10 Remaining-Variance Forecasts Across the Session

Question. How well does each one-bar model extend to the variance remaining until the close, at every entry clock, using only what is known at the clock? Three regressions for the log of the remaining variance, M1 to M3 of Table 24, are fitted per entry clock on strictly earlier days (at least 63) and rescaled by the running mean of realized over forecast; M0 uses no model. Each is built for the HAR OLS baseline and for the block-diagonal ridge, on the full panel: 5,582 days from 2002-01-23 to 2024-04-30.

Table 24: QLIKE of the rest-of-day forecasts, pooled over the entry clocks 10:00–15:30 (the target runs to the 16:00 close), on all 5,582 days and on the 2,089 days from 2016. DM t compares the block-diagonal ridge with the HAR OLS baseline in the same construction (negative: the ridge is better).

construction	all days HAR OLS	all days block ridge	all days DM t	from 2016 HAR OLS	from 2016 block ridge	from 2016 DM t
M0: per-clock mean of past $\log RV^{\text{rem}}$ (no model)	0.8626	—	—	0.8993	—	—
M1: log of the next-bar forecast	0.1600	0.1435	−5.23	0.2137	0.1877	−3.24
M2: M1 + realized variance so far	0.1563	0.1407	−6.68	0.2106	0.1858	−4.17
M2, one fit with clock dummies	0.1565	0.1410	−6.71	0.2099	0.1856	−4.15
M3: M2 + yesterday's RV^{rem}	0.1507	0.1367	−7.19	0.1997	0.1784	−4.34

Finding. The block-diagonal ridge beats the baseline in every construction and era ($|t|$ from 3.2 to 7.2); M1 already has 0.1435 against 0.1600, and M3 is the best, 0.1367, against 0.863 for the model-free clock mean. The loss drops between the 14:00 and 14:30 entries (M1, block ridge: 0.196 to 0.122) because the 14:00–14:30 bar, the FOMC statement bar, leaves the window. On the option days the same maps do not trade better (Table 14): M3, the best here, trades worst there.

Why it is not in the main text. The paper's rest-of-session sections are parked.

D.11 Sizing by the Size of the Forecast Gap

Question. sign(s) bets the same amount every day. Does a position that grows with the gap do better? Three sizes, each set from earlier days only: (a) s_d over the root mean square of s on earlier days (unbounded); (b) sign(s_d) times the share of earlier days with a smaller $|s|$; (c) twice the share of earlier days with a smaller s, minus one, which can take the side opposite to sign(s). The first block scores the days after a year of history, the second every day from the second.

Table 25: Sized rules against sign(s) on the same days. The ridge columns give the block-diagonal ridge's sized Sharpe ratio and its difference from sign(s) with a 95% circular block-bootstrap interval (blocks of 21 days). The count columns report, over the 15 forecasts, how many differences are positive, and how many intervals lie wholly above and wholly below zero.

size	sample	days	ridge Sharpe (mid)	ridge: minus sign(s) (mid) [95%]	15 forecasts (mid): >0 / CI>0 / CI<0	median (mid)	15 forecasts (crossed): >0 / CI>0 / CI<0
(a) proportional, s_d/rms_d	after 252 days	614	0.76	−0.78 [−2.09, 0.62]	0 / 0 / 0	−0.73	0 / 0 / 0
(b) rank of $\lvert s_d\rvert$, sign kept	after 252 days	614	1.58	+0.04 [−0.73, 0.86]	8 / 0 / 0	+0.03	8 / 0 / 0
(c) centred rank, $2\hat{F}_d(s_d)-1$	after 252 days	614	1.14	−0.39 [−1.36, 0.51]	0 / 0 / 0	−0.38	0 / 0 / 0
(a) proportional, s_d/rms_d	from day 2	865	−0.70	−2.05 [−3.05, −0.26]	0 / 0 / 11	−1.84	0 / 0 / 10
(b) rank of $\lvert s_d\rvert$, sign kept	from day 2	865	1.13	−0.23 [−0.90, 0.49]	2 / 0 / 0	−0.13	2 / 0 / 0
(c) centred rank, $2\hat{F}_d(s_d)-1$	from day 2	865	0.62	−0.74 [−1.52, 0.14]	0 / 0 / 5	−0.54	0 / 0 / 3

Finding. After a year of history, proportional sizing loses to sign(s) for all 15 forecasts (block-diagonal ridge 1.54 for sign(s) on those 614 days); the rank with the sign kept ties it

(8 of 15 above, median +0.03); the centred rank is below it for all 15. No sized rule has an interval above zero for any forecast, block or fill.

Why it is not in the main text. It is a negative result; the main text states only that the position uses the sign.

D.12 The Trade Beside the S&P 500 and Other Markets

Question. An investor already holds the S&P 500. How much of the sign(s) trade belongs beside it, and is the trade a disguised bet on another market? The weight is the maximum-Sharpe mix of the two risk-scaled daily returns, re-estimated every day from earlier days only (252-day warm-up, then expanding; no short positions), at the midpoint.

Table 26: The trade beside the S&P 500, 614 days from 2021-08-11. Correlations are those of the trade's daily return with the S&P 500 over the trade's own window (15:30 to the close), with the size of that move, and close to close. The risk share is the trade's share of the risk budget; the dollar column is the median premium held per $100 of index. The last column is the combination's Sharpe ratio minus that of the S&P 500 alone (0.40), with a 95% paired block-bootstrap interval.

forecast	corr. S&P 15:30→close	corr. |S&P| 15:30→close	corr. S&P close-to-close	mean risk share	premium $ per $100 S&P	Sharpe combined	minus S&P alone [95%]
baseline (HAR + calendar OLS)	0.04	−0.23	0.00	0.63	3.42	1.00	+0.60 [−0.67, 1.77]
block-diagonal ridge	−0.04	−0.14	−0.01	0.73	4.32	1.44	+1.04 [−0.46, 2.55]
block-diagonal ridge, without the FOMC columns	−0.06	−0.14	−0.01	0.68	3.82	1.40	+1.00 [−0.32, 2.30]
LightGBM	−0.07	−0.08	−0.01	0.81	6.82	1.37	+0.96 [−0.70, 2.52]
XGBoost	−0.04	−0.11	−0.01	0.81	7.11	1.48	+1.08 [−0.51, 2.57]
lasso (causally tuned)	−0.05	−0.09	−0.04	0.69	4.28	1.68	+1.28 [−0.14, 2.66]
lasso ($\alpha = 10^{-4}$)	−0.05	−0.08	−0.04	0.76	5.02	1.58	+1.18 [−0.33, 2.66]
elastic net (causally tuned)	0.00	−0.14	−0.03	0.66	3.22	1.08	+0.67 [−0.74, 2.02]
per-bar ridge (HAR + calendar)	0.08	−0.12	0.02	0.73	4.52	1.36	+0.95 [−0.29, 2.06]
per-bar ridge (all features)	−0.05	−0.16	−0.04	0.80	5.66	1.43	+1.03 [−0.41, 2.45]
per-bar lasso (all features)	−0.05	−0.21	−0.02	0.79	5.87	1.34	+0.93 [−0.57, 2.38]
per-bar elastic net (all features)	−0.05	−0.19	−0.02	0.81	5.68	1.28	+0.87 [−0.66, 2.31]
per-bar ridge (live-feasible set)	−0.11	−0.17	0.01	0.78	5.52	1.49	+1.09 [−0.43, 2.54]
per-bar lasso (live-feasible set)	−0.06	−0.16	0.03	0.75	5.12	1.63	+1.23 [−0.31, 2.74]
per-bar elastic net (live-feasible set)	−0.08	−0.17	0.01	0.77	5.38	1.71	+1.31 [−0.11, 2.71]

Table 27: Correlation of the trade's daily return (midpoint) with same-day moves of other markets, on the 866 days wherever both exist, with a 95% circular block-bootstrap interval (blocks of 21 days).

same-day move	days	block-diagonal ridge [95%]	per-bar ridge, live-feasible [95%]
S&P 500 return, 15:30→close	866	−0.060 [−0.173, 0.019]	−0.068 [−0.181, 0.017]
S&P 500 |return|, 15:30→close	866	−0.109 [−0.206, −0.001]	−0.091 [−0.200, 0.018]
S&P 500 return, close-to-close	866	0.001 [−0.059, 0.050]	−0.001 [−0.058, 0.066]
S&P 500 |return|, close-to-close	866	−0.046 [−0.088, 0.010]	0.001 [−0.046, 0.059]
VIX change, close-to-close (points)	861	0.012 [−0.038, 0.067]	0.009 [−0.056, 0.055]
VIX change, 15:30→16:00 (panel)	795	0.062 [−0.027, 0.155]	0.040 [−0.038, 0.143]
Nasdaq-100 return, close-to-close	866	−0.019 [−0.079, 0.033]	−0.010 [−0.073, 0.057]
Russell 2000 return, close-to-close	866	0.005 [−0.059, 0.059]	0.022 [−0.043, 0.093]
10-year Treasury yield change, close-to-close (points)	856	0.027 [−0.035, 0.086]	−0.017 [−0.086, 0.051]
long Treasuries (TLT) total return, close-to-close	864	−0.031 [−0.093, 0.035]	0.027 [−0.036, 0.089]
high-yield credit (HYG) total return, close-to-close	858	−0.017 [−0.059, 0.034]	−0.002 [−0.062, 0.073]
dollar index return, close-to-close	856	−0.041 [−0.100, 0.016]	−0.043 [−0.112, 0.023]
realized minus implied variance, 15:30→16:00	866	−0.013 [−0.065, 0.101]	−0.030 [−0.084, 0.090]

Finding. The trade barely moves with the index's direction (close-to-close correlation −0.04 to +0.03 across the 15 forecasts) and moves against the size of the last half hour's move (−0.23 to −0.08), since the rule sells on most days. Earlier days put most of the risk budget in it: for the block-diagonal ridge a mean risk share of 0.73, a median $4.32 of premium per $100 of index, and a combined Sharpe ratio of 1.44 against 0.40 — a gain of 1.04 [−0.46, 2.55]. Every forecast's interval includes zero (highest lower bound −0.11). Across the 13 same-day series only 1 interval excludes zero for the block-diagonal ridge, the size of the S&P 500's move from 15:30 to the close (−0.11, [−0.206, −0.001]), and 0 for the per-bar ridge.

Why it is not in the main text. The paper is about the forecast, not portfolio construction, and on this sample the gain is not distinguishable from none.

D.13 Strategy variations of the last-30-min trade

The straddle — the nearest out-of-the-money call and the nearest out-of-the-money put, same-day expiry, one position — is one choice of instrument for the sign(s) rule. Table 28 writes down the standard alternatives and the way the rule maps onto each: a variation is written as the structure L bought when $s > 0$, and it is sold (the position is $-L$) when $s < 0$; always short sells it every day. The straddle with a wing hedge is the one exception: it holds the plain straddle on buying days and the short iron butterfly on selling days. All structures are same-day expiry SPXW options entered at the 15:30 quotes and cash-settled at the official close, on the same 866 days as the straddle. Strikes are the live 15:30 strikes (K_c the smallest call strike at or above the index S, K_p the largest put strike at or below it); a wing is the nearest live strike at least $w \in \{10, 25, 50\}$ index points beyond the body strike. Fills are the midpoint of every leg, or crossed (every leg bought at the ask and sold at the bid). The return is the day's profit and loss in index points divided by that day's straddle premium at the same fill, so every row shares the straddle row's denominator and a paired difference isolates the structure; the straddle row is the straddle trade's return. Intervals

are 95% circular block bootstrap intervals (block 21 days, 2,000 draws), paired against the straddle on the same days, rule and fill.

Table 28: The strategy variations. $C(K)$, $P(K)$: a call, a put at strike K; + bought, − sold in the buy-side form L. K_c^+, K_p^-: the next live strike beyond K_c, K_p.

variation	bought when $s > 0$ (L)	sold when $s < 0$ ($-L$)	own notional	purpose
straddle (baseline)	$+C(K_c) + P(K_p)$	short straddle	premium	convexity both ways at the money
strangle, one strike wide	$+C(K_c^+) + P(K_p^-)$	short strangle	premium	all time value; cheaper, less gamma at the money
strangle x OTM, $x \in \{0.5, 1, 2\}\%$	$+C(\geq S(1+x))+P(\leq S(1-x))$	short strangle	premium	tail-only exposure; sold, a small premium for the tails
iron butterfly, w	$+C(K_c) + P(K_p) - C(\geq K_c + w) - P(\leq K_p - w)$	short straddle + long wings (credit)	larger wing gap	tail cap on the selling side
iron condor, w	body: the strangle one strike wide; wings w beyond it	short strangle + long wings (credit)	larger wing gap	cheaper body, tail cap
put vertical, w	$+P(K_p) - P(\leq K_p - w)$	credit put vertical	wing gap	premium from the put side alone, capped
call vertical, w	$+C(K_c) - C(\geq K_c + w)$	credit call vertical	wing gap	premium from the call side alone, capped
call (put) butterfly, w	short butterfly $-C(K_0 - w) + 2C(K_0) - C(K_0 + w)$; K_0 the straddle strike nearer S	long butterfly (debit; pays on a pin at K_0)	w	defined-risk selling with a pin payoff; one wing in the money
put (call) ratio 1×2, w	backspread $-P(K_p)+2P(\leq K_p - w)$	front ratio $+P(K_p) - 2P(\leq K_p - w)$, one leg uncovered	none	tail convexity when bought, tail premium when sold
straddle + wing hedge, w	the straddle	the short iron butterfly	none (two units)	cap the selling side's tail, keep the buying side's convexity

What the variations show. Under sign(s) with the per-bar ridge (Table 29), the highest Sharpe ratio at the midpoint is the iron butterfly, w50 row (1.94; the straddle 1.90), and at crossed fills the straddle row (1.44). The mid-fill lead of the iron butterfly, w50 row over the straddle is 0.04 [−0.04, 0.14]. Wings cost the rule at crossed fills: 6 of the 6 iron-butterfly and wing-hedge rows lie below the straddle with an interval that excludes zero, against 2 of 6 at the midpoint, where the cost shrinks as the wing moves out. The median entry cost (crossed minus mid) is 2.9% of the straddle's own premium, 6.1% (w10) falling to 4.0% (w50) of the iron butterfly's, 10.4% falling to 7.0% of the iron condor's, and the single-type butterflies, with one wing in the money, cost 9.7%–13.5% of the straddle's premium against 2.9% for the straddle itself.

Table 29: Strategy variations under sign(s) with the per-bar ridge on the live-feasible set (16:00-bar recalibration), 866 days (861 for the butterflies at $w = 25, 50$). Sharpe ratios annualized by $\sqrt{252}$ per trade day, return per straddle premium; ΔSharpe: the variation minus the straddle, paired. Worst day: mid fill, in straddle premiums. Cost share: the median crossed-minus-mid entry cost over the structure's own absolute midpoint premium, averaged over the buying and the selling side.

	mid fill			crossed fill			worst	cost
variation	Sharpe	ΔSharpe	[95% interval]	Sharpe	ΔSharpe	[95% interval]	day	share
straddle	1.90	—		1.44	—		−5.42	0.03
strangle, one strike wide	1.81	−0.10	[−0.54, 0.31]	1.37	−0.07	[−0.53, 0.34]	−4.68	0.04
strangle 0.5% OTM	1.19	−0.71	[−1.94, 0.52]	0.33	−1.11	[−2.37, 0.09]	−2.87	0.25
strangle 1% OTM	0.04	−1.87	[−3.15, 0.38]	−1.42	−2.86	[−5.41, −1.67]	−1.77	0.67
strangle 2% OTM	−0.57	−2.47	[−3.69, 0.06]	−3.01	−4.45	[−13.44, −3.16]	−1.43	1.00
iron butterfly, w10	1.66	−0.24	[−0.73, 0.23]	0.58	−0.86	[−1.39, −0.37]	−2.43	0.06
iron butterfly, w25	1.84	−0.06	[−0.23, 0.12]	1.13	−0.31	[−0.48, −0.13]	−5.44	0.04
iron butterfly, w50	1.94	0.04	[−0.04, 0.14]	1.31	−0.13	[−0.21, −0.03]	−5.44	0.04
iron condor, w10	1.69	−0.21	[−0.67, 0.21]	0.56	−0.88	[−1.36, −0.46]	−2.53	0.10
iron condor, w25	1.82	−0.09	[−0.53, 0.36]	1.06	−0.38	[−0.84, 0.06]	−4.70	0.08
iron condor, w50	1.85	−0.05	[−0.49, 0.34]	1.18	−0.26	[−0.72, 0.14]	−4.70	0.07
put vertical, w10	1.09	−0.82	[−1.71, 0.10]	0.44	−1.00	[−1.93, −0.05]	−2.96	0.07
put vertical, w25	1.31	−0.59	[−1.36, 0.19]	0.88	−0.56	[−1.35, 0.24]	−4.03	0.05
put vertical, w50	1.36	−0.54	[−1.29, 0.21]	0.98	−0.46	[−1.23, 0.32]	−3.98	0.04
call vertical, w10	0.83	−1.08	[−2.00, −0.13]	0.22	−1.22	[−2.16, −0.23]	−2.58	0.06
call vertical, w25	0.87	−1.03	[−1.88, −0.24]	0.45	−0.99	[−1.86, −0.17]	−5.71	0.04
call vertical, w50	0.96	−0.94	[−1.78, −0.17]	0.57	−0.87	[−1.73, −0.08]	−5.71	0.04
call butterfly, w10	1.07	−0.83	[−1.71, 0.01]	−1.33	−2.77	[−3.72, −1.85]	−3.84	0.13
call butterfly, w25	1.48	−0.37	[−0.97, 0.18]	−0.68	−2.07	[−2.92, −1.37]	−4.92	0.04
call butterfly, w50	1.68	−0.27	[−0.80, 0.21]	−0.68	−2.17	[−2.99, −1.46]	−4.92	0.02
put butterfly, w10	1.05	−0.85	[−1.73, −0.01]	−1.50	−2.94	[−3.92, −2.01]	−3.84	0.14
put butterfly, w25	1.51	−0.35	[−0.93, 0.19]	−0.81	−2.20	[−3.05, −1.45]	−4.91	0.04
put butterfly, w50	1.74	−0.22	[−0.74, 0.27]	−1.13	−2.62	[−3.45, −1.90]	−4.89	0.02
put ratio 1×2, w10	−0.28	−2.18	[−3.82, −0.53]	−1.13	−2.57	[−4.21, −0.92]	−4.79	0.14
put ratio 1×2, w25	−1.22	−3.12	[−4.92, −1.16]	−1.77	−3.21	[−5.02, −1.27]	−7.21	0.06
put ratio 1×2, w50	−1.37	−3.27	[−5.01, −1.40]	−1.84	−3.28	[−5.03, −1.42]	−10.72	0.05
call ratio 1×2, w10	−0.43	−2.34	[−4.06, −0.66]	−1.34	−2.78	[−4.47, −1.15]	−4.60	0.11
call ratio 1×2, w25	−0.79	−2.69	[−4.48, −0.84]	−1.38	−2.81	[−4.62, −0.97]	−4.69	0.06
call ratio 1×2, w50	−0.99	−2.89	[−4.70, −1.06]	−1.51	−2.95	[−4.80, −1.10]	−5.14	0.05
straddle + wing hedge, w10	1.44	−0.46	[−0.85, −0.10]	0.73	−0.71	[−1.12, −0.36]	−2.43	0.04
straddle + wing hedge, w25	1.69	−0.21	[−0.38, −0.01]	1.10	−0.34	[−0.51, −0.12]	−5.44	0.04
straddle + wing hedge, w50	1.85	−0.05	[−0.12, 0.06]	1.30	−0.14	[−0.22, −0.03]	−5.44	0.03

Table 30: As Table 29, with sign(s) on the block-diagonal ridge (the paper's forecast, with the paper's recalibration over the session bars).

	mid fill			crossed fill			worst	cost
variation	Sharpe	ΔSharpe	[95% interval]	Sharpe	ΔSharpe	[95% interval]	day	share
straddle	1.34	—		0.87	—		−5.42	0.03
strangle, one strike wide	1.31	−0.03	[−0.47, 0.40]	0.86	−0.01	[−0.45, 0.42]	−4.68	0.04
strangle 0.5% OTM	0.66	−0.67	[−1.84, 0.58]	−0.19	−1.06	[−2.25, 0.13]	−2.87	0.25
strangle 1% OTM	−0.36	−1.70	[−2.94, 0.63]	−1.78	−2.64	[−4.96, −1.45]	−1.77	0.67
strangle 2% OTM	−0.47	−1.81	[−2.96, 0.79]	−2.93	−3.80	[−12.80, −2.42]	−1.43	1.00
iron butterfly, w10	1.23	−0.11	[−0.61, 0.41]	0.15	−0.72	[−1.25, −0.21]	−3.88	0.06
iron butterfly, w25	1.32	−0.02	[−0.21, 0.23]	0.60	−0.27	[−0.46, −0.01]	−5.44	0.04
iron butterfly, w50	1.37	0.03	[−0.04, 0.12]	0.73	−0.14	[−0.21, −0.05]	−5.44	0.04
iron condor, w10	1.45	0.11	[−0.43, 0.64]	0.32	−0.55	[−1.10, −0.01]	−2.53	0.10
iron condor, w25	1.35	0.01	[−0.46, 0.47]	0.59	−0.28	[−0.75, 0.16]	−4.70	0.08
iron condor, w50	1.34	0.00	[−0.43, 0.44]	0.67	−0.20	[−0.65, 0.22]	−4.70	0.07
put vertical, w10	1.47	0.13	[−0.71, 0.98]	0.83	−0.04	[−0.91, 0.85]	−2.96	0.07
put vertical, w25	1.51	0.17	[−0.56, 0.89]	1.08	0.21	[−0.53, 0.96]	−4.35	0.05
put vertical, w50	1.51	0.18	[−0.57, 0.90]	1.14	0.27	[−0.48, 1.01]	−4.33	0.04
call vertical, w10	−0.08	−1.42	[−2.37, −0.54]	−0.68	−1.55	[−2.51, −0.66]	−4.63	0.06
call vertical, w25	0.02	−1.32	[−2.27, −0.40]	−0.40	−1.26	[−2.21, −0.35]	−5.71	0.04
call vertical, w50	0.07	−1.27	[−2.21, −0.37]	−0.32	−1.19	[−2.13, −0.29]	−5.71	0.04
call butterfly, w10	1.11	−0.23	[−1.06, 0.57]	−1.29	−2.16	[−3.04, −1.36]	−3.84	0.13
call butterfly, w25	1.23	−0.05	[−0.55, 0.45]	−0.92	−1.73	[−2.37, −1.14]	−5.37	0.04
call butterfly, w50	1.38	−0.03	[−0.48, 0.39]	−0.97	−1.91	[−2.51, −1.36]	−5.32	0.02
put butterfly, w10	1.09	−0.25	[−1.07, 0.56]	−1.46	−2.32	[−3.23, −1.50]	−3.84	0.14
put butterfly, w25	1.25	−0.03	[−0.54, 0.50]	−1.04	−1.86	[−2.51, −1.24]	−5.68	0.04
put butterfly, w50	1.39	−0.02	[−0.48, 0.43]	−1.46	−2.40	[−3.10, −1.76]	−5.68	0.02
put ratio 1×2, w10	−0.86	−2.20	[−3.81, −0.62]	−1.73	−2.60	[−4.22, −1.02]	−4.79	0.14
put ratio 1×2, w25	−1.46	−2.80	[−4.70, −0.97]	−2.02	−2.89	[−4.81, −1.05]	−7.21	0.06
put ratio 1×2, w50	−1.53	−2.86	[−4.73, −1.02]	−2.01	−2.88	[−4.73, −1.04]	−10.72	0.05
call ratio 1×2, w10	0.24	−1.09	[−2.81, 0.54]	−0.65	−1.52	[−3.25, 0.11]	−2.80	0.11
call ratio 1×2, w25	0.01	−1.33	[−3.09, 0.35]	−0.57	−1.44	[−3.25, 0.26]	−4.16	0.06
call ratio 1×2, w50	−0.09	−1.43	[−3.19, 0.23]	−0.60	−1.47	[−3.29, 0.21]	−4.64	0.05
straddle + wing hedge, w10	0.98	−0.36	[−0.79, 0.04]	0.25	−0.62	[−1.08, −0.21]	−3.88	0.04
straddle + wing hedge, w25	1.16	−0.18	[−0.36, 0.05]	0.57	−0.30	[−0.49, −0.07]	−5.44	0.04
straddle + wing hedge, w50	1.28	−0.06	[−0.13, 0.04]	0.72	−0.15	[−0.23, −0.05]	−5.44	0.03

Table 31: As Table 29, always short (no forecast): every structure sold every day.

	mid fill			crossed fill			worst	cost
variation	Sharpe	ΔSharpe	[95% interval]	Sharpe	ΔSharpe	[95% interval]	day	share
straddle	0.20	—		−0.27	—		−10.32	0.03
strangle, one strike wide	0.63	0.43	[−0.01, 0.93]	0.15	0.43	[0.00, 0.93]	−9.47	0.04
strangle 0.5% OTM	1.78	1.57	[0.23, 3.99]	0.83	1.10	[−0.10, 3.20]	−3.65	0.25
strangle 1% OTM	1.40	1.20	[−0.57, 16.83]	−0.20	0.08	[−1.19, 7.23]	−1.77	0.67
strangle 2% OTM	2.64	2.44	[0.39, 32.95]	−0.55	−0.28	[−1.18, 4.33]	−1.43	1.00
iron butterfly, w10	−0.40	−0.61	[−1.19, −0.06]	−1.47	−1.19	[−1.74, −0.69]	−5.66	0.06

Table 31: (continued)

	mid fill			crossed fill			worst	cost
variation	Sharpe	ΔSharpe	[95% interval]	Sharpe	ΔSharpe	[95% interval]	day	share
iron butterfly, w25	−0.11	−0.31	[−0.56, −0.04]	−0.82	−0.55	[−0.78, −0.29]	−6.87	0.04
iron butterfly, w50	0.06	−0.15	[−0.22, −0.05]	−0.59	−0.31	[−0.39, −0.22]	−10.33	0.04
iron condor, w10	0.07	−0.14	[−0.59, 0.32]	−1.07	−0.79	[−1.22, −0.37]	−4.38	0.10
iron condor, w25	0.32	0.12	[−0.30, 0.60]	−0.45	−0.18	[−0.56, 0.27]	−7.34	0.08
iron condor, w50	0.43	0.23	[−0.19, 0.72]	−0.26	0.01	[−0.38, 0.46]	−9.48	0.07
put vertical, w10	−0.79	−1.00	[−1.99, −0.11]	−1.41	−1.14	[−2.10, −0.28]	−5.98	0.07
put vertical, w25	−0.48	−0.69	[−1.58, 0.14]	−0.90	−0.63	[−1.50, 0.19]	−7.20	0.05
put vertical, w50	−0.29	−0.50	[−1.32, 0.33]	−0.67	−0.40	[−1.21, 0.42]	−10.71	0.04
call vertical, w10	0.35	0.15	[−0.79, 1.05]	−0.27	0.01	[−0.90, 0.87]	−4.63	0.06
call vertical, w25	0.38	0.18	[−0.66, 1.05]	−0.05	0.22	[−0.58, 1.05]	−5.71	0.04
call vertical, w50	0.40	0.19	[−0.63, 1.07]	−0.01	0.27	[−0.52, 1.10]	−5.71	0.04
call butterfly, w10	−0.29	−0.50	[−1.28, 0.25]	−2.63	−2.36	[−3.11, −1.63]	−4.23	0.13
call butterfly, w25	0.02	−0.22	[−0.74, 0.28]	−2.12	−1.88	[−2.55, −1.29]	−5.95	0.04
call butterfly, w50	0.24	0.01	[−0.43, 0.42]	−2.07	−1.83	[−2.53, −1.24]	−9.61	0.02
put butterfly, w10	−0.30	−0.51	[−1.28, 0.22]	−2.79	−2.51	[−3.28, −1.78]	−4.31	0.14
put butterfly, w25	0.02	−0.22	[−0.73, 0.27]	−2.25	−2.01	[−2.70, −1.41]	−5.95	0.04
put butterfly, w50	0.28	0.04	[−0.39, 0.46]	−2.53	−2.28	[−2.99, −1.66]	−9.69	0.02
put ratio 1×2, w10	1.42	1.21	[−0.13, 2.49]	0.51	0.79	[−0.59, 2.05]	−5.64	0.14
put ratio 1×2, w25	0.81	0.61	[−1.03, 2.00]	0.24	0.52	[−1.12, 1.89]	−1.22	0.06
put ratio 1×2, w50	0.42	0.21	[−1.46, 1.63]	−0.05	0.23	[−1.47, 1.66]	−0.85	0.05
call ratio 1×2, w10	−0.10	−0.31	[−1.81, 1.09]	−1.00	−0.72	[−2.26, 0.69]	−2.49	0.11
call ratio 1×2, w25	−0.30	−0.50	[−2.22, 0.94]	−0.88	−0.60	[−2.31, 0.84]	−1.74	0.06
call ratio 1×2, w50	−0.34	−0.55	[−2.32, 0.96]	−0.85	−0.58	[−2.35, 0.96]	−0.83	0.05
straddle + wing hedge, w10	−0.40	−0.61	[−1.19, −0.06]	−1.47	−1.19	[−1.74, −0.69]	−5.66	0.04
straddle + wing hedge, w25	−0.11	−0.31	[−0.56, −0.04]	−0.82	−0.55	[−0.78, −0.29]	−6.87	0.04
straddle + wing hedge, w50	0.06	−0.15	[−0.22, −0.05]	−0.59	−0.31	[−0.39, −0.22]	−10.33	0.03

D.14 Where the Headline's P&L Comes From

Question. Which days carry the P&L of the last-30-min trade under the headline forecast of Section 5.4, and does the rule's advantage over always short come from a calendar effect? The series are the headline's sign(s) portfolio and always short, under the 16:00-bar recalibration of Table 7, on its 866 days at the midpoint. P&L is in premium units: one unit of premium staked every day, daily returns summed and never compounded. Intervals are 95% circular block bootstrap intervals (blocks of 21 days, 2,000 draws) on the same resampled days for every series, a cell's sum recomputed on the resampled days that fall in it. The VIX terciles place the previous session's VIX close against the terciles of every close since 1990 up to that session.

Table 32: The headline's sign(s) P&L by cell, in premium units at the midpoint, against always short on the same days; the difference is the headline minus always short, zero on every day the rule sells. Buys: the days the rule is long.

cell	days	buys	sign(s), headline	always short	difference
all days	866	348	+116.2 [+50.2, +183.0]	+12.5 [−41.5, +65.8]	+103.7 [+10.7, +200.1]
month-end, last session	52	12	−6.0 [−26.1, +12.7]	−24.5 [−43.5, −7.3]	+18.5 [−1.2, +41.3]
month-end, the session before	35	13	−2.6 [−13.8, +7.8]	+6.7 [−4.3, +16.5]	−9.3 [−19.7, +1.2]
month-end, the session after	44	19	+8.0 [−4.6, +21.1]	+10.6 [−2.3, +22.4]	−2.6 [−21.4, +19.0]
more than one session from a month-end	735	304	+116.9 [+59.8, +174.1]	+19.7 [−27.3, +64.6]	+97.2 [+12.4, +184.1]
FOMC statement day	33	12	+1.3 [−9.9, +11.1]	−5.0 [−16.2, +5.2]	+6.3 [−2.2, +15.3]
previous VIX close: low tercile	172	105	+15.4 [−23.1, +56.9]	−14.0 [−43.5, +12.3]	+29.4 [−20.7, +89.4]
previous VIX close: middle tercile	299	120	+77.1 [+36.0, +122.7]	+5.1 [−32.3, +44.2]	+72.0 [+7.4, +139.5]
previous VIX close: high tercile	395	123	+23.7 [−8.2, +57.8]	+21.4 [−10.2, +53.0]	+2.3 [−39.0, +48.3]
2020	158	69	+21.0 [−4.2, +51.1]	−4.8 [−25.2, +14.6]	+25.8 [−8.6, +64.4]
2021	158	54	+20.4 [−9.4, +56.3]	−6.4 [−32.7, +17.7]	+26.8 [−19.4, +82.9]
2022	219	49	+2.0 [−18.3, +23.7]	+22.3 [+0.6, +47.7]	−20.2 [−44.8, +2.5]
2023	248	124	+47.0 [+6.9, +96.8]	+15.8 [−9.8, +44.0]	+31.3 [−15.5, +88.9]
2024 (to 30 April)	83	52	+25.8 [+0.3, +57.1]	−14.3 [−31.7, −0.5]	+40.1 [+5.9, +80.9]

Finding. The P&L sits in a few long days. The headline makes 116.2 premium units at the midpoint (87.4 crossed); its 10 best days carry 42% of that and its 20 best 68% (86% crossed). All 20 best days are buys and all 20 worst are sells; without the 20 best days the other days make +37.6 (Sharpe 0.76), and 40 days — 4.6% of the sample — add up to the whole total. The rule owns more of the straddle's largest payoffs than its buy rate gives: of the 20 days the straddle paid most it bought 14, against 8.0 expected (hypergeometric $p = 0.006$). Month-ends are not the source: the long straddle pays on them on average (always short makes −24.5 [−43.5, −7.3] on the 52 last sessions of a month), the rule buys 12 of them and makes −6.0 [−26.1, +12.7] there, against +116.9 [+59.8, +174.1] on the 735 days more than one session away. The difference to always short, +103.7 [+10.7, +200.1], falls to +6.0 [−59.8, +72.7] without its 10 best days, and in the high-VIX tercile the two make the same (+2.3 [−39.0, +48.3] over 395 days).

Why it is not in the main text. The main text keeps the three facts that change its reading — a tail trade, month-ends not the source, the difference to always short and its interval; the cells are here.

D.15 Parked from Section 5.4: The Paper's Forecast under the Session-Bar Recalibration

Why it is parked. Until 29 September 2026, Section 5.4 read the paper's eight forecasts under the session-bar recalibration of Section 4.5 and qualified the block-diagonal ridge's sign(s) portfolio with the paragraphs below. Section 5.4 now scores every forecast under the 16:00-bar recalibration of Table 7. These paragraphs belong to the other recalibration and to the paper's forecast rather than the headline; none of their numbers is comparable with Table 7 or set beside it. They are kept as they stood, because three of them answer questions a reader will ask of any same-day trade — whether the forecast is dated where

it claims to be (the shift test), how much of the trade's sign a variance forecast can reach at all, and why QLIKE and the trade rank forecasts differently — and they have not been re-run under the 16:00-bar recalibration.

Where the level comes from. Two sample splits qualify the level rather than the sign. Dropping the 39 expiration days through 31 March 2020 — the February–March 2020 spike — *raises* the block-diagonal ridge portfolio to Sharpe 1.51 ($t = 2.73$ on the remaining 827 days): the full-sample figure is dragged, not flattered, by that quarter. Splitting instead at 16 May 2022, when SPXW listings became daily — a market-structure date, not a threshold chosen on the portfolio — the same portfolio scores Sharpe 0.98 ($t = 1.20$) on the 377 days before and 1.63 ($t = 2.27$) on the 489 after: the direction edge is positive on both tapes but earns its significance on the daily-0DTE era. The split is not uniform across columns — the two tree forecasts score higher before the date than after — which is one more reason to read the panel as one rule rather than as eight results.

Settlement pins carry a disproportionate share. The straddle expires worthless — $R = -1$ exactly — on 193 of the 866 days, 22.3 percent, whenever the close settles inside the one-strike window. The always-short control collects $+1$ on those days against -0.27 on all others; the sign(s) portfolio averages $+0.18$ on them against $+0.07$ elsewhere, so pin days supply 43 percent of its total return from 22.3 percent of the days. The portfolio monetizes a direction call and the pin premium together, and its P&L remains tilted toward days the index settles inside the strikes.

Eight rows are one or two bets. The eight daily sign(s) portfolios are correlated (mean pairwise correlation 0.70, from 0.58 to 0.93; their buy-day sets overlap at a mean pairwise Jaccard index of 0.70), an effective count of 1.8 independent portfolios by the participation ratio $(\sum_i \lambda_i)^2 / \sum_i \lambda_i^2$ of their correlation matrix. Agreement down the eight columns is evidence that the rule is insensitive to which forecast drives it — not eight independent confirmations of the rule.

What the forecasts are scored on, and what the trade uses. The eight columns cannot be ordered by the trade, and the paper's accuracy measure does not order them on it either. The reason is structural. QLIKE (Equation (5)) is consistent for the conditional mean of the variance [Patton, 2011], and every consistent scoring function for a mean can be taken apart by threshold. For a threshold $\theta > 0$ let

$$S_\theta(f, y) \;=\; |y - \theta|\, \mathbf{1}\{\min(f, y) \le \theta < \max(f, y)\}, \tag{20}$$

the loss of someone who uses the forecast f to decide whether the outcome y will exceed θ: nothing when f and y fall on the same side of θ, and the distance by which y landed from θ when they do not. Every consistent scoring function for the mean is, up to a term that does not involve the forecast, a mixture of these elementary scores over θ with a nonnegative weight [Ehm et al., 2016]; for a Bregman loss generated by a convex function ϕ the weight is $\phi''(\theta)\, d\theta$ in the normalization of Equation (20). QLIKE is the Bregman loss of $\phi(u) = -\log u$,

so for every forecast and outcome

$$\frac{RV}{\widehat{RV}} - \log\frac{RV}{\widehat{RV}} - 1 \;=\; \int_0^\infty S_\theta\big(\widehat{RV},\, RV\big)\,\theta^{-2}\,d\theta. \tag{21}$$

The identity is a two-line calculation — when $RV > \widehat{RV}$ the integral is $\int_{\widehat{RV}}^{RV}(RV-\theta)\,\theta^{-2}\,d\theta = RV/\widehat{RV} - 1 - \log(RV/\widehat{RV})$, and the other case is its mirror image — and it was also checked numerically, day by day, for the block-diagonal ridge on the scored days. QLIKE is therefore an average of threshold decisions, weighted by θ^{-2} toward small thresholds.

The sign(s) rule takes one of those decisions and no other: it is long exactly when $\widehat{RV}_t > \mathrm{IV}^2_{30,t}$, the decision at $\theta = \mathrm{IV}^2_{30,t}$. With the threshold measured in units of the day's implied variance, $\theta = c\,\mathrm{IV}^2_{30,t}$, the right-hand side of Equation (21) becomes $\int_0^\infty e_t(c)\,c^{-2}\,dc$, where $e_t(c)$ is the elementary score of $\widehat{RV}_t/\mathrm{IV}^2_{30,t}$ against $RV_t/\mathrm{IV}^2_{30,t}$ at the threshold c; the trade reads that score at $c = 1$ only,

$$e_t(1) \;=\; \left|\frac{RV_t}{\mathrm{IV}^2_{30,t}} - 1\right| \mathbf{1}\big\{\widehat{RV}_t \text{ and } RV_t \text{ fall on opposite sides of } \mathrm{IV}^2_{30,t}\big\}. \tag{22}$$

A single elementary score is itself consistent for the mean, though not strictly, so it ranks variance forecasts in the same sense QLIKE does; it spends all of its weight at the threshold the trade uses. Table 33 reports it beside QLIKE. QLIKE remains the paper's measure of accuracy everywhere else.

Table 33: The 15:30 forecasts scored two ways, on the 866 expiration days of Section 5.4. QLIKE is Equation (5) applied to the 16:00 row; the score at the threshold is Equation (22), averaged over days (lower is better for both). "Same side as RV" is the percentage of days on which $\widehat{RV}_t$ and RV_t fall on the same side of $\mathrm{IV}^2_{30,t}$; "same side as R" the percentage on which the sign(s) position has the sign of the settlement return. The last column is the annualized Sharpe ratio of the sign(s) portfolio at the crossed spread. The last row is a rule told RV_t in advance. †: the four columns still fitted on the earlier design panel (Section 4.5).

	QLIKE	score at $\theta = \mathrm{IV}^2_{30}$	same side as RV	same side as R	$\mathrm{Sharpe}_{\mathrm{ann}}$, crossed
baseline (HAR + calendar OLS)	0.108	0.102	68.5	55.2	0.49
block-diagonal ridge	0.110	0.109	65.9	54.5	0.87
block-diagonal ridge, without the FOMC columns	0.110	0.111	64.7	54.2	0.80
LightGBM†	0.101	0.094	68.7	54.7	0.98
XGBoost†	0.103	0.097	68.8	55.3	1.02
lasso (causally tuned)†	0.104	0.106	66.3	55.1	0.93
lasso ($\alpha = 10^{-4}$)	0.111	0.107	65.7	54.0	1.04
elastic net (causally tuned)†	0.104	0.109	66.5	54.6	0.54
realized variance known in advance	0	0	100.0	62.9	4.07

The Murphy diagram. Averaging $e_t(c)$ over the scored days and plotting it against c, one line per forecast, gives the Murphy diagram of Ehm et al. [2016] (Figure 2). QLIKE is the area under each line weighted by c^{-2}; the trade reads each line at $c = 1$. The lines cross within a factor of two of the implied variance. At every threshold on the grid up to 0.71 times it the lowest score belongs to a penalized linear forecast — the block-diagonal ridge,

one of the two lassos or the elastic net. From 0.77 to 1.41 times it the two tree forecasts lead, LightGBM at $c = 1$ itself with a score of 0.094. Between 1.54 and 2.18 times it the baseline leads at three of the five thresholds, and from 2.38 times upward the best score is shared exactly by forecasts of different families, because few forecasts ever reach so high a threshold and those that do not are scored identically. At the trade's own threshold the block-diagonal ridge ranks sixth of the eight, with 0.109, and the same estimator without the FOMC columns ranks eighth, with 0.111.

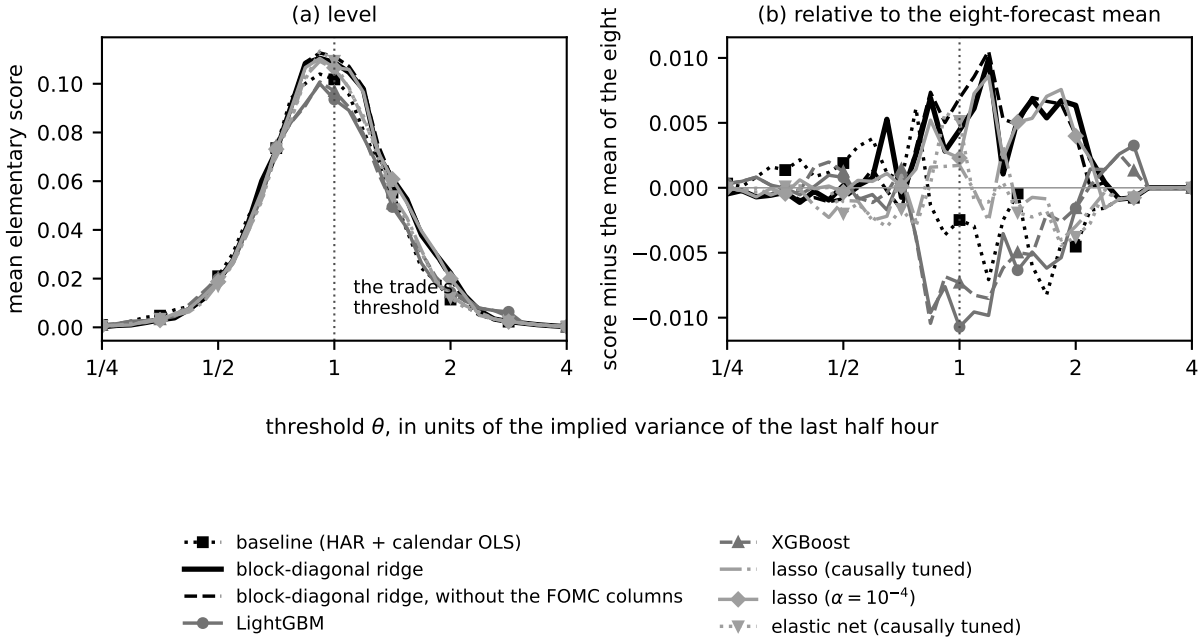


Figure 2: Murphy diagram of the eight 15:30 forecasts on the 866 expiration days. The horizontal axis is the threshold θ in units of the day's implied variance of the last half hour, $\mathrm{IV}^2_{30,t}$, on a logarithmic scale from 0.25 to 4; the dotted vertical line at 1 is the threshold the sign(s) rule uses. (a) The mean of the elementary score $e_t(c)$ (lower is better); QLIKE is the area under each line weighted by c^{-2}. (b) Each line minus the mean of the eight at the same threshold, which makes the crossings visible.

What a variance forecast can reach. Even the right threshold decision does not settle the trade. The straddle's settlement value depends on where the index closes relative to the two strikes — on the net move of the last half hour — and not on its realized variance, which counts every swing whether or not it is undone before the close. A rule told RV_t in advance and trading $\mathrm{sign}(RV_t - \mathrm{IV}^2_{30,t})$ takes the side of the payoff's sign on only 62.9 percent of days, for an annualized Sharpe ratio of 4.53 at the mid and 4.07 across the spread, against 17.3 across the spread for a rule told the sign of the payoff itself. On 37.1 percent of days — about a third — the trade's sign is therefore beyond any variance forecast, however accurate. The eight forecasts take the side of the payoff's sign on 54.0 to 55.3 percent of days, and the side of $RV_t - \mathrm{IV}^2_{30,t}$ on 64.7 to 68.8 percent.

Neither score orders the forecasts on this trade. The fixed lasso ranks eighth of the eight on QLIKE and fifth at the trade's threshold, and has the highest crossed-spread Sharpe ratio of the eight (1.04). Across the eight columns the rank correlation between the crossed-spread Sharpe ratio and minus the loss is 0.12 for QLIKE (95% interval from a day-block bootstrap -0.69 to 0.76) and 0.40 for the score at the threshold (-0.62 to 0.90); both intervals contain zero. Adding fifteen deliberately improved and degraded copies of the ridge forecast — blends toward the realized value, and multiplicative noise — lifts both correlations, to 0.95 and 0.97, but that says only that a much better or much worse forecast trades better or worse: every loss orders those copies perfectly, so they cannot choose between losses. The score at the threshold is reported because it isolates the one decision the trade takes, not because it ranks the eight columns here; with eight forecasts that take the same side on most days, and a payoff whose sign a variance forecast can reach only in part, no variance score does.

The signal's content is a same-day sign, and the right test is the same-day one. The signal is fixed at 15:30 and the return settles at 16:00 on the same expiration day, so pairing s_t with R_t involves nothing observed after the decision. On that pairing the content is a shift in the conditional mean by sign: when the forecast exceeds the implied variance the straddle returns $+0.10$ on average, and -0.09 when it does not (difference 0.19, heteroskedasticity-robust $t = 2.32$ for the block-diagonal ridge; t between 1.63 and 2.61 across the eight columns). That is the block-diagonal ridge's sign(s) portfolio under the session-bar recalibration, read as a regression, and Figure 3 shows it. A straight line is the wrong instrument for the same relation: fitted by least squares the slope of R_t on the raw signal comes out *negative* for every one of the eight columns (heteroskedasticity-robust t from -3.9 to -0.2) — the opposite sign to the step the split finds — because R has a point mass at -1 on 22.3 percent of days and a fat right tail that a few extreme values of s_t can lever. The effect lives in the conditional means of the two sides, which a sign split sees and a slope does not.

Magnitude carries information; sizing on it does not. Within either side of the split the binned means of the figure show no trend in the level of the signal, but a line fitted *inside* a side does find one, and it runs the wrong way for a trader. Restricted to the sell side the heteroskedasticity-robust slope of R_t on s_t is negative and resolved for all eight columns ($|t|$ from 2.2 to 5.7; 5.4 for the block-diagonal ridge): the further below zero the signal, the larger the settlement return, so the short earns *less* on precisely the days its signal is most extreme. On the buy side the slope is negative too, but resolved for only two of the eight. The precise claim is therefore narrower than "magnitude adds nothing to the sign": the conditional mean does vary with $|s|$ inside a side. What does not follow is a sizing rule, and none is retained — scaling the position by the rank of $|s_t|$ raised no column's Sharpe ratio, and that overlay is parked (Section 4.5). Only the sign is used.

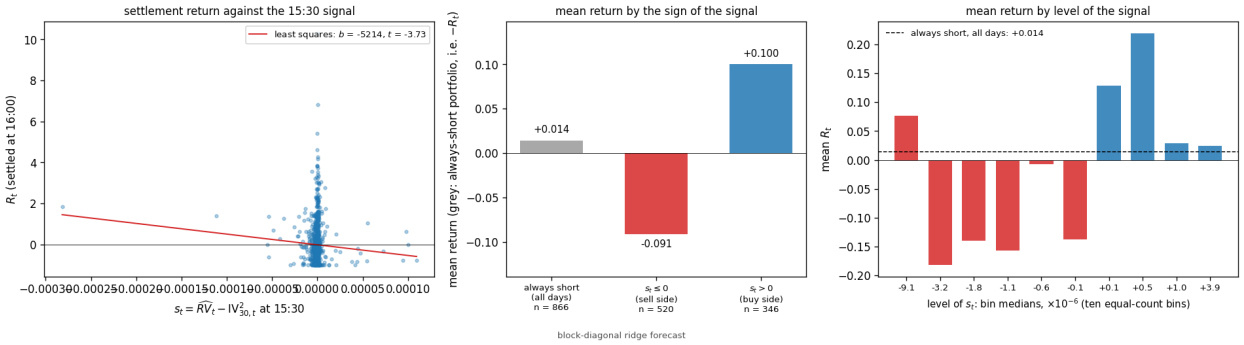


Figure 3: The 15:30 signal against the 16:00 settlement return, block-diagonal ridge forecast, 866 expiration days. Left: the settled return R_t against the raw signal s_t with the least-squares line. Middle: the mean of R_t on days with $s_t \leq 0$ and $s_t > 0$ (means only, no error bars) beside the always-short portfolio's mean over all days (grey bar), the benchmark throughout. Right: the mean of R_t in ten equal-count bins of the signal's level, each labelled by its median value of s_t in units of 10^{-6} (thirty-minute variance); the dashed line is the always-short mean. The relation is a step at the sign, and the binned means inside either half show no trend in the level: what the signal carries is which side of the quoted variance the forecast stands on.

The forecast is dated where it claims to be. A same-day trade driven by a same-day forecast invites the obvious suspicion, so the shift test is reported rather than asserted. Re-running the whole portfolio on the forecast row stamped k bars away from the one the rule uses gives an annualized Sharpe of 0.48 at $k = -1$ (a forecast issued one bar early, for a window that has already closed), 1.34 at $k = 0$ (the row actually used), and 3.11 at $k = +1$ — one bar of genuine look-ahead — against a ceiling of 4.53 for a portfolio told the last half hour's realized variance in advance. The profile is a step exactly at $k = 0$: flat to its left and far higher to its right. A leak would have lifted the bars left of $k = 0$ or flattened the step, and it does neither. One column is worth flagging: the elastic net's best stale shift, at $k = -2$, is 1.04 against its own 1.02 at $k = 0$, so that one forecast is not separated from its own stale copies by this test. Every other column's best stale shift sits below its trade.

Compounded at a fixed three percent of wealth, the $\operatorname{sign}(s)$ **portfolio grows; the control does not.** Betting the fixed fraction $f = 0.03$ of Section 4.5, the block-diagonal ridge $\operatorname{sign}(s)$ portfolio turns one unit of wealth into 7.2 over the sample (annualized log-growth +0.57), with a maximum wealth drawdown of 42 percent and a worst single day that removes 16 percent of wealth; the ruin bound $1/|\min_t R'_t|$ is 0.18 for this portfolio and 0.10 for the control, so three percent sits well inside both. All eight $\operatorname{sign}(s)$ portfolios compound positively under the same fraction (terminal wealth 3.6 to 9.9), and the headline portfolio's annualized log-growth is positive in four of the five calendar years (2022 is −0.04). The always-short control ends at 0.86 — its premium does not compound. A defined-risk variant is parked and not reported here.

Table 34: The shift (dating) test of the headline forecast under the 16:00-bar recalibration: the 15:30 sign(s) rule with the forecast row shifted by k half-hour bars, every row placed on the traded bar's scale, on the 865 deck days for which every shifted row exists. $k < 0$: the same session's earlier row; $k = 0$: the trade; $k > 0$: the next session's row (the headline has session rows only), whose lags contain the traded bar. Δ is against $k = 0$ with the paired day-block 95 % interval.

row	row read	Sharpe mid	Sharpe crossed	buy %	agree %	Δ mid [95 %]	Δ crossed [95 %]
k =-12	10:00, same	0.12	-0.35	49	69	-1.73 [-2.91, -0.57]	-1.74 [-2.92, -0.57]
k =-11	10:30, same	-0.31	-0.78	48	69	-2.17 [-3.34, -1.03]	-2.17 [-3.35, -1.02]
k =-10	11:00, same	-0.03	-0.50	48	66	-1.88 [-3.07, -0.67]	-1.89 [-3.08, -0.67]
k =-9	11:30, same	0.35	-0.13	43	68	-1.50 [-2.70, -0.28]	-1.51 [-2.71, -0.28]
k =-8	12:00, same	0.42	-0.05	46	70	-1.44 [-2.52, -0.36]	-1.44 [-2.53, -0.36]
k =-7	12:30, same	0.51	0.03	42	69	-1.35 [-2.47, -0.22]	-1.36 [-2.49, -0.23]
k =-6	13:00, same	0.64	0.17	44	69	-1.21 [-2.46, +0.07]	-1.22 [-2.48, +0.07]
k =-5	13:30, same	0.95	0.47	38	70	-0.91 [-1.92, +0.11]	-0.91 [-1.93, +0.11]
k =-4	14:00, same	0.93	0.45	37	71	-0.93 [-2.11, +0.21]	-0.94 [-2.11, +0.21]
k =-3	14:30, same	1.29	0.80	26	72	-0.57 [-1.67, +0.46]	-0.59 [-1.69, +0.45]
k =-2	15:00, same	0.50	0.02	25	71	-1.35 [-2.72, -0.01]	-1.37 [-2.74, -0.02]
k =-1	15:30, same	-0.00	-0.47	32	75	-1.86 [-2.96, -0.78]	-1.86 [-2.96, -0.79]
k =+0	16:00, same	1.85	1.39	40	100	+0.00 [+0.00, +0.00]	+0.00 [+0.00, +0.00]
k =+1	10:00, next	0.88	0.41	44	68	-0.97 [-2.22, +0.31]	-0.97 [-2.24, +0.32]
k =+2	10:30, next	0.93	0.46	44	68	-0.92 [-1.91, +0.12]	-0.92 [-1.91, +0.13]
k =+3	11:00, next	1.00	0.53	41	66	-0.85 [-2.12, +0.45]	-0.86 [-2.12, +0.46]
k =+4	11:30, next	0.83	0.36	42	66	-1.02 [-2.30, +0.26]	-1.03 [-2.31, +0.26]
k =+5	12:00, next	1.22	0.75	41	65	-0.63 [-1.93, +0.63]	-0.64 [-1.95, +0.63]
k =+6	12:30, next	1.51	1.04	42	63	-0.34 [-1.66, +0.89]	-0.35 [-1.67, +0.90]
realized (target)	16:00 (traded bar)	4.48	4.02	27	67	+2.63 [+1.21, +4.17]	+2.63 [+1.20, +4.17]
realized (unclipped)	16:00 (traded bar)	4.50	4.03	27	67	+2.64 [+1.22, +4.19]	+2.65 [+1.21, +4.19]
always short		0.27	-0.22	0	60	-1.58 [-2.98, -0.13]	-1.60 [-3.00, -0.16]

D.16 The Shift (Dating) Test of the Headline Forecast

What is tested. The last-30-min trade reads the per-bar ridge's row labelled 16:00 — the forecast of the 15:30–16:00 bar, issued at 15:30. The test re-scores the sign(s) rule with the forecast *row* shifted by k half-hour bars and everything else unchanged. For $k < 0$ the rule reads the same session's earlier row (the 15:30 row at $k = -1$, down to the 10:00 row at $k = -12$): a stale forecast, still free of look-ahead. For $k > 0$ the headline, a per-bar model with session rows only, has no row after the close; the rule reads the *next* session's row (10:00 at $k = +1$, to 12:30 at $k = +6$), the first rows whose lag ladder contains the traded bar. Because another row forecasts another bar, every shifted row is placed on the traded bar's own scale, $(\hat{y}_k^2 + \hat{s}_{16:00})\, B_{16:00}$, with the smear $\hat{s}$ and profile B the trade itself uses, so the only thing that changes with k is what the model had seen; at $k = 0$ this is the trade's forecast (gated). The star is the traded bar's realized variance in place of the forecast. Scorer, trade, fills and intervals are those of Table 7; the 865 deck days for which every shifted row exists are scored (the last deck day has no next session, and two sessions lack the 10:00–11:00 rows), and at $k = 0$ on all 866 days the rule reproduces Table 7 (1.90 mid, 1.44 crossed).

Reading. Table 34 gives every row. One bar stale, the rule falls from 1.85 to -0.00 (Δ -1.86 [-2.96, -0.78]); its position agrees with the trade's on 75 % of days. Of the twelve stale rows, 0 sit above $k = 0$ in point estimate, 8 have an interval below zero and 0 an interval

above; the best stale row is $k = -3$ at 1.29 (Δ -0.57 [-1.67, +0.46]). The next session's rows, placed on the traded bar's scale, score 0.83 to 1.51 at the midpoint (0 of six intervals above zero; $k = +1$: 0.88 mid, 0.41 crossed, Δ -0.97 [-2.22, +0.31], agreement 68 %), and the realized variance itself scores 4.48 mid, 4.02 crossed, against always short at 0.27 on the same days. The next-session rows are a different probe from the notebook's after-hours rows (Section D.15): each is a separate regression for its own bar, so the traded bar enters through that regression's own lag weights rather than through the traded row's, and the two tests are not compared.

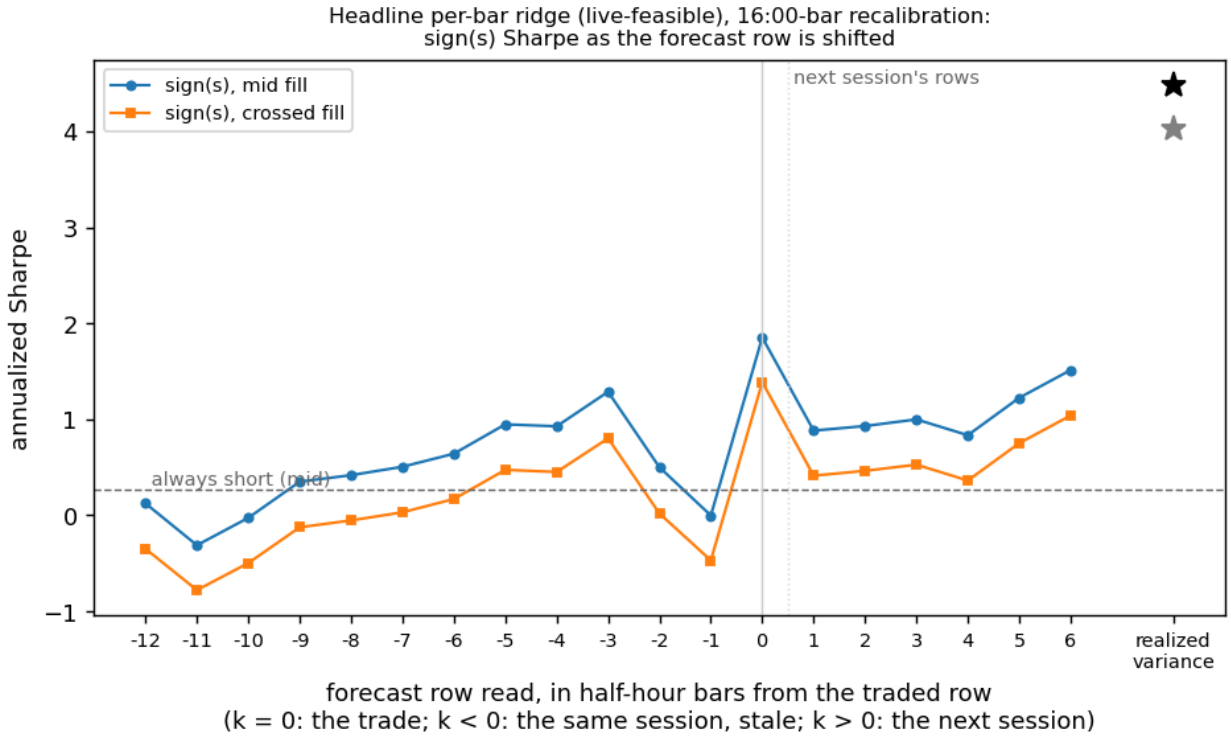


Figure 4: The headline forecast's sign(s) Sharpe as the forecast row is shifted, on the same 865 days (source: `results/close_shift_test/shift_test_headline.csv`).

Table 35: The comparison set scored two ways on the 866 deck days under the 16:00-bar recalibration: QLIKE against the scorer's target, the elementary score at the trade's threshold (the 15:30 implied variance), the economic loss $|R|\,\mathbf{1}\{q \neq \text{sign}(R)\}$, the share of days the forecast falls on the same side of the threshold as the realized variance (same side as RV) and the share on which the position has the sign of the settlement return (same side as R), and the sign(s) Sharpe at both fills. The last two rows are rules told the realized variance, and the payoff's sign, in advance.

forecast	QLIKE	ES(c=1)	EL	side of RV %	side of R %	Sharpe mid	Sharpe crossed
per-bar ridge [live_feasible]	0.101	0.100	0.349	67.4	54.7	1.90	1.44
block-diagonal ridge	0.106	0.100	0.380	67.2	55.0	1.00	0.53
baseline (HAR + calendar OLS)	0.117	0.099	0.374	69.7	56.6	1.17	0.69
per-bar lasso [live_feasible]	0.096	0.096	0.351	67.8	55.1	1.83	1.36

Table 35: (continued)

forecast	QLIKE	ES(c=1)	EL	side of RV %	side of R %	Sharpe mid	Sharpe crossed
per-bar ridge [all_features]	0.100	0.095	0.357	68.5	54.8	1.66	1.20
per-bar LightGBM [live_feasible]	0.099	0.112	0.360	65.0	54.6	1.59	1.12
tuned per-bar LightGBM [all_features], MSE-selected	0.103	0.106	0.357	66.4	55.1	1.67	1.20
per-bar LSTM [live_feasible]	0.119	0.113	0.384	63.9	54.2	0.91	0.43
per-bar LightGBM, daily refit [live_feasible]	0.096	0.106	0.366	67.1	55.5	1.42	0.95
Optuna per-bar LightGBM, TUNE_PER 1, best of 50 [live_feasible]	0.103	0.113	0.385	65.9	54.4	0.87	0.40
intraday-sequence LSTM [live_feasible]	0.110	0.099	0.362	67.8	55.1	1.53	1.06
oracle: told RV (sign(RV - slice))	–	–	0.262	100.0	62.8	4.52	4.06
oracle: told the payoff's sign	–	–	0.000	62.8	100.0	17.35	17.28

Table 36: Sample splits of the headline's sign(s) return (midpoint fill) under the 16:00-bar recalibration, beside the paper's block-diagonal ridge scored the same way and always short. $t = \sqrt{n}\,\bar{x}/s_x$; the Sharpe ratio is annualized with its day-block bootstrap 95 % interval (circular blocks of 21 days, 2000 draws); the P&L share is the split's sum of daily returns over the full-sample sum.

series	split	n	mean	t	Sharpe	[95 %]	P&L share %
headline sign(s)	all days	866	+0.134	+3.53	1.90	[+0.89, +2.94]	100
headline sign(s)	through 2020-03-31 (the COVID quarter)	39	+0.174	+0.73	1.85	[-0.10, +4.02]	6
headline sign(s)	after 2020-03-31	827	+0.132	+3.46	1.91	[+0.86, +2.99]	94
headline sign(s)	before 2022-05-16	377	+0.122	+2.04	1.67	[+0.23, +2.90]	39
headline sign(s)	from 2022-05-16 (daily SPXW listings)	489	+0.144	+2.92	2.09	[+0.62, +3.67]	61
headline sign(s)	settlement pins (R = -1)	193	+0.254	+3.64	4.16	[+1.57, +7.33]	42
headline sign(s)	off pins	673	+0.100	+2.24	1.37	[-0.08, +2.71]	58
reference sign(s) (this scorer)	all days	866	+0.071	+1.86	1.00	[-0.03, +2.03]	100
reference sign(s) (this scorer)	through 2020-03-31 (the COVID quarter)	39	-0.116	-0.48	-1.22	[-4.86, +0.78]	-7
reference sign(s) (this scorer)	after 2020-03-31	827	+0.080	+2.08	1.15	[+0.10, +2.18]	107
reference sign(s) (this scorer)	before 2022-05-16	377	+0.063	+1.05	0.86	[-0.80, +2.45]	38
reference sign(s) (this scorer)	from 2022-05-16 (daily SPXW listings)	489	+0.077	+1.56	1.12	[-0.11, +2.44]	62
reference sign(s) (this scorer)	settlement pins (R = -1)	193	+0.285	+4.12	4.71	[+1.23, +8.47]	89
reference sign(s) (this scorer)	off pins	673	+0.010	+0.22	0.13	[-1.20, +1.29]	11
always short	all days	866	+0.014	+0.38	0.20	[-0.62, +1.16]	100
always short	through 2020-03-31 (the COVID quarter)	39	-0.302	-1.28	-3.25	[-6.76, +0.59]	-94
always short	after 2020-03-31	827	+0.029	+0.76	0.42	[-0.34, +1.41]	194
always short	before 2022-05-16	377	-0.033	-0.55	-0.45	[-1.52, +0.84]	-98
always short	from 2022-05-16 (daily SPXW listings)	489	+0.051	+1.02	0.73	[-0.35, +1.93]	198
always short	settlement pins (R = -1)	193	+1.000	–	–		1540
always short	off pins	673	-0.268	-6.15	-3.77	[-4.64, -2.85]	-1440

D.17 The Headline Forecast Scored at the Trade's Threshold, and Its Sample Splits

The score at the threshold. Section D.15 takes QLIKE apart by threshold (Equations (20)–(22)) and reads the trade at the one threshold it uses, the day's implied variance of the last half hour. Table 35 repeats that scoring under the 16:00-bar recalibration for a comparison set of 11 forecasts — the headline, the paper's block-diagonal ridge scored the same way, the HAR + calendar least squares, the headline's lasso twin, the all-features ridge, an untuned LightGBM refit every tenth session and the same model refit every session, a

LightGBM tuned by random search and one tuned by Optuna at every session, the per-bar LSTM and the intraday-sequence LSTM, each tree and LSTM forecast being the one the master table scores — and `scores_by_forecast.csv` carries every table-A forecast of the master table. The overall ranks and rank correlations below are over its 220 distinct forecasts (the rest-of-day check rows, exact copies of their per-bar twins, are left out). QLIKE is against the scorer's target; the identity between QLIKE and the θ^{-2} mixture of elementary scores is checked day by day for the headline, and the identity $\overline{qR} = \overline{|R|} - 2\,\overline{EL}$ for every forecast. The headline scores 0.101 on QLIKE and 0.100 at the threshold, ranking 5 and 5 of 11 in the set (63 and 40 of 220 overall; its crossed-spread Sharpe ratio ranks 2 of 220). It falls on the realized variance's side of the threshold on 67 % of days and on the payoff's side on 55 %, and it is long on 14 of the 20 best long-straddle days.

The Murphy diagram. Figure 5 plots the mean elementary score against the threshold, in units of the implied variance, from 0.25 to 4 on a logarithmic grid of 33 points. The headline has the lowest score of the set at 11 of the 33 thresholds and a linear forecast at 20; at the trade's threshold the leader is per-bar ridge [all_features] and the headline ranks 5 of 11.

What a variance forecast can reach. A rule told the realized variance in advance takes the payoff's side on 63 % of days (Sharpe 4.52 at the midpoint, 4.06 across the spread), against 17.28 across the spread for a rule told the payoff's sign; on 37 % of days the trade's sign is beyond any variance forecast.

Which score orders the forecasts on the trade. Across the comparison set the rank correlation of minus the loss with the crossed-spread Sharpe ratio is +0.49 for QLIKE (day-block bootstrap 95 % interval -0.10 to +0.81) and +0.56 for the score at the threshold (-0.23 to +0.83); across all 220 distinct forecasts of the master table it is +0.56 (+0.07 to +0.65) and +0.63 (+0.20 to +0.75). The economic loss is affine in the midpoint P&L and correlates perfectly by construction; it is reported as the ceiling.

Sample splits. Table 36 repeats the splits of Section D.15 for the headline's sign(s) return, beside the paper's forecast scored the same way and always short. Over all 866 days the headline's annualized Sharpe ratio is 1.90 ($t = 3.53$). Dropping the 39 expiration days through 31 March 2020 leaves 1.91 ($t = 3.46$, interval 0.86 to 2.99) on the remaining 827 days. Split at 16 May 2022, when SPXW listings became daily, it is 1.67 ($t = 2.04$, 0.23 to 2.90) on the 377 days before and 2.09 ($t = 2.92$, 0.62 to 3.67) on the 489 days after; the paper's forecast scored the same way reads 0.86 and 1.12. The straddle expires worthless on 193 of the days (22.3 %): always short collects +1.00 on them against -0.27 elsewhere, and the headline averages +0.25 on them against +0.10 off them, so pin days supply 42 % of its total return. Read as a same-day sign split, the settlement return averages +0.15 on the 348 days the headline buys and -0.12 on the 518 days it sells (difference 0.27, heteroskedasticity-robust $t = 3.28$).

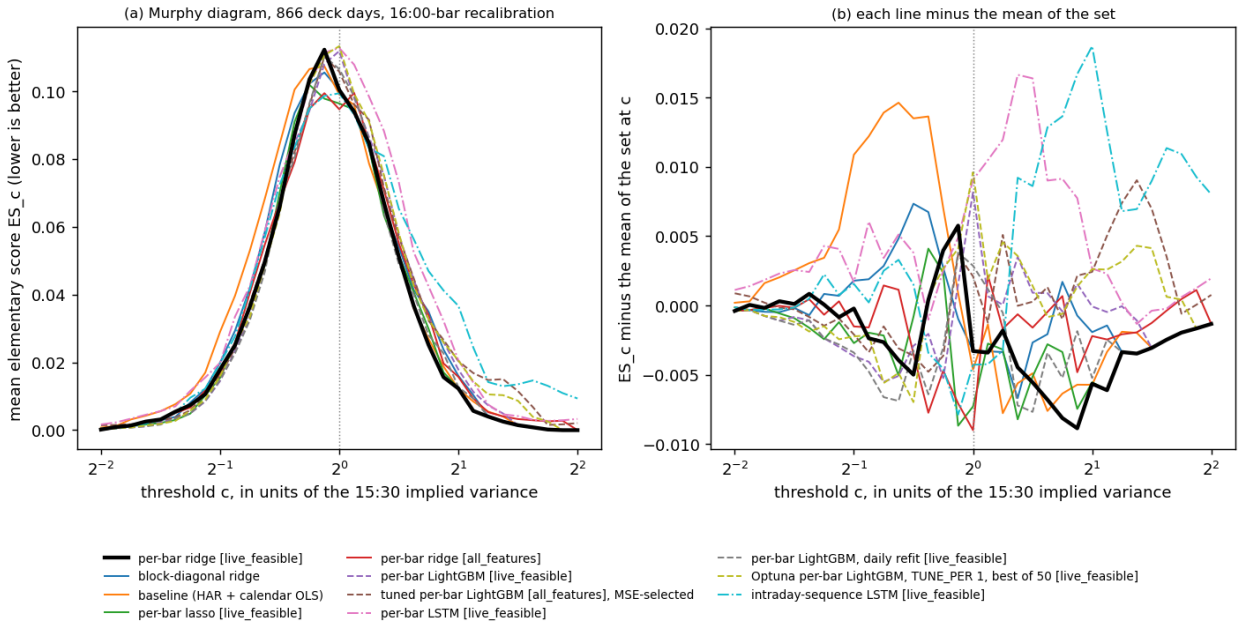


Figure 5: Murphy diagram of the comparison set on the 866 deck days under the 16:00-bar recalibration. The horizontal axis is the threshold in units of the day's implied variance of the last half hour, on a logarithmic scale from 0.25 to 4; the dotted line at 1 is the threshold the sign(s) rule uses. (a) The mean elementary score (lower is better); QLIKE is the area under each line weighted by c^{-2}. (b) Each line minus the mean of the set at the same threshold (source: `results/close_murphy_splits/murphy_grid.csv`).

Table 37: Compounding at a fixed $f = 0.03$ of wealth per day under the 16:00-bar recalibration, 866 days: terminal wealth $W_T = \prod_t (1 + f\, q_t R_t)$, annualized log-growth g, the largest fall of wealth from its running peak, the worst single-day wealth factor $1 + f \min_t R'_t$, and the ruin bound $1/|\min_t R'_t|$ (the sample's worst day). The headline, the paper's block-diagonal ridge scored the same way, and always short, at both fills.

series	fill	W_T	g	max DD	worst day	ruin bound
headline sign(s)	mid	20.18	+0.87	34 %	0.837	0.18
headline sign(s)	crossed	8.55	+0.62	38 %	0.832	0.18
reference sign(s) (this scorer)	mid	3.89	+0.40	42 %	0.837	0.18
reference sign(s) (this scorer)	crossed	1.62	+0.14	50 %	0.832	0.18
always short	mid	0.86	-0.04	55 %	0.691	0.10
always short	crossed	0.33	-0.32	74 %	0.671	0.09

D.18 The Headline Forecast Compounded at a Fixed Fraction of Wealth

What is computed. The fixed-fraction exercise of Section 4.5 — a share $f = 0.03$ of current wealth deployed as straddle premium every day, the same on every day and for every rule, wealth $W_T = \prod_t (1 + f\, q_t R_t)$, the annualized mean of $\log(1 + f R'_t)$ as the growth rate,

every wealth factor asserted positive — repeated for the headline's sign(s) positions under the 16:00-bar recalibration, on the 866 days and at both fills, beside the paper's block-diagonal ridge scored the same way and always short (Table 37, Figure 6). The midpoint frame is the one Section D.15 reports for the paper's forecast under the other recalibration; the crossed frame pays the ask and hits the bid on every day.

Reading. At the midpoint the headline turns one unit of wealth into 20.2 over the sample (annualized log-growth +0.87), with a maximum wealth drawdown of 34 percent and a worst single day that removes 16 percent of wealth; the ruin bound $1/|\min_t R'_t|$ is 0.18 for the headline and 0.10 for always short, so 0.03 sits inside both. Across the spread it ends at 8.5 (log-growth +0.62, drawdown 38 percent). The paper's forecast scored the same way ends at 3.9 at the midpoint (+0.40) and 1.6 across the spread; always short ends at 0.86 (-0.04) and 0.33. Of the 220 distinct forecasts of the master table (the rest-of-day check rows, exact copies of their per-bar twins, left out), 218 compound above one unit at the midpoint (terminal wealth 0.6 to 20.4; the headline ranks 2) and 208 across the spread (0.2 to 8.6; the headline ranks 2). By calendar year, annualized from the days traded in each, the headline's midpoint log-growth is positive in 4 of 5 years (2020: +0.87, 2021: +0.79, 2022: −0.01, 2023: +1.29, 2024: +2.15) and across the spread in 4 (2020: +0.44, 2021: +0.48, 2022: −0.19, 2023: +1.11, 2024: +1.96).

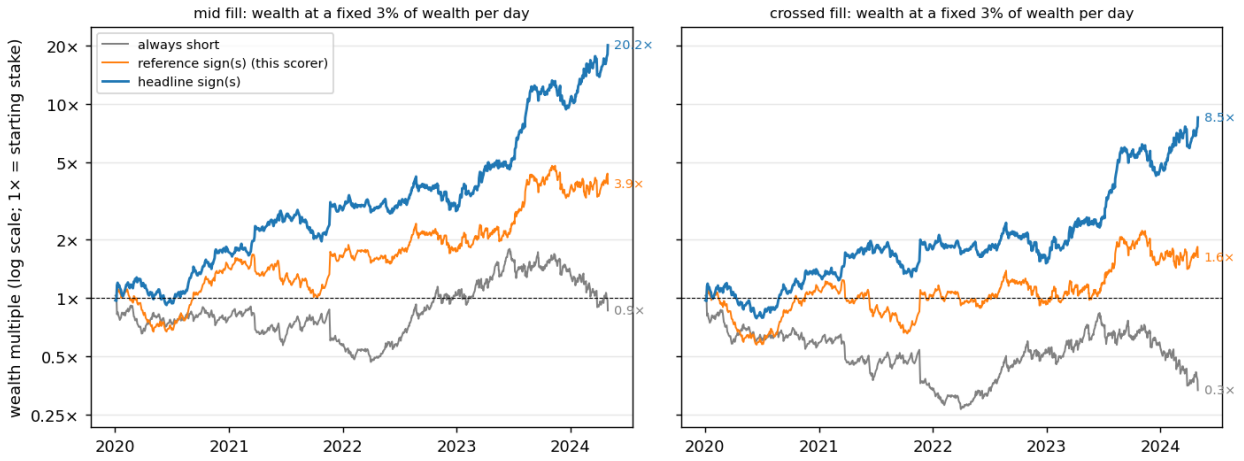


Figure 6: Wealth at a fixed 0.03 of wealth per day, 866 days, under the 16:00-bar recalibration: the headline's sign(s), the paper's block-diagonal ridge scored the same way, and always short; midpoint fill (left) and crossed spread (right), log scale (source: `results/close_compounding/fixedfrac_headline.csv`).

D.19 The 16:00 Models Re-run: Design, Mask, Cadence, Tuning

Every per-bar model that forecasts the 16:00 bar (the 15:30–16:00 half hour, issued at 15:30, the bar the last-30-min trade is decided on) was re-run with one change at a time: the design (D.19.1), a per-window column mask for the trees and the LSTM (D.19.2), the refit cadence (D.19.3), the tuning (D.19.4); an intraday-sequence LSTM was added (D.19.5). The section also records what the re-run showed about reproducibility across CPUs (D.19.6), the

master table before and after (D.19.7), the importance and density studies on the new design (D.19.8) and the compute used (D.19.9). Nothing is discarded: a rung that did worse is reported beside one that did better.

Every comparison uses the research scorer (the 16:00 bar recalibrated on its own), the same 866 trade days, and the sign(s) rule on the straddle (nearest out-of-the-money call and put, same-day expiry, one position). A difference is $a - b$; a QLIKE difference is given in percent of b's QLIKE, with its 95 % paired day-block bootstrap interval in the same units (negative = a has the lower loss); a Sharpe difference is the midpoint-fill sign(s) Sharpe ratio of a minus that of b, with its interval. In the grid tables each cell holds the QLIKE difference on its first line and the Sharpe difference on its second. Input sets: `baseline` (the HAR ladder and the calendar columns), `live_feasible` (the inputs a 15:30 forecaster can rebuild live) and `all_features`. Key to the design columns named below: `har_ma_*` is the bar's realized variance averaged over the last 1, 5, 25, 125, 625 and 3125 bars (the HAR ladder, `har_ma_k` one rung); `har_ma_k_x_open` and `har_ma_k_x_close` are that rung times the session-open and session-close indicators (the session-edge interactions).

D.19.1 The design: the session-edge columns

The session-edge interactions were built for the pooled 48-bar models. In a one-bar-per-session series at the 16:00 bar the 6 `har_ma_k_x_open` columns are identically zero and the 6 `har_ma_k_x_close` columns are exact copies of `har_ma_k`. The 12 columns were dropped from the per-bar design and kept in the pooled models: `baseline` goes from 34 to 22 columns, `live_feasible` from 244 to 232, `all_features` from 640 to 628 (Table 38).

The linear arms' identifiability mask — at every penalty tune, drop the columns constant on the tuning window and the exact copies of an earlier kept column — had already removed all of them: in 75 of the 75 masked linear forecasts the number of masked columns per tune falls by exactly the number of dropped columns, and the columns each fit keeps are the same before and after (last column of Table 38). The fits solve the same problem, and the 76 linear forecasts of the master table move by rounding only (Table 39): OLS by at most 8×10^{-15} and ridge by at most 4×10^{-10} (relative change of the 16:00 forecast). The lasso and the elastic net move by up to 9×10^{-2} and 9×10^{-2}, on the rows of one penalty-tune period, in 7 of 25 and 10 of 25 forecasts; positions change on 9 of the 76×866 forecast-days, and no Sharpe ratio moves by more than 0.06. A control on the 17 arms that moved attributes this to the solvers' floating-point path: the pre-change code re-run on the same CPU architecture reproduces its first run bit for bit in 17 of 17; the new code against the pre-change code on one architecture moves 13 of them, by 5.5×10^{-4} to 1.7×10^{-2}; the pre-change code alone, moved to another machine, moves 12, by 7.3×10^{-4} to 8.1×10^{-2}. The LSTM, which had no mask at that step, moved more (median relative change of the 16:00 forecast 2.5–7.6 % over the three input sets; a new input width also redraws its initial weights).

The rest-of-day model's 117 arms were re-run on the same design; their mask fell by exactly 12 columns per tune in 117 of them. The master table's 9 check rows (the rest-of-day model at 15:30, which should equal its per-bar 16:00 twin) had differed from their twins by up to 1.7×10^{-2} (the same position on as few as 865 of 866 days); they are now bit-identical to their twins in 9 of 9, on every one of the 1,469 16:00 rows.

Table 38: The per-bar design before and after the session-edge columns were dropped, and the linear arms' identifiability mask per penalty tune (range over tunes) for the per-bar ridge: the columns it masked fall by exactly the columns removed, so the columns kept are unchanged.

input set	columns before	columns after	removed	masked per tune, before	masked per tune, after	kept per tune
`baseline`	34	22	12	17	5	17
`live_feasible`	244	232	12	106–108	94–96	136–138
`all_features`	640	628	12	262–280	250–268	360–378

Table 39: The master table's per-bar linear forecasts, new design minus old, by estimator: the largest and the median (over forecasts) of the per-forecast maximum and median relative change of the 16:00 forecast, the forecasts whose maximum exceeds the reproduction gate's bound, the trade positions changed (forecast-days, of 866 per forecast) and the largest change of a sign(s) Sharpe ratio (midpoint).

estimator	forecasts	max rel. change	median rel. change	moved ($> 1 \times 10^{-8}$)	positions changed	max $\lvert\Delta\text{Sharpe}\rvert$ mid
OLS	1	8.4×10^{-15}	4.4×10^{-16}	0	0	0.00
ridge	25	4.3×10^{-10}	1.1×10^{-14}	0	0	0.00
lasso	25	9.1×10^{-2}	2.8×10^{-15}	7	6	0.06
elastic net	25	9.5×10^{-2}	3.0×10^{-15}	10	3	0.04

D.19.2 One mask for every model

The trees and the LSTM were then given the linear arms' rule. At every fit — a refit on its training window, or a tuning point on the window it tunes on — every column constant on that window and every exact copy of an earlier kept column is dropped, first occurrences kept in column order; the LSTM, whose network is rebuilt at every refit, recomputes it at every refit. Importance and TreeSHAP are mapped back to all columns with zero for a dropped column. The rule looks at the window only, so every model of an input set keeps the same columns at the same refit: over the 1,469 daily refits `baseline` keeps 17 of 22 columns at every refit, `live_feasible` 138 of 232 (median; range 136–145), and `all_features` 369 of 628 (360–378); the Optuna runs below keep the same counts.

Table 40 sets each masked rung against the same rung unmasked (both on the new design). Rungs: T10 and T1 are the shipped tree configurations refit every 10 sessions and every session; RS10 and RS1 the random search, refit every 10 sessions and every session; LSTM10 the per-bar LSTM refit every 10 sessions. Over the 36 tree rows the median absolute QLIKE change is 0.7 %; 5 QLIKE intervals lie above zero (the masked rung has the higher loss) and 0 below; 1 Sharpe interval lies below zero and 0 above. For the LSTM 1 of 3 QLIKE intervals lie below zero (`all_features`, −17.1 % [−33.2, −4.2]) and 0 above. The masked rungs ran on one CPU class; the unmasked rungs did not pin the class (the class of

their chunks was not recorded), so a masked-minus-unmasked difference also contains any difference of CPU class, whose size alone is recorded in D.19.6.

Table 40: The per-window mask, masked minus unmasked, per rung, model and input set, same design, 866 days. Each cell: QLIKE difference in % of the unmasked QLIKE [95 % interval] over the sign(s) Sharpe difference at the midpoint [95 % interval]. The unmasked rungs were not pinned to a CPU class.

model	T10 masked − unmasked	T1 masked − unmasked	RS10 masked − unmasked	RS1 masked − unmasked	LSTM10 masked − unmasked
baseline					
	+0.5 [−0.6, +1.5]	0.0 [−0.9, +0.9]	−0.2 [−1.1, +0.6]	−0.2 [−1.1, +0.6]	
LightGBM	−0.18 [−0.80, +0.30] +0.4 [−0.3, +1.0]	−0.25 [−0.74, +0.15] +0.3 [−0.3, +0.9]	+0.08 [−0.09, +0.27] −0.2 [−1.9, +1.4]	−0.04 [−0.13, +0.04] −0.7 [−2.3, +1.0]	
XGBoost	−0.01 [−0.21, +0.16] +0.4 [−0.3, +1.2]	−0.07 [−0.30, +0.11] 0.0 [−0.7, +0.7]	−0.03 [−0.48, +0.41] −0.1 [−1.5, +1.4]	+0.16 [−0.06, +0.40] +0.4 [−1.1, +1.9]	
random forest	0.00 [−0.25, +0.25]	+0.06 [−0.16, +0.31]	+0.16 [−0.46, +0.79]	+0.32 [−0.26, +0.88]	
					+0.2 [−2.9, +4.2]
LSTM					−0.03 [−0.55, +0.49]
live_feasible					
	+1.0 [−0.4, +2.3]	−1.0 [−2.4, +0.3]	−0.9 [−2.7, +0.8]	−0.1 [−2.1, +1.8]	
LightGBM	−0.17 [−0.80, +0.39] +0.3 [−0.5, +1.3]	−0.06 [−0.49, +0.37] +1.3 [+0.5, +2.3]	−0.12 [−0.67, +0.43] −1.1 [−3.6, +1.2]	+0.01 [−0.55, +0.58] −1.1 [−4.0, +1.4]	
XGBoost	−0.05 [−0.37, +0.33] +0.2 [−0.8, +1.1]	−0.36 [−0.73, −0.06] +0.1 [−0.9, +1.1]	+0.16 [−0.21, +0.60] +12.8 [−0.7, +32.8]	−0.01 [−0.29, +0.28] +14.7 [−0.2, +37.1]	
random forest	−0.08 [−0.36, +0.20]	+0.10 [−0.17, +0.44]	+0.37 [−0.48, +1.24]	−0.09 [−0.77, +0.71]	
					+0.3 [−3.0, +3.4]
LSTM					−0.05 [−0.69, +0.57]
all_features					
	−0.7 [−2.3, +1.1]	+1.5 [−0.3, +3.4]	+2.7 [−0.4, +6.2]	+0.7 [−3.5, +4.2]	
LightGBM	+0.09 [−0.34, +0.53] +0.9 [−0.2, +2.1]	−0.16 [−0.73, +0.35] +0.1 [−0.7, +1.0]	+0.06 [−0.59, +0.63] +6.5 [+1.8, +11.6]	−0.30 [−0.95, +0.34] +6.2 [+1.6, +11.3]	
XGBoost	−0.36 [−0.87, +0.08] +0.1 [−0.8, +1.0]	+0.33 [−0.11, +0.80] +0.9 [0.0, +1.7]	−0.44 [−0.92, +0.04] +9.4 [+0.9, +23.1]	−0.36 [−0.85, +0.09] +10.1 [+0.6, +26.5]	
random forest	−0.27 [−0.62, +0.06]	−0.17 [−0.70, +0.29]	+0.31 [−0.29, +0.94]	+0.16 [−0.29, +0.72]	
					−17.1 [−33.2, −4.2]
LSTM					−0.01 [−0.84, +0.83]

D.19.3 Refit cadence

With the mask in place, the trees refit every session, as the per-bar linear arms do, and every 10th session as before (first two columns of Table 41). For the shipped configuration, T1 minus T10 ranges from −3.6 % to −0.8 % in QLIKE, with 3 of 9 intervals below zero and 0 above; 1 Sharpe interval lies above zero and 0 below. For the random search, RS1 minus RS10 ranges from −2.2 % to +0.2 %, with 1 of 9 QLIKE intervals below zero and 0 above; no Sharpe interval excludes zero (0 below, 0 above). The per-bar LSTM refit every session against every 10th: −1.9 % [−3.5, −0.3] (`baseline`), −2.5 % [−7.8, +3.2] (`live_feasible`), −10.1 % [−18.0, −2.0] (`all_features`); 2 of 3 intervals below zero, and no Sharpe interval excludes zero (0 below, 0 above).

Against the per-bar ridge, none of the 36 masked tree rows has a QLIKE interval below zero (0) and 18 have one above; the lowest relative QLIKE is T1 LightGBM `live_feasible` at −5.0 % [−11.5, +0.4]; no Sharpe interval excludes zero (0 below, 0 above). The daily-refit LSTM minus the ridge is +6.8 % in QLIKE on `baseline`, +18.6 % on `live_feasible` and +30.0 % on `all_features`, with 3 of 3 intervals above zero.

Table 41: The masked ladder: refit every session minus every 10th (shipped configuration; random search; the per-bar LSTM in the first column), and random search minus the shipped configuration, both refit every session. Cells as in Table 40, the second forecast of each pair being the comparator.

model	T1 − T10 (refit 1 vs 10)	RS1 − RS10 (refit 1 vs 10)	RS1 − T1 (search vs shipped)
baseline			
	−0.8 [−2.3, +0.7]	+0.1 [−0.8, +1.0]	+7.2 [+2.1, +14.0]
LightGBM	+0.18 [−0.19, +0.52]	−0.29 [−0.76, +0.09]	−0.28 [−0.89, +0.30]
	−0.8 [−1.6, 0.0]	−0.8 [−1.7, +0.2]	+5.2 [+0.2, +12.8]
XGBoost	+0.03 [−0.16, +0.29]	+0.10 [−0.11, +0.38]	+0.27 [−0.18, +0.76]
	−1.3 [−3.3, +0.8]	−0.7 [−1.7, +0.3]	+4.6 [−2.2, +14.1]
random forest	+0.27 [−0.42, +1.07]	+0.19 [−0.20, +0.57]	+0.08 [−0.55, +0.67]
	−1.9 [−3.5, −0.3]		
LSTM (1 vs 10)	+0.09 [−0.17, +0.34]		
live_feasible			
	−3.6 [−7.4, −0.8]	+0.2 [−1.5, +1.6]	+4.1 [−0.2, +8.9]
LightGBM	−0.17 [−1.11, +0.59]	+0.16 [−0.19, +0.58]	+0.21 [−0.61, +1.07]
	−0.9 [−3.0, +0.8]	−1.9 [−5.2, +0.1]	+6.5 [+1.2, +13.0]
XGBoost	+0.26 [−0.05, +0.61]	−0.13 [−0.52, +0.29]	−0.19 [−0.77, +0.36]
	−3.0 [−6.5, +0.2]	−2.2 [−4.9, +0.3]	+17.9 [+2.2, +41.5]
random forest	−0.19 [−0.88, +0.48]	+0.16 [−0.32, +0.62]	+0.09 [−0.66, +0.91]
	−2.5 [−7.8, +3.2]		
LSTM (1 vs 10)	−0.17 [−0.80, +0.48]		
all_features			
	−2.5 [−7.2, +0.9]	−1.7 [−5.1, +0.7]	+3.6 [−0.4, +8.5]
LightGBM	−0.18 [−1.14, +0.77]	−0.27 [−0.61, +0.06]	−0.26 [−1.05, +0.61]
	−2.9 [−5.7, −1.0]	−1.8 [−3.6, −0.4]	+10.7 [+4.6, +17.5]
XGBoost	+0.52 [+0.10, +1.02]	+0.04 [−0.34, +0.45]	−0.47 [−1.06, +0.11]
	−2.0 [−5.5, +1.1]	−2.1 [−5.7, +0.3]	+12.6 [+1.4, +30.0]
random forest	+0.23 [−0.29, +0.73]	−0.14 [−0.61, +0.28]	+0.17 [−0.59, +0.98]
	−10.1 [−18.0, −2.0]		
LSTM (1 vs 10)	+0.37 [−0.39, +1.22]		

D.19.4 Tuning: random search and Optuna

Random search against the shipped configuration. The random search re-chooses each tree's configuration from a fixed set of candidates on the validation tail of its window. Both refit every session, RS1 minus T1 ranges from +3.6 % to +17.9 % in QLIKE, with the interval above zero in 6 of 9 rows and below zero in 0; no Sharpe interval excludes zero (0 below, 0 above; last column of Table 41).

Optuna. The random search was replaced by Optuna's TPE sampler: at every tuning point a fresh study of 50 sequential trials (trial 1 the shipped configuration, trials 1–10 TPE's random start-up) minimises validation MSE on the tail of the tuning window after an embargo; the trees refit every session with the configuration in force; a new tuning point is drawn every TUNE_PER sessions (TUNE_PER 1, 5, 25 or 250), and at TUNE_PER 1 and 25 the best of the first 10 and the first 25 trials is kept beside the best of all 50. Every Optuna run is masked and ran on one CPU class.

Are 50 trials enough? Over the 1,469 tuning points of each arm (Table 42), the best of the 50 trials falls in trials 41–50 at 27–40 % of the points, against 20 % if the trials were

exchangeable draws, and the shipped configuration is the best at no more than 1.1 % of them. The last 25 trials lower the best validation MSE by 1.1–2.1 % on average (medians 0.4–1.2 %) and the last 10 by 0.3–0.7 % (medians at most 0.0 %); the pick of the first 25 differs from the pick of all 50 at 61–74 % of the points. Out of sample (Table 43), of the 54 budget comparisons (best of 50 against 10, against 25, and 25 against 10, at TUNE_PER 1 and 25) 0 have a QLIKE interval below zero and 1 above; 1 Sharpe interval lies below zero and 0 above.

How often to tune. Against a new tuning point every 250 sessions, tuning every 1, 5 or 25 sessions (every path refit every session; Table 44) has 8 of 27 QLIKE intervals below zero — LightGBM `baseline` (TUNE_PER 5, 25); random forest `live_feasible` (TUNE_PER 1, 5, 25); random forest `all_features` (TUNE_PER 1, 5, 25) — and 0 above; no Sharpe interval excludes zero (0 below, 0 above). Choosing the configuration by validation QLIKE instead of MSE (at TUNE_PER 25) has 2 of 9 QLIKE intervals below zero and 0 above, 1 Sharpe interval below zero and 0 above.

Optuna against the other rungs. Table 45 counts the intervals. Against the shipped configuration refit every session (T1), 19 of 36 Optuna QLIKE intervals lie above zero and 0 below. Against the random search refit every session (RS1), 8 lie below zero — XGBoost `live_feasible` (TUNE_PER 250); XGBoost `all_features` (TUNE_PER 250); random forest `live_feasible` (TUNE_PER 1, 5, 25); random forest `all_features` (TUNE_PER 1, 5, 25) — and 3 above. Against the per-bar ridge, 0 lie below zero and 14 above, with 2 Sharpe intervals below zero and 0 above.

Where the picks sit. Table 46 gives, per axis, the share of the best-of-50 picks in the outermost twentieth of the search space at either end (on the log scale for a log axis; for the forest's depth, the smallest and the largest choice), beside the share of TPE's random start-up draws there (the prior). The penalties `lambda_l2` and `reg_lambda` are picked at their upper end in no more than 0.0 % of points against a prior of at least 5.1 %; LightGBM's `min_child_samples` sits at its lower end in 14–23 % of points, and the forest picks unbounded depth in 20–28 % (prior 17 %).

Table 42: Is the Optuna budget enough? Per arm, over all tuning points: the median trial that holds the best of 50, the share of points where it is the shipped configuration and where it falls in trials 41–50 (20 % if exchangeable), the relative gain in the best validation MSE from the first 10, 25 and 40 trials to all 50 (mean / median), and the share of points whose best of 25 differs from the best of 50.

arm	best trial (median)	best = shipped (%)	best in 41–50 (%)	val. MSE gain 10→50 (%)	val. MSE gain 25→50 (%)	val. MSE gain 40→50 (%)	pick of 25 ≠ of 50 (%)
LightGBM `baseline`	35	0.1	36.8	5.22 / 4.42	1.66 / 0.99	0.50 / 0.00	72
LightGBM `live_feasible`	34	0.3	34.5	5.94 / 4.93	2.04 / 1.22	0.63 / 0.00	70
LightGBM `all_features`	33	0.7	31.9	5.96 / 4.86	2.15 / 1.13	0.69 / 0.00	66
XGBoost `baseline`	36	0.2	40.2	4.92 / 3.94	1.82 / 1.13	0.56 / 0.00	74
XGBoost `live_feasible`	34	0.2	34.7	5.54 / 4.70	1.94 / 1.14	0.62 / 0.00	71
XGBoost `all_features`	33	1.0	33.1	5.24 / 4.29	1.95 / 1.10	0.63 / 0.00	68
random forest `baseline`	32	1.1	26.8	4.76 / 3.41	1.06 / 0.41	0.29 / 0.00	61
random forest `live_feasible`	32	0.4	30.7	4.64 / 3.76	1.42 / 0.82	0.40 / 0.00	67
random forest `all_features`	33	0.3	31.8	5.05 / 4.16	1.42 / 0.82	0.43 / 0.00	67

Table 43: The budget out of sample: best of 50 trials minus best of 10 and of 25, at TUNE_PER 1 and 25. Cells as in Table 40.

arm	TUNE_PER 1: 50 vs 10	TUNE_PER 1: 50 vs 25	TUNE_PER 25: 50 vs 10	TUNE_PER 25: 50 vs 25
	−0.3 [−2.7, +2.2]	0.0 [−1.9, +2.0]	−0.6 [−3.0, +1.8]	+1.4 [−0.1, +3.0]
LightGBM `baseline`	+0.04 [−0.45, +0.48] +0.4 [−2.8, +3.8]	+0.09 [−0.19, +0.37] +0.7 [−1.8, +3.0]	+0.17 [−0.24, +0.59] +3.2 [−0.6, +7.5]	+0.08 [−0.30, +0.44] +0.6 [−2.4, +4.0]
LightGBM `live_feasible`	−0.64 [−1.48, +0.22] −1.5 [−5.4, +1.6]	−0.23 [−0.90, +0.35] −0.1 [−2.4, +2.0]	+0.02 [−0.66, +0.65] +0.3 [−3.1, +3.4]	−0.12 [−0.65, +0.39] +2.3 [−0.6, +5.8]
LightGBM `all_features`	−0.24 [−0.92, +0.42] +3.4 [+0.5, +6.5]	−0.34 [−0.76, +0.07] +1.3 [−0.2, +3.1]	+0.10 [−0.49, +0.66] +1.4 [−1.0, +3.9]	+0.58 [−0.14, +1.45] +1.1 [−0.7, +3.1]
XGBoost `baseline`	−0.13 [−0.86, +0.50] −1.5 [−4.9, +1.3]	−0.09 [−0.79, +0.47] −0.9 [−2.9, +0.9]	−0.07 [−0.62, +0.48] −0.7 [−4.9, +3.0]	+0.06 [−0.45, +0.68] −0.5 [−2.3, +1.3]
XGBoost `live_feasible`	−0.42 [−1.40, +0.50] +0.2 [−2.9, +2.8]	+0.02 [−0.44, +0.54] −0.3 [−2.5, +1.6]	+0.28 [−0.52, +1.17] +2.5 [−2.3, +7.8]	+0.49 [−0.18, +1.20] +1.4 [−2.6, +5.3]
XGBoost `all_features`	+0.05 [−0.46, +0.56] −0.2 [−2.8, +2.3]	+0.11 [−0.35, +0.60] +0.9 [−0.8, +2.9]	−0.64 [−1.35, +0.03] +0.8 [−1.5, +3.0]	−0.38 [−1.15, +0.28] +1.0 [−0.7, +2.7]
random forest `baseline`	−0.03 [−0.51, +0.50] +0.2 [−2.0, +2.5]	−0.07 [−0.52, +0.35] −0.4 [−2.0, +1.1]	−0.15 [−0.63, +0.33] +2.8 [−2.4, +10.1]	+0.11 [−0.26, +0.50] −0.7 [−2.1, +0.7]
random forest `live_feasible`	−0.25 [−0.68, +0.13] −1.9 [−4.5, +0.4]	−0.27 [−0.63, +0.07] −1.1 [−2.8, +0.4]	−0.57 [−1.05, −0.07] +1.2 [−2.3, +6.1]	−0.12 [−0.63, +0.39] −0.4 [−1.9, +1.0]
random forest `all_features`	+0.33 [−0.19, +0.89]	−0.02 [−0.55, +0.43]	+0.30 [−0.33, +0.96]	+0.12 [−0.20, +0.48]

Table 44: How often to tune: Optuna (best of 50) with a new tuning point every 1, 5 and 25 sessions minus every 250 sessions, every path refit every session; the second column is the QLIKE at TUNE_PER 250. Cells as in Table 40.

arm	QLIKE TUNE_PER 250	TUNE_PER 1 vs 250	TUNE_PER 5 vs 250	TUNE_PER 25 vs 250
		−4.9 [−11.4, +0.4]	−5.8 [−12.3, −0.7]	−4.4 [−9.3, −0.3]
LightGBM `baseline`	0.1142	+0.25 [−0.41, +0.83] −2.8 [−9.2, +2.9]	−0.08 [−0.68, +0.49] −3.4 [−9.6, +1.8]	+0.09 [−0.61, +0.72] +1.3 [−3.2, +6.5]
LightGBM `live_feasible`	0.1055	−0.55 [−1.42, +0.30] +0.4 [−4.9, +4.7]	−0.13 [−0.66, +0.38] −0.3 [−6.1, +4.8]	−0.10 [−0.56, +0.39] +5.4 [−1.0, +14.5]
LightGBM `all_features`	0.1041	−0.13 [−0.72, +0.45] −1.7 [−8.3, +4.0]	−0.31 [−1.23, +0.62] −3.4 [−9.4, +1.3]	+0.05 [−0.56, +0.68] −4.0 [−10.7, +1.0]
XGBoost `baseline`	0.1101	−0.34 [−1.11, +0.33] −1.6 [−7.5, +3.6]	−0.32 [−0.84, +0.13] −1.0 [−6.8, +4.3]	−0.13 [−0.59, +0.31] +0.1 [−5.0, +5.3]
XGBoost `live_feasible`	0.1024	−0.67 [−1.62, +0.32] +2.1 [−3.3, +6.9]	−0.62 [−1.51, +0.30] −0.2 [−5.4, +4.5]	−0.04 [−0.55, +0.54] +2.3 [−0.5, +5.1]
XGBoost `all_features`	0.1032	−0.31 [−1.26, +0.59] −0.5 [−8.4, +5.8]	+0.05 [−0.51, +0.67] −2.7 [−10.5, +2.7]	−0.46 [−1.24, +0.27] −1.0 [−6.3, +3.2]
random forest `baseline`	0.1074	+0.01 [−0.63, +0.74] −7.5 [−16.7, −1.1]	+0.13 [−0.61, +0.90] −7.1 [−16.3, −0.7]	−0.18 [−0.82, +0.41] −4.8 [−10.5, −0.6]
random forest `live_feasible`	0.1083	−0.06 [−0.56, +0.48] −13.7 [−33.8, −1.1]	−0.15 [−0.57, +0.30] −13.6 [−32.5, −1.5]	−0.62 [−1.44, +0.09] −11.9 [−27.9, −1.3]
random forest `all_features`	0.1184	−0.22 [−0.74, +0.27]	−0.04 [−0.50, +0.38]	+0.15 [−0.22, +0.57]

Table 45: Optuna (best of 50, each TUNE_PER) minus the shipped trees refit every session (T1), the random search refit every session (RS1) and the per-bar ridge: the number of model × input-set arms whose 95 % interval lies wholly below or above zero.

Optuna −	TUNE_PER	arms	QLIKE below 0	QLIKE above 0	Sharpe mid below 0	Sharpe mid above 0
T1	1	9	0	5	0	0
T1	5	9	0	5	0	0
T1	25	9	0	4	0	0
T1	250	9	0	5	0	0
all		36	0	19	0	0
RS1	1	9	2	0	0	0
RS1	5	9	2	0	1	0
RS1	25	9	2	2	0	0
RS1	250	9	2	1	0	0
all		36	8	3	1	0
per-bar ridge	1	9	0	3	0	0
per-bar ridge	5	9	0	3	1	0
per-bar ridge	25	9	0	3	1	0
per-bar ridge	250	9	0	5	0	0
all		36	0	14	2	0

Table 46: Where the best-of-50 picks sit: share of picks in the outermost twentieth of each axis at its low and high end, per input set (`baseline` / `live_feasible` / `all_features`), beside the share of the random start-up draws there (the prior), and the median pick. †A pseudo-axis: the early-stopped number of rounds at the cap set by the learning rate.

axis	search space	picks at low bound (%) base / live / all	picks at high bound (%) base / live / all	prior low (%)	prior high (%)	median pick base / live / all
LightGBM						
`num_leaves`	[2, 128] log int	20.6 / 7.6 / 5.7	2.0 / 3.3 / 3.3	11.1	4.8	6 / 14 / 15
`min_child_samples`	[1, 256] log int	19.1 / 22.7 / 14.2	0.7 / 0.5 / 0.8	18.3	4.2	5 / 5 / 12
`feature_fraction`	[0.05, 1]	1.4 / 5.9 / 5.0	7.9 / 5.4 / 5.3	5.2	5.3	0.646 / 0.486 / 0.553
`bagging_fraction`	[0.3, 1]	5.2 / 4.8 / 4.1	5.0 / 3.8 / 4.6	4.7	4.9	0.638 / 0.644 / 0.666
`lambda_l2`	[0.001, 10000] log	5.4 / 5.2 / 5.6	0.0 / 0.0 / 0.0	5.5	5.1	0.633 / 0.594 / 0.637
`learning_rate`	[0.005, 0.1] log	3.1 / 4.4 / 3.7	6.3 / 4.2 / 4.7	4.9	5.2	0.0256 / 0.02 / 0.0203
`rounds_cap`†	early-stopped rounds ≤ cap(lr)	— / — / —	1.1 / 0.5 / 0.6	—	16.9–17.9	272 / 324 / 294
XGBoost						
`max_depth`	[1, 8] int	21.4 / 8.8 / 7.4	6.1 / 9.5 / 7.4	12.2	12.5	3 / 4 / 4
`min_child_weight`	[0.1, 256] log	6.1 / 5.6 / 4.9	0.4 / 0.6 / 1.0	5.3	4.8	1.45 / 1.92 / 2.42
`subsample`	[0.15, 1]	6.9 / 5.4 / 3.1	3.9 / 4.2 / 4.8	5.2	5.3	0.573 / 0.598 / 0.618
`colsample_bytree`	[0.05, 1]	0.2 / 4.5 / 3.3	8.3 / 4.3 / 7.0	4.7	4.9	0.691 / 0.538 / 0.586
`reg_lambda`	[0.0001, 10000] log	5.5 / 6.3 / 5.9	0.0 / 0.0 / 0.0	5.5	5.1	0.101 / 0.125 / 0.144
`learning_rate`	[0.005, 0.1] log	3.5 / 4.2 / 3.6	6.3 / 4.2 / 3.9	4.9	5.2	0.0241 / 0.0194 / 0.0213
`rounds_cap`†	early-stopped rounds ≤ cap(lr)	— / — / —	2.0 / 1.6 / 1.8	—	18.5–20.4	313 / 354 / 323
random forest						
`max_features`	[0.02, 1]	0.3 / 2.6 / 2.9	3.6 / 4.3 / 5.9	5.1	4.8	0.516 / 0.382 / 0.395
`min_samples_leaf`	[1, 256] log int	17.5 / 29.0 / 14.1	0.0 / 0.0 / 0.1	17.7	4.4	4 / 3 / 7
`max_depth`	{2, 4, 8, 16, 32, None}	0.1 / 0.0 / 0.3	20.1 / 28.4 / 25.3	17.0	16.5	16 / 32 / 32

D.19.5 The intraday-sequence LSTM

The per-bar LSTM reads a sequence of the last sessions' 16:00-row vectors of the per-bar design, the HAR ladders among them. The intraday LSTM reads instead the last N half-hour bars of the near-24-hour panel ending with the 15:30 bar, one step per bar, with bar-level inputs — the bar's adjusted target, the adjusted exogenous values of the input set, their availability and activity flags, the calendar columns and a half-hour clock — so that the network, not the ladders, does the aggregation over time. N is tuned in $\{13, 48, 96\}$ with the per-bar LSTM's grid, selection rule, window, seed average and daily refits, and the per-window mask acts on the step columns (kept at the refits: `baseline` 19 of 19, range 16–19; `live_feasible` 40 of 54, 37–41; `all_features` 80 of 120, 74–82). The chosen N moves over time: over the 18 tuning points of the three input sets (the selection rule of record) it is 13 at 3, 48 at 9 and 96 at 6.

Against the per-bar LSTM (the masked, daily-refit table), the intraday LSTM's QLIKE interval lies below zero in 1 of 3 input sets and above zero in 0; against the per-bar ridge it lies above zero in 3 of 3. No Sharpe interval excludes zero (against the per-bar LSTM 0 below and 0 above; against the ridge 0 below and 0 above; Table 47).

Table 47: The intraday-sequence LSTM (selection rule of record) beside the per-bar LSTM and the per-bar ridge: QLIKE and sign(s) Sharpe (midpoint) levels, and the paired differences, cells as in Table 40.

input set	QLIKE intraday	QLIKE per-bar LSTM	QLIKE ridge	Sharpe mid intraday	Sharpe mid per-bar LSTM	Sharpe mid ridge	intraday − per-bar LSTM	intraday − ridge
							+2.2 [−3.4, +9.0]	+9.2 [+2.5, +17.2]
`baseline`	0.1073	0.1049	0.0982	0.81	1.26	1.22	−0.45 [−1.07, +0.20]	−0.41 [−1.18, +0.41]
							−7.5 [−20.9, +4.7]	+9.7 [+0.1, +20.0]
`live_feasible`	0.1103	0.1193	0.1005	1.53	0.91	1.90	+0.62 [−0.26, +1.54]	−0.37 [−1.33, +0.60]
							−14.3 [−26.9, −2.0]	+11.5 [+2.2, +22.5]
`all_features`	0.1119	0.1305	0.1004	0.97	1.33	1.66	−0.37 [−1.42, +0.76]	−0.70 [−1.74, +0.35]

D.19.6 Reproducibility across CPUs

The tree and LSTM numbers depend on the CPU's vector class. The executor builds the design matrix X with numpy's runtime-dispatched loops, and X comes out as one of two byte strings: one on every machine with AVX-512 and another on every machine without it (12 machines probed on two clusters). The two differ in 28 of 232 `live_feasible` columns and 94 of 628 `all_features` columns, by at most 2.6×10^{-11} in absolute value; the target and the kept-column sets are the same on every machine. Within a class the Optuna trials and forecasts reproduce bit for bit across clusters (900/900 trials, 1,620/1,620 forecasts). Across classes the fits part: from the same stage-1 records the random forest's forecasts are identical in 360/360, XGBoost's in 344/360 (largest relative difference 0.033) and LightGBM's in 5/360 (0.103); re-running 20 tuning points on the other class changes LightGBM's best-of-50 pick at 18 of them (XGBoost 0, random forest 0), while a re-run on the same class is bit-identical at all 20. A whole masked LightGBM `live_feasible` run on the other class moves the forecast by up to 4.5 % (T10) and 5.5 % (T1), QLIKE by -0.1 % [-0.8, $+0.5$] and -0.1 % [-0.8, $+0.8$], and the Sharpe ratio by -0.26 [-0.68, $+0.11$] and $+0.07$ [-0.30, $+0.42$] (Table 48).

So every canonical chunk ran on one CPU class: the census finds 1 class among the masked ladder's 327 chunks and 1 among Optuna's 214 chunk-arms (the masked ladder's 6 class-gate chunks ran on the other class on purpose), and the intraday LSTM, whose first attempt on mixed nodes was stopped by its tuning cache's fingerprint, was re-run on one class (the mixed attempt agrees on 17/18 tuning points and 19/20 refit chunks). Optuna's first pass had landed on several classes; it is kept as the class-mixed run beside the one-class run: 35 of its 81 forecast tables differ, with 2 QLIKE intervals below zero and 3 above (one class minus mixed) and 0 Sharpe intervals excluding zero.

Table 48: CPU-class checks: what was compared and what reproduced. "Bit-identical a/b": a of b items equal bit for bit; relative differences are of the forecast; QLIKE and Sharpe differences (other class minus the fleet's class, in % of QLIKE and Sharpe units) with 95 % paired intervals.

check	result
design X, machines of the two vector classes	`live_feasible`: 28 of 232 columns differ; `all_features`: 94 of 628; largest absolute difference 2.6×10^{-11}; target and kept-column sets identical
Optuna, same class on two clusters	stage-1 trials bit-identical 900/900 (replicate 900/900); stage-2 forecasts 1,620/1,620
Optuna, across classes on one cluster (same inputs)	stage-1 trials bit-identical: LightGBM 100/300, XGBoost 211/300, random forest 283/300; stage-2 forecasts from the same records: 5/360 (max rel. 0.103), 344/360 (0.033), 360/360
Optuna, first 20 tuning points re-run on the other class	best-of-50 pick changes: LightGBM 18, XGBoost 0, random forest 0; XGBoost forecast max rel. 7.3×10^{-3}; same-class re-run bit-identical at 20 of 20 points for all three
masked LightGBM `live_feasible`, whole run on the other class	T10 / T1: max rel. forecast difference 4.5 / 5.5 %; QLIKE −0.1 [−0.8, +0.5] / −0.1 [−0.8, +0.8] %; ΔSharpe mid −0.26 [−0.68, +0.11] / +0.07 [−0.30, +0.42]
intraday LSTM, mixed-class attempt vs one class	tuning points bit-identical 17/18; refit chunks 19/20 (the others ran on the other class)
Optuna, one class vs class-mixed first pass	35 of 81 tables differ (max rel. LightGBM 0.65, XGBoost 0.17, random forest 0.0025); QLIKE intervals below / above 0: 2 / 3; Sharpe mid intervals excluding 0: 0

D.19.7 The master table before and after

The master table of the closing strategy was rebuilt from the new tables, on the same 866 days and with the same resampled days for every interval. The headline forecast, the per-bar ridge on `live_feasible`, is unchanged: QLIKE 0.1005 before and 0.1005 after, sign(s) Sharpe 1.90 at the midpoint and 1.44 crossed, positions changed on 0 of 866 days (largest relative change of its recalibrated forecast 1.0×10^{-12}). Table 49 summarises the 136 forecasts present before and after, by family; 114 forecasts are new (the daily-refit and random-search-daily trees, the Optuna rungs and the intraday LSTM). The own Sharpe ratio of 3 forecasts moved with the interval excluding zero: per-bar XGBoost [live_feasible]: 1.70 → 1.27, −0.43 [−0.94, −0.03]; per-bar XGBoost [baseline]: 1.12 → 0.92, −0.21 [−0.42, −0.04]; per-bar LSTM [baseline]: 0.61 → 1.26, +0.65 [+0.05, +1.27].

Against the headline, of the 219 other forecasts 0 have a Sharpe-difference interval wholly above zero at the midpoint and 0 crossed, and 43 have one wholly below zero at the midpoint. On forecast accuracy 3 forecasts have a QLIKE interval against it wholly below zero, from the family HAR-ladder variants, none of them with a Sharpe interval above zero: per-bar elastic net [live_feasible], HAR ladder base3: QLIKE 0.0934 (−7.1 %, interval on the daily difference [−0.0144, −0.0019]), ΔSharpe mid −0.33 [−0.91, +0.34]; per-bar elastic net [live_feasible], HAR ladder base2: QLIKE 0.0944 (−6.1 %, interval on the daily difference

[−0.0122, −0.0012]), ΔSharpe mid −0.15 [−0.82, +0.59]; per-bar lasso [live_feasible], HAR ladder base2: QLIKE 0.0948 (−5.8 %, interval on the daily difference [−0.0126, −0.0001]), ΔSharpe mid −0.12 [−0.76, +0.55]. 37 have a QLIKE interval wholly above zero.

Table 49: The master table before and after the campaign, per family of the forecasts present in both: what changed, the QLIKE change in % and the sign(s) Sharpe change (midpoint), median and range over the family, the number of paired Sharpe-change intervals wholly above / below zero, and the trade positions changed (of 866 days).

family	keys	what changed	QLIKE change (%) median [min, max]	ΔSharpe mid median [min, max]	ΔSharpe interval above / below 0	positions changed median [max]
paper	9	nothing	0.00 [0.00, 0.00]	0.00 [0.00, 0.00]	0 / 0	0 [0]
per-bar linear	10	design de-dup	0.00 [0.00, +0.05]	0.00 [0.00, 0.00]	0 / 0	0 [0]
pooled twin	9	nothing	0.00 [0.00, 0.00]	0.00 [0.00, 0.00]	0 / 0	0 [0]
VIX-only family	21	design de-dup	0.00 [−0.04, +0.09]	0.00 [−0.05, +0.06]	0 / 0	0 [2]
per-bar tree (untuned)	9	design de-dup + per-window mask + CPU-class pinning	+0.63 [+0.26, +1.64]	−0.10 [−0.43, +0.49]	0 / 2	26 [49]
per-bar tree (tuned)	18	design de-dup + per-window mask + CPU-class pinning	+0.94 [−5.79, +16.14]	+0.07 [−0.27, +0.48]	0 / 0	48 [80]
LSTM	6	design de-dup + per-window mask + refit cadence 10 → 1 + CPU-class pinning	−4.40 [−17.46, −1.68]	+0.40 [−0.16, +0.80]	1 / 0	128 [172]
direct rest-of-day at 15:30 (check)	9	design de-dup + CPU-architecture pinning to the per-bar twin	0.00 [0.00, 0.00]	0.00 [−0.01, 0.00]	0 / 0	0 [1]
implied-vol representations	15	design de-dup	0.00 [0.00, +0.06]	0.00 [0.00, 0.00]	0 / 0	0 [0]
HAR-ladder variants	9	design de-dup	0.00 [−0.01, +0.01]	0.00 [0.00, 0.00]	0 / 0	0 [1]
chain-period buckets	21	design de-dup	0.00 [0.00, 0.00]	0.00 [0.00, 0.00]	0 / 0	0 [0]

D.19.8 Feature importance and density on the new design

The importance study of the 16:00 forecast (MDI, split count, permutation, TreeSHAP) and the dense-versus-sparse study were re-run on the new design with every tree fit masked, with the first pass's refits, held-out tails, permutation scheme and seeds. In the trees (Table 50) the `har_ma_*` ladder still leads 52 of 54 input-set × model × measure rows at the series level. The table gives its MDI, TreeSHAP and permutation shares before and after, beside the part of the old ladder's MDI that sat on the `har_ma_k_x_close` copies (column five). The rank correlation of the series rankings, old against new, has median 0.90 over the tree rows (lowest 0.66). No interval is computed for these share changes; the yardstick is the ridge, which solves the same problem in both runs, and whose permutation importance of the ladder moves by at most 1.7 points of the tail loss between the two sets of permutation draws.

In the density table (Table 51) the linear forecasts are the first pass's to 3.3×10^{-12}, so their effective number of inputs N_{eff} does not move; the trees' N_{eff} is given for the first pass, the de-duplicated design without the mask and with it, beside the share of the first pass's TreeSHAP that sat on the session-edge columns (last column). 20 of the first pass's 23 statements read the same on the new runs. LightGBM on all columns, new minus first pass: QLIKE +0.0004 [−0.0012, +0.0020] and Sharpe +0.04 [−0.30, +0.39] (`live_feasible`); QLIKE +0.0002 [−0.0012, +0.0016] and Sharpe +0.20 [−0.27, +0.77] (`all_features`).

Table 50: The `har_ma_*` ladder's importance in the trees, first pass (old design, unmasked) to the new design with the mask: MDI and TreeSHAP shares, permutation (P2, QLIKE) in % of the model's tail loss, the part of the old ladder's MDI that sat on the `_x_close` copies, and the ladder's MDI rank among the series.

input set	model	MDI share (%) old → new	TreeSHAP share (%) old → new	perm. P2 QLIKE (% of tail loss)	old `_x_close` share of the ladder's MDI (%)	MDI rank old → new
`live_feasible`	LightGBM	62 → 57	56 → 52	+268 → +219	29	1 → 1
`live_feasible`	XGBoost	73 → 69	63 → 59	+252 → +210	22	1 → 1
`live_feasible`	random forest	67 → 67	64 → 64	+331 → +320	49	1 → 1
`all_features`	LightGBM	57 → 52	47 → 44	+237 → +200	22	1 → 1
`all_features`	XGBoost	69 → 64	55 → 51	+224 → +187	22	1 → 1
`all_features`	random forest	62 → 62	58 → 58	+290 → +283	54	1 → 1

Table 51: Signal density before and after: effective number of inputs $N_{\text{eff}} = 1/\sum_j s_j^2$ of each forecast's mean absolute contributions (first pass; new design without the mask, trees only; new design with the mask), the fewest columns reproducing 95 % of the input-driven forecast's variance (k_{95}), the columns with any weight, and the first pass's share of the trees' mean |TreeSHAP| on the session-edge columns.

input set	model	N_{eff} first pass	N_{eff} de-dup, no mask	N_{eff} de-dup + mask	k_{95} old → new	columns with weight	first pass: TreeSHAP on session-edge (%)
`live_feasible`	ridge	40.7		40.7	22 → 22	138 → 138	
`live_feasible`	lasso	10.8		10.8	7 → 7	86 → 86	
`live_feasible`	elastic net	7.7		7.7	5 → 5	52 → 52	
`live_feasible`	LightGBM	19.1	13.9	14.1	12 → 10	150 → 139	13.0
`live_feasible`	XGBoost	12.7	9.2	9.7	10 → 8	146 → 138	12.8
`live_feasible`	random forest	13.2	6.8	6.8	6 → 4	174 → 147	29.1
`all_features`	ridge	124.2		124.2	64 → 64	379 → 379	
`all_features`	lasso	21.3		21.3	6 → 6	204 → 204	
`all_features`	elastic net	39.4		39.4	20 → 20	238 → 238	
`all_features`	LightGBM	25.8	20.4	20.4	16 → 12	411 → 367	8.6
`all_features`	XGBoost	16.4	11.9	13.0	15 → 11	392 → 358	11.9
`all_features`	random forest	16.4	8.5	8.4	9 → 6	566 → 388	27.6

D.19.9 Compute used

Table 52 lists the compute each stage reported. Cluster A is the batch cluster that ran the tree and LSTM fleets (Slurm); cluster B the one that ran the linear and rest-of-day arms (SGE). Allocated CPU-hours count every core a job held for its whole run; used CPU-hours the busy share. The largest allocation is the masked Optuna run's (1422.6 allocated CPU-hours, at up to 900 CPUs at once).

Table 52: Compute used by the campaign's stages, as each stage recorded it (— = not reported). For cluster B, slot-hours are wall-clock hours of the job slots.

stage	cluster	allocated CPU-h	used CPU-h	peak CPUs	wall-clock (first start–last end)
trees T10 / T1 / RS10 / RS1, no mask	A	107.3	54.0	360	09-29 19:39–09-29 21:02 (1 h 23 min)
Optuna, no mask (stopped; not used)	A	928.3	—	1260	09-29 20:15–09-29 22:28 (2 h 13 min)
trees and LSTM with the mask, one CPU class	A	186.4	102.5	408	09-29 22:09–09-30 00:50 (2 h 41 min)
Optuna with the mask, all jobs	A	1422.6	—	900	09-29 22:07–09-30 04:05 (5 h 58 min)
of which the one-class re-run	A	274.3	—	860	09-30 02:44–09-30 04:05 (1 h 21 min)
intraday-sequence LSTM	A	197.3	159.2	340	09-29 23:31–09-30 03:46 (4 h 15 min)
cross-cluster check (Optuna)	A	18.2	—	73	not reported
cross-cluster check (Optuna)	B	29.1 slot-h	6.7	58 slots	not reported
rest-of-day arms, all entry clocks	B	2.06 slot-h	1.58	not reported	21:47–21:59 (11.2 min)
attribution control	B	0.21 slot-h	0.18	not reported	22:35–22:50 (14.7 min)
feature importance re-run	A	10.8	7.1	45	21:50–22:11
dense-vs-sparse re-run	A	4.1	—	20	18 min of job time
its local part	local	—	4.6	4 processes	69 min

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